

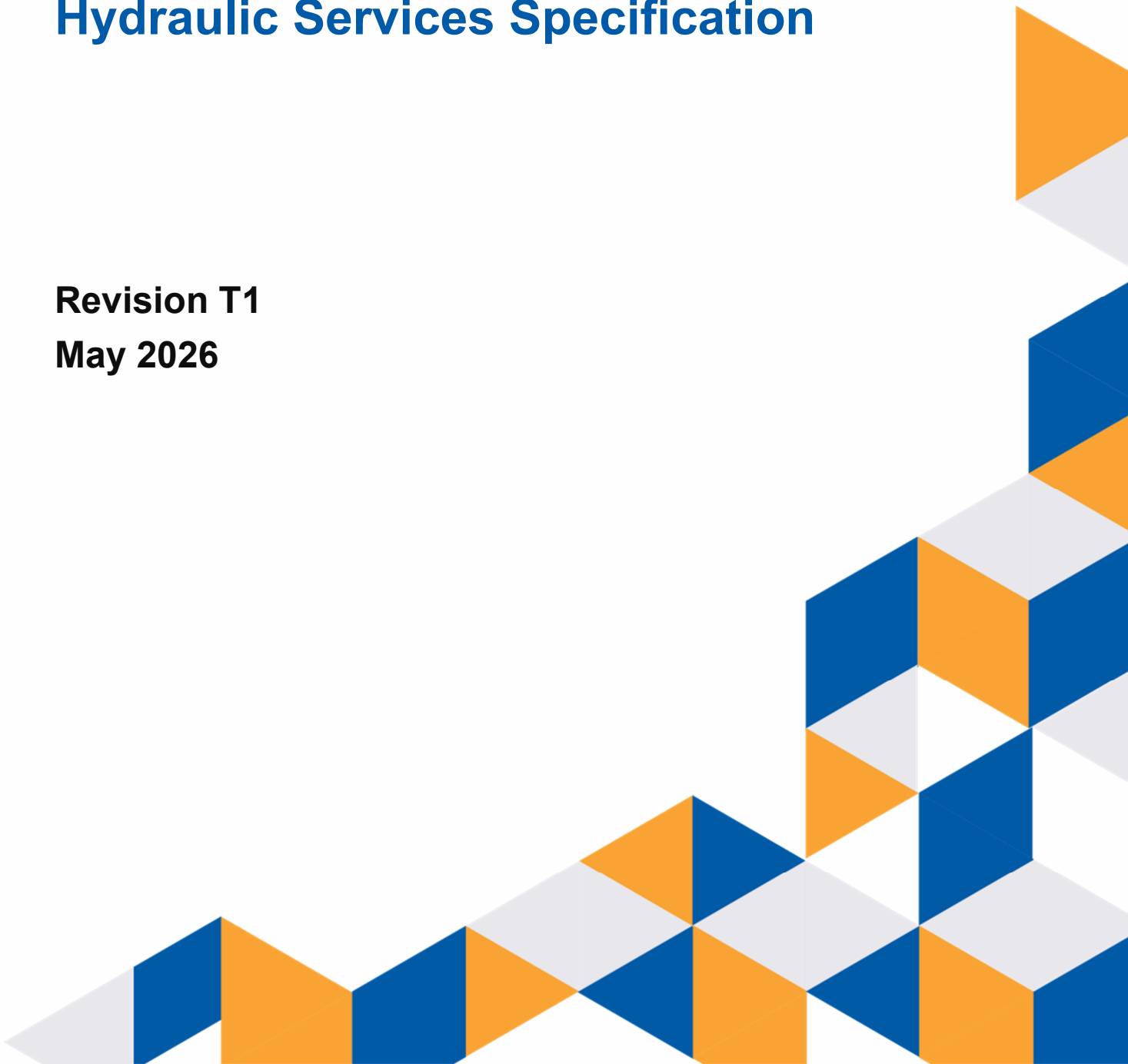


25015 – Rockhampton Sports Precinct Stage 1

Hydraulic Services Specification

Revision T1

May 2026



HK Solutions project references No:

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1. Introduction

1.1 Introduction

The following is a brief description only of the proposed Hydraulic services for the site:

The project involves the construction of a 2 single level building with external site works.

- A new sewer connection is required.
- A new water connection is required.
- A new hydrant system is being installed
- A grease arrestor and drainage is required
- A trade waste application will be required for the Grease Arrestor
(Refer to guideline at Rear of spec). This will need to be completed by the plumbing contractor in conjunction with the client.

1.2 Prior to the commencement of work on-site

It is strongly recommended that the Plumbing Contractor attend a meeting with the HK Solutions design and construction administration team to discuss the project.

Any options to replace the nominated fixtures, fittings, or materials nominated in this specification must be approved by the Superintendent, including pumps. Any non-notification will result in removing the items from the site, and the plumbing contractor will bear the costs associated with redundant work. Attached at the rear of this specification is a material change submission form. This form must be completed and approved by the client before any changes to this specification can occur. Completed forms will be emailed to HK Solutions, who will then forward them to the client.

The contractor must ensure that the builder is regularly notified of completed works to ensure all required hydraulic inspections are carried out at the appropriate times.

1.3 Limitations of drawings

The drawings referenced in this specification are not to be used for installation purposes. Any drawing labelled "Construction" is used by the plumbing contractor to prepare detailed shop drawings. The plumbing contractor is responsible for inspecting the site, obtaining all relevant measurements and other site-specific requirements.

It is the plumbing contractors responsibility to ensure the shop drawings are coordinated with the latest Architectural, structural and services drawings during the construction phase.

Refer to Section 4 of the specification for additional requirements.

2. Hydraulic inspections

2.1 Purpose of hydraulic review

The installation of hydraulic services and related items is subject to periodic reviews from the hydraulic Consultant on behalf of the client.

The intent of these reviews is to:

Provide the client with an overview of the quality and progression of the hydraulic installation,
Ensure an acceptable standard of workmanship is maintained,
Compliance is met with Australian standards and the Local Authority's requirements,
Ensure the installation of services is congruent to the hydraulic documentation and suitable for its intended purpose.

The results of the reviews will be documented in a report that is issued to the client, which will outline any inconsistencies with the hydraulic documentation and/or Australian standards.

2.2 Attendance at inspections

It is the responsibility of the hydraulic contractor to ensure that they have a company representative who is suitably qualified to present with the hydraulic consultant for the entirety of the inspection.

2.3 Frequency of inspections

As project sizes and scopes vary, the number of inspections required also changes.

Generally, an inspection will be required to review the following:

- In-ground stormwater drainage before backfilling
- In-ground sanitary drainage before backfilling
- In-wall pressure services before plaster is applied to the wall
- Hydrant system pressure and flow testing
- Final inspection before occupation certificate

2.4 Response to report

Upon receipt of the report, it is the responsibility of the hydraulic Contractor to:

- Acknowledge the receipt of the report within 7 days,
- Remediate each item within the report as indicated,
- Provide photographic evidence and certification proving the rectification work has occurred in a manner deemed satisfactory by the hydraulic Consultant.

Note: If the evidence of rectification works provided by the hydraulic Contractor is deemed unsatisfactory, additional inspections may be charged and recuperated from retention monies.

3. Preliminaries

3.1 Extent of work.

The work under this section covers the complete supply, installation and testing of the work listed in the index, specified hereinafter and shown on drawings listed here:

Drawing Number	Drawing Title
25015-H001	COVER SHEET, LOCALITY PLAN, SYMBOLS & LEGEND
25015-H002	SPECIFICATION
25015-H103	STAGE 1 SITE RETICULATION
25015-H300	DETAILS
25015 - QCPS32VC/AO1	Q-MAX FORTIS 3200 PUMP STATION GENERAL LAYOUT (SECTIONAL PLAN VIEW)
25015 - QCPS32VC/AO2	Q-MAX FORTIS 3200 PUMP STATION GENERAL LAYOUT (SECTIONAL ELEVATION)
25015-H-PAM-000	COVER SHEET
25015-H-PAM-001	ABBREVIATIONS, LEGEND & SYMBOLS
25015-H-PAM-002	GENERAL NOTES
25015-H-PAM-100	GROUND FLOOR SANITARY PLUMBING LAYOUT
25015-H-PAM-101	ROOF LEVEL STORM WATER LAYOUT
25015-H-PAM-200	GROUND FLOOR WATER RETICULATION LAYOUT
25015-H-PAM-300	DETAILS
25015-H-PAM-301	DETAILS
25015-H-NBC-000	COVER SHEET
25015-H-NBC-001	LEGEND AND SYMBOLS
25015-H-NBC-002	HYDRAULIC NOTES
25015-H-NBC-100	GROUND FLOOR SANITARY PLUMBING LAYOUT - PART 1
25015-H-NBC-101	GROUND FLOOR SANITARY PLUMBING LAYOUT - PART 2
25015-H-NBC-102	GROUND FLOOR SANITARY PLUMBING LAYOUT - PART 3
25015-H-NBC-103	ROOF LEVEL STORM WATER LAYOUT
25015-H-NBC-200	GROUND FLOOR WATER RETICULATION LAYOUT-PART 1
25015-H-NBC-201	GROUND FLOOR WATER RETICULATION LAYOUT-PART 2
25015-H-NBC-202	GROUND FLOOR WATER RETICULATION LAYOUT-PART 3
25015-H-NBC-300	DETAILS
25015-H-NBC-301	DETAILS

3.2 Definition of terms

Drawing Number	Drawing Title
Client	Sporteng™
Principle	The Client or Client's stakeholders
Architect	MODE
PCA	Principal Certifier
Superintendent	The appointed representative of the Principal managing the Contract.
Builder/Building Contractor	The main Contractor to whom the Contractor is engaged by and who has entered into a Contract of Works with the Principal.
Plumbing Contractor	Contractor carrying out the hydraulic services installation under this section of work.
Contract	The agreement encompasses the Contract of Works to be undertaken by the Builder for the Principal.
Authority/Authorities	A person or organisation having statutory, political, administrative, technical or regulatory control over a project, process, equipment or Installation.
Approved/Approval	Deemed acceptable or fit for purpose by the Superintendent
Provide	Supply and Install and make available for use
Supply	Make available to others on-site, ready for Installation.
Install/Installation/Installed	Place or fix the item (supplied by others) in position ready for use, including the provision of all required ancillary equipment
Submit/Submitted/Submission	Subject to the Superintendent for processing, review and Approval
Inspected/Reviewed	Inspected/Reviewed by the Superintendent and the approved Authority.
Works	Provide complete operational systems and tasks as outlined within the Contract documents

3.3 Authorities & Standards

The whole of the work shall be carried out by or under the full supervision of a fully licensed Plumbing Contractor in accordance with the drawings and specifications, reviewed by the Superintendent and approved by all relevant authorities.

All work shall be at least equal or better than the appropriate current Australian Standard. Any proposal to install alternative items to those specified shall be accompanied by a written confirmation by the Manufacturer that the proposed article complies with the relevant Australian Standard in each and every respect.

The submission shall be made, and the approval obtained before ordering of materials and commencement of the respective section of work.

Where some doubts exist as to the appropriate standard, a decision shall be made by the Superintendent before the commencement of any work on or off the site. If any doubt exists as to whether a section of the design can comply with the relevant authorities' regulations, the Superintendent shall be notified before the commencement of any work. No consideration of a claim for redundant work shall be given if the Superintendent is not notified.

The relevant Authorities shall include but not be limited to the following:

Water/Sewer Authority	Fitzroy Water
Gas Authority	Origin Energy
Council	Rockhampton City Council
Fire Authority	Fire and Rescue Of the state
WorkCover	WorkCover Authority of the state
Building and Construction Regulator	Building Commission of the state
Health	State Health Body
Roads and Maritime Services	Roads and Maritime Services of that state

It is the responsibility of the Plumbing Contractor to carry out all liaison and coordination with all Authorities and to ensure the satisfaction of their requirements.

The designer is to check the gas standard in the table below to ensure the correct code is referenced at the time of documentation issue.

AS 5601.1:2022 was published 30.09.2022

All works shall comply with the National Construction Code (NCC), Queensland Development Code (QDC), and referenced Australian Standards as adopted and in force in Queensland at the time of building approval or as otherwise required by the relevant regulatory authority, including any applicable transition arrangements. Design, supply and install all works described for this project to be in accordance with the relevant standards but not limited to the following:

Standard	Year	Description
EP&A		Environmental Planning and Assessment Act 1979
NCC Volume 1	2022	Building Class 2 - 9
NCC Volume 3	2022	Plumbing Code of Australia
AS 2419.1	2021	Fire hydrant installations Part 1: System design, installations and commissioning
AS 2441	2005	Installation of fire hose reels
AS 2444	2001	Portable fire extinguishers and fire blankets – selection and location
AS 3500.0	2021	Plumbing and drainage – Glossary of terms
AS 3500.1	2021	Plumbing and drainage – Water services
AS 3500.2	2021	Plumbing and drainage – Sanitary plumbing and drainage
AS 3500.3	2021	Plumbing and drainage – Stormwater drainage
AS 3500.4	2021	Plumbing and drainage – Heater water services
AS 5601.1	2022	Gas Installations – Part 1: General installations

The standards referenced in the table above are primary references only. Refer to the relevant standard for secondary references as necessary.

3.4 Construction Sustainability Coordinator

The Contractor must have personnel on their project team or engage a Sustainability Coordinator (separate from the Contract Administrator's ESD Consultant) who holds a current GSAP qualification. This coordinator will serve as the single point of contact for the Contract Administrator's ESD Consultant and will coordinate with trades and suppliers to provide compliant documentation for submission. The Sustainability Coordinator will be responsible for managing day-to-day Green Star-related activities, collecting the required evidence, and reviewing and collating it before transferring to the ESD Consultant for formal certification with the GBCA. The role of the Contract Administrator's ESD Consultant is to provide general high-level assistance and liaise with the Contractor's Sustainable Coordinator throughout the process.

3.5 Tender price

The tender price shall be based on current standards and associated amendments as of the date the tender is submitted/lodged.

3.6 Applications, fees and charges

The Plumbing Contractor shall be responsible for making application and paying all fees and charges rendered by the respective Authorities in reference to the project, which shall include but not be limited to the following:

- Commencement of Work fees
- Inspection Fees
- Document inspection fees

- Road Opening fees
- Restoration Charges
- Water main connection Fees
- Sewer Connection Fees
- Gas main connection fees
- Stormwater connection fees
- Service Connection Charges

3.7 Certification of works

At the completion of the works and prior to the submission for final payment, the Plumbing Contractor shall make all necessary applications, pay all fees, and obtain and issue to the Superintendent Certificates indicating that the works comply with the current regulations and requirements of the relevant Authority.

Where applicable, the relevant Authority shall issue the Certificate. Where this is not standard practice, the Plumbing Contractor shall provide a Certificate or Letter of Certification, which will guarantee that the works comply with the relevant Authorities' regulations, requirements and conditions.

3.8 Maintenance

The Plumbing Contractor shall allow to maintain and service the works for a period of 12 Months from the date of practical completion of the works. This shall include all pipework, pumps, filters, valves etc. and providing test results at the end of that period for all required services.

3.9 Service drawings

The accompanying drawings indicate the approximate positions and number of sanitary fixtures and other items requiring connection to the hydraulic services. Allow for all necessary diversions and minor adjustments to pipework and equipment as may be required to carry out the work and complete the works. Refer to architectural drawings for equipment location and coordinate with other services where applicable.

3.10 Dimensions and levels

All invert levels on the drawings are recommended only and must be checked on site before excavation or installation of pipework to ensure connection to supply services and correct cover and fall. Advise the Superintendent of any apparent discrepancies before the commencement of any work; claims for redundant work due to failing to do so shall not be considered valid.

Note: Ascertain the depth, position and suitability of the sewer & stormwater connection points before the commencement of any work (Pipe laying & excavation) and ensure that the sewer & stormwater can gravitate to the relevant connection points. The Superintendent shall be advised immediately should any adjustment be required to execute the work.

No claims for redundant work will be considered due to failure to comply with this requirement. No pipe laying or excavation shall be undertaken until this has been undertaken.

3.11 Testing

Allow for the cost of carrying out all tests set out in the drawings or specifications, or as required by the respective authorities. Supply all plugs, apparatus, and other materials necessary for the tests. Underground or enclosed work shall not be covered or concealed from view until it has been inspected, tested and sighted by the Superintendent and approved by the Authority concerned.

On completion of the works included under this part of the specification, the Plumbing Contractor shall carry out any procedure required to prove that the respective systems are operational under normal working conditions, as requested by the Superintendent and authorities.

Provide a minimum of two (2) working days' notice to the Superintendent before the commencement of testing.

Maintain a separate set of drawings on site to record the progress of testing. The Superintendent shall initial the drawings of each section of work that has passed a satisfactory test. Remedy any defects in the piping found during testing and re-test as specified under each section of work.

3.12 Piping

All piping shall conform to the standards and descriptions and be in accordance with materials and sizes shown on the drawings and/or detailed in the specification.

Pipelines shall be installed using the longest practicable length of tube/pipe to eliminate unnecessary jointing. Building up lengths from short off-cuts shall not be permitted.

3.13 Capping off

During construction, leave all unfinished work in a safe condition and protect the works against damage or loss through any cause whatsoever. At all times, the open ends of pipes shall be sealed off in such a manner as to prevent the entry of foreign matter into the lines. Plugs of rags, paper or wood will not be acceptable for this purpose. Drainage risers shall be temporarily sealed off with patented plastic caps.

3.14 Fixing & supporting of pipes

3.14.1 General

Where required, as per the seismic restraint section of the specification, all pipework shall be provided with seismic restraints complying with AS1170.4, providing stabilisation both transversely and longitudinally. Hanger rods shall also be provided with rod stiffeners to prevent buckling during seismic events. Supports and fixings to AS1170.4 are to be designed by suitably qualified persons and be suitable for the application and material to be supported.

Pipework and fittings shall be:

- a. Only fixed in approved locations.
- b. Adequately secured to the structure to support the pipework under full load conditions with a safety margin of 2:1.
- c. Kept clear of structure and other services.
- d. Provided with 0.75mm galvanised sheet metal sleeves where passing through structure. Pack space between pipe and sleeve with fire rated material equal to fire rating of structure through which pipe passes.
- e. Installed to allow for adequate expansion and contraction without causing stress on pipes or joints.
- f. Fixed on hanger brackets to allow adjustment for fall.
- g. Not be fixed with explosive power tools.
- h. Cleaned of all cement dropping on completion.

3.14.2 Materials

- a. Use purpose-made galvanised mild steel channel equal to "Unistrut P1000T", complete with purpose-made fittings for multiple parallel pipes. Provide plastic end caps equal to "Unistrut A1280" on exposed channels.

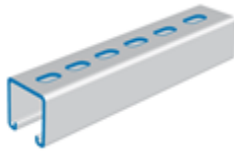


Figure 1 P1000T Channel

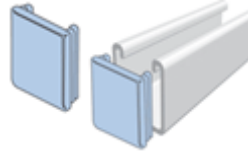


Figure 2 A1280 End Cap

- b. Use two-piece brackets equal to “Unistrut Pipe Clamps” to secure pipework to Channel and rubber cushioning equal to “Unistrut Unicushion” between copper pipes and steel brackets where Insulation/Lagging is not required.

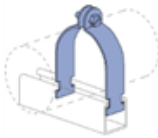


Figure 3 Two Piece Pipe Clamps

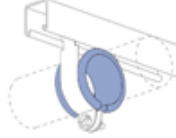


Figure 4 Unicushion

- c. Use Timber Blocks between pipework and bracketing where Insulation/Lagging is installed.



Figure 5 Timber Block

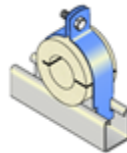


Figure 6 Typical Timber Block Installation

- d. Flexistrut

- i. Use brackets equal to “Flexistrut Nut Clamp” for singular pressure service pipes where Insulation/Lagging is installed.
- ii. Use clips equal to “Flexistrut Acoustic Pipe Clips ‘S’ Series” for singular pressure service pipes where Insulation/Lagging is not required.

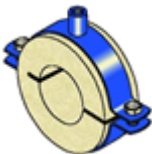


Figure 7 Nut Clamp



Figure 8 Acoustic Pipe Clips ‘S’ Series

- e. Use slide guide equal to “Flexistrut Heavy Duty Slide Guide and Timber Ferrule” for all hot/warm water flow and return pipework.

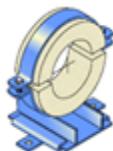


Figure 9 Heavy Duty Slide Guide and Timber Ferrule

- f. Use slide Cantilever Bracket to “UniStrut PCL Range” for gravity drainage pipework greater than 100mm in diameter when installed against walls.

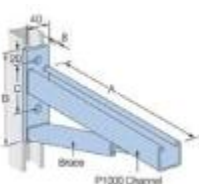


Figure 10 Heavy Duty Slide Guide and Timber Ferrule

- g. Use galvanised bolts and fixings of adequate size.
- h. Obtain approval for any alternative fixing methods before the commencement of work.
All mild steel must be hot-dipped galvanised.
- i. Use patented masonry anchors to fix masonry elements.
- j. PVC-coated brackets shall not be used
- k. Pex pipework - Use only pipe brackets and clips that are Suitable for plastic pipes that are the correct size (to allow easy gliding of pipe without pulling the rubber lining out) and are free of burrs.

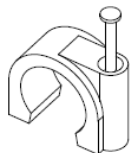


Figure 11 Open Clip

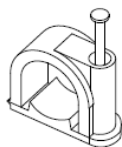


Figure 12 Closed Clip

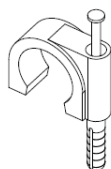


Figure 13 Anchor Clip

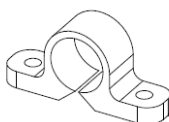


Figure 14 Saddle Clip



Figure 15 Support Clip

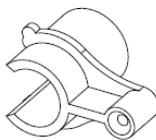


Figure 16 Stud Clip

3.14.3 Spacing

Fix piping at the following maximum intervals:

- a. Sanitary, Vents & Downpipes (Non-Pressure Services)
Ref:AS3500.2:2021 Section 10
Ref:AS3500.3:2021 Clause 4.96

Pipework Material	Maximum spacing of supports (m)	
	Vertical Pipes	Graded Pipes
Cast Iron	3.0 m	3.0 m
Ductile Iron	3.0 m	3.0 m
Copper & Brass	3.0 m	3.0 m
Stainless Steel	3.0 m	3.0 m
FRC	4.0 m	4.0 m
PVC-U DN 40 -50	2.0 m	1.0 m
PVC-U DN 65 - 150	2.5 m	1.2 m
PVC-U DN > 150	3.0 m	1.5 m
PP	2.0 m	1.0 m
PE	3.0 m	3.0 m

In the case of rubber ring jointed pipes, there shall be a fixing at least at each collar or pipefitting. Brackets shall be adequate to restrain the effect of the internal forces of piping.

All pipework is to be installed according to manufacturer guidelines for specific installation requirements; the above table indicates the maximum intervals for support systems.

b. Pressure Services

Ref:AS3500.1:2021 Clause 5.7.4

Nominal Pipe Size	Copper, Copper alloy & Stainless Steel pipes	Galvanised steel & ductile iron pipes	PVC-U, PVC-C, PVC-M, PVC-O, PE, PE-X, PP & PB pipes	
			Horizontal or graded pipes	Vertical Pipes
15	1.5 m	2.0 m	0.60 m	1.20 m
16	-	-	0.60 m	1.20 m
18	1.5 m	-	0.60 m	1.20 m
20	1.5 m	2.0 m	0.70 m	1.40 m
22	-	-	0.70 m	1.40 m
25	2.0 m	2.0 m	0.75 m	1.50 m
32	2.5 m	2.5 m	0.85 m	1.70 m
40	2.5 m	2.5 m	0.90 m	1.80 m
50	3.0 m	3.0 m	1.05 m	2.10 m
63	-	-	1.10 m	2.20 m
65	3.0 m	3.0 m	1.20 m	2.40 m
75	-	-	1.30 m	2.60 m
80	3.0 m	4.0 m	1.35 m	2.70 m
90	3.0 m	-	1.40 m	2.80 m
100	3.0 m	4.0 m	1.50 m	3.00 m
110	-	-	1.50 m	3.00 m
125	3.0 m	4.0 m	1.70 m	3.00 m
140	-	-	1.70 m	3.40 m
150	3.0 m	4.0 m	2.00 m	4.00 m
160	-	-	2.00 m	4.00 m
200	3.0 m	4.0 m	2.00 m	4.00 m

Note: Refer to manufacturers' guidelines for specific requirements.

Hoop Iron will not be acceptable for pipe hangers or Brackets.

3.14.4 Hydrant Systems

The fixing supporting and bracketing of hydrant systems to be in accordance with AS 2419.1:2021

Ref:AS2419.1:2021 Section 10

Pipework Material	Pipe Size DN	Maximum spacing of supports (m)	
		Vertical Pipes	Graded Pipes
Copper	All	4.0 m	4.0 m
Galvanised Steel	<= 150	4.0 m	4.0 m
Galvanised Steel	> 150	5.0 m	5.0 m

3.15 Seismic restraint of non-structural components

3.15.1 Requirement

All pipework shall be provided with seismic restraints complying with AS1170.4, providing transversely and longitudinally stabilisation. Hanger rods shall also be provided with rod stiffeners to prevent buckling during seismic events.

3.15.2 Earthquake Design Category:

2

3.15.3 Spacing Generally

Component hanger spacing will typically be less than the required seismic spacing.

3.15.4 Ductile Material Spacing

The maximum allowable brace spacing for piping constructed of ductile materials (e.g. steel, copper, aluminium) are:

- a. 12m for transverse bracing (piping larger than 65 mm dia.);
- b. 24m for longitudinal bracing (piping larger than 65 mm dia.);
- c. 9m for transverse bracing (piping smaller than 65 mm dia.);
- d. 18m for longitudinal bracing (piping smaller than 65 mm dia.).

3.15.5 Non-ductile Material Spacing

The maximum allowable brace spacing limits for piping non-ductile materials (e.g. cast iron, plastic) are:

- a. 6 m for transverse bracing;
- b. 12 m for longitudinal bracing.

3.15.6 Installation:

- a. All seismic restraint systems shall be installed in strict accordance with the manufacturer's seismic restraint guidelines manual and all certified submittal data;
- b. Transverse restraint for one pipe section may also act as a longitudinal restraint for a pipe section of the same size connected perpendicular to it if the restraint is installed within 600 mm of the elbow or tee or combined stresses are within allowable limits at longer distances;
- c. Hold down clamps must be used to attach pipe to all trapeze members before applying restraints;
- d. Branch lines may not be used to restrain main lines;
- e. Provide reinforced clevis bolts when required;
- f. Piping crossing building seismic or expansion joints, passing from building to building, or supported from different portions of the building shall be installed to allow differential support displacements without damaging the pipe, equipment connections, or support connections. Pipe offsets, loops, anchors, and guides shall be installed as required to provide specified motion capability and limit motion of adjacent piping;
- g. Do not brace a system to two independent structures such as ceiling and wall;
- h. Provide appropriately sized openings in walls, floors, and ceilings for anticipated seismic movement. Provide fire seal systems in fire-rated walls.

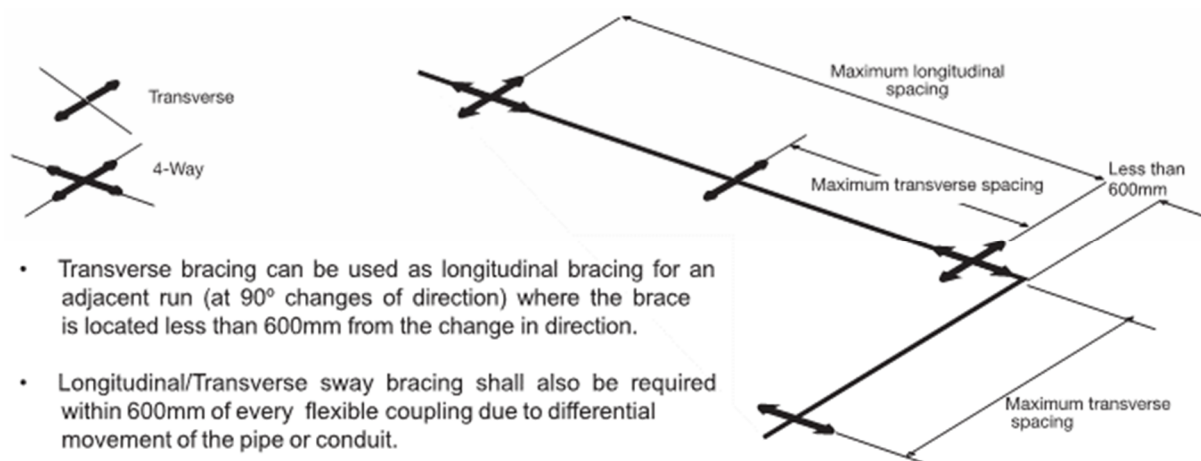


Figure 17 Change in Direction Bracing

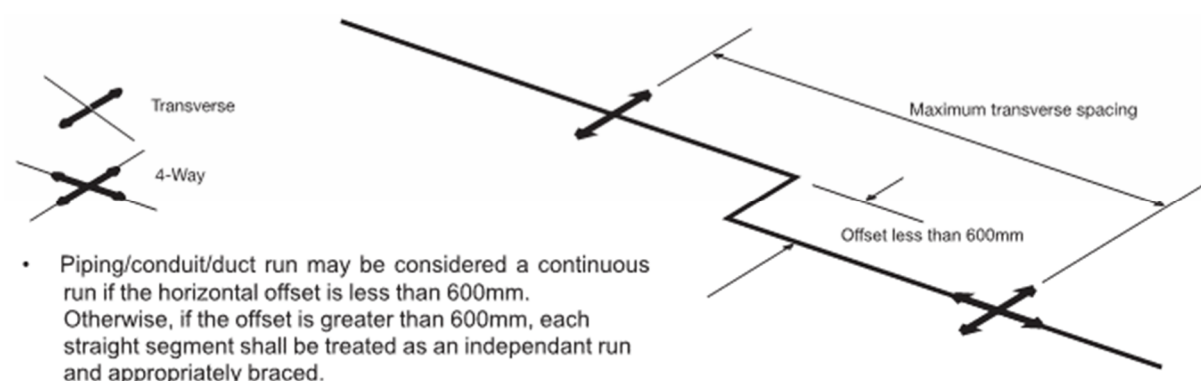


Figure 18 <600 mm Offset Bracing

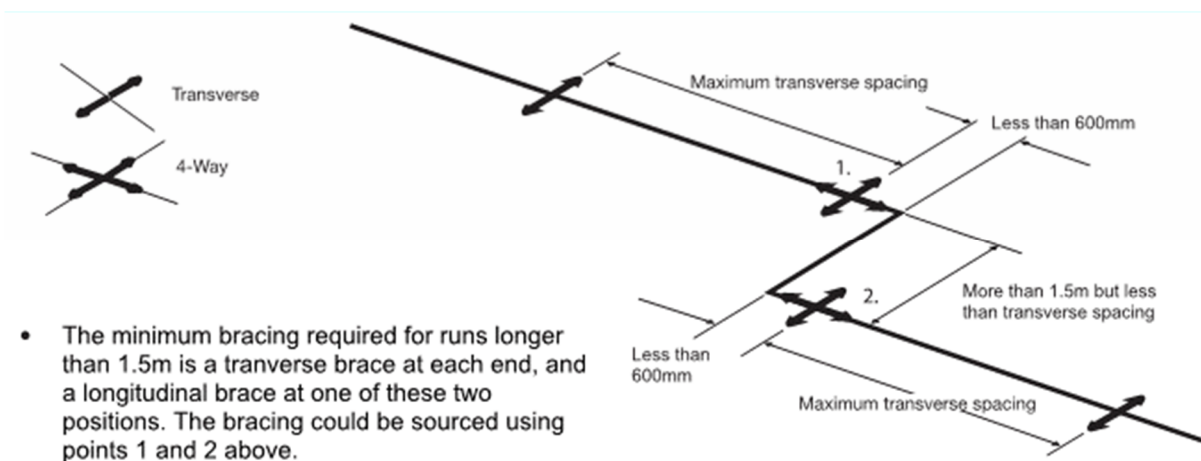


Figure 19 >1.5 m Offset Bracing

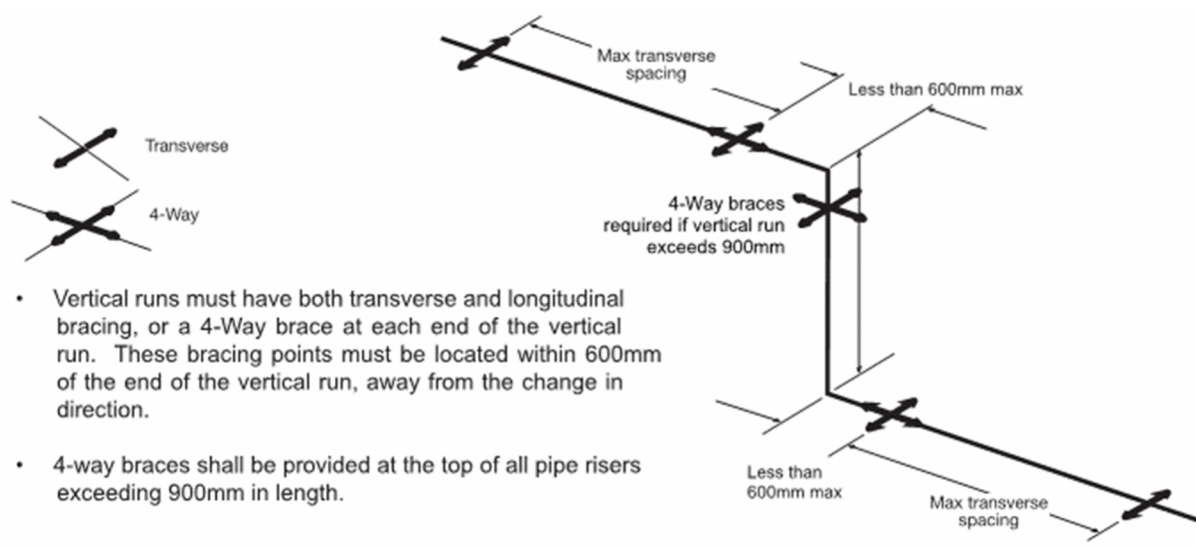


Figure 20 Vertical Transition Bracing

3.16 Exposed pipework treatment

Except as otherwise specified or directed, all internal exposed piping adjacent to plumbing fixtures, including traps and fittings, shall be chromium-plated finished and, where passing through a finished wall, floor or ceiling, shall be fitted with approved chrome cover plates.

All other exposed piping shall be cleaned free of cement droppings.

3.17 Painting

All exposed piping situated in plantrooms, corridors, fire stairs, car park areas and ducts and, where not chrome plated, shall be painted continuously in the colours specified below and, in addition, shall be fitted with identification labels.

Painting to pipework shall be in accordance to the colour coding as indicated in the schedule of colours for identification of piping in AS1345 - Identification of the Contents of piping, Conduits and Ducts.

Description	Colour	Sample
Soil wastes and vents	Black	
Potable water	Emerald Green No. 20	
Non potable cold water	Lilac	
Bore water	Mauve	
Gas	Light Beige No. 366	
Vacuum	Light Blue	
Stormwater	Jade Green	
Fire	Red	

Where potable water pipes are laid in parallel with other pipes of similar colour, provide additional blue identification banding to the potable pipe.

Label each service with proprietary labels to accurately identify the service and its direction of flow.

Where pipe work is not required to be painted in locations such as in ceilings, service ducts and under floors, it shall be marked with colour bands 450mm long and at 3000mm spacing. Fit supplementary markers at all colour bands similar to '3M Safetyman'.

Painting shall be carried out by experienced Painters.

Where piping is specified to be painted, the supports and brackets shall be similarly painted in selected colours.

Copper pipes shall be painted with a suitable etching paint before applying finishing enamel. Paint cast iron pipes with all metal undercoat and finishing enamel. All steel pipes shall be painted with one coat of zinc chromate primer and undercoat before application of finishing enamel.

Vents – All vents penetrating the roof shall be painted to match the roof colour. Where the roof is a Colourbond colour or similar, the paint batch number shall be matched and applied to the vents.

3.18 Identification plates for underground pipework services

Provide and install Identification Plates for underground service pipes. Different identification plates shall be provided for each service. Plates shall be 3mm thick stainless-steel screw fixed to concrete with stainless steel expansion fastenings. Engrave details onto plates. Include Service, Size and Depth

3.19 Identification

Labels on pipework shall be installed in accordance with AS 1345 - Identification of the Contents of Pipes, Conduits and Ducts - and be equal to the Safetyman pattern positioned on each side of valves, bends and junctions and along the length of the pipeline at a maximum of 3000 mm centres. Where pipes are installed within ducts or false ceiling areas additional labels shall be installed so as to be visible at the access panel position. The labels shall indicate the direction of flow.

Text examples	
COLD WATER	< With directional arrows >
HOT WATER	< trimmed to suit direction of flow >
NON-POTABLE WATER	
RAINWATER	
RECYCLED WATER	
REVERSE OSMOSIS WATER	
SOLAR HOT WATER FLOW	
SOLAR HOT WATER RETURN	
STORM WATER	
WARM WATER	
L.P. GAS	
NATURAL GAS	
FIRE HOSE REEL	
FIRE HYDRANT	
FIRE SPRINKLER	
GREASE	
GREASE VENT	
PUMPED DRAINAGE	
SEWAGE	
TRADE WASTE	
TRADE WASTE VENT	
VENT	
	
Recommend Seton – Phone: 1800 531 492 Design Your Own Quality Pipe Markers. Choose the Pipe Marker size and colour combination and enter your own custom text. Self-Adhesive Outdoor Grade Vinyl.	

3.20 Underground pipe warning tape

During the backfilling process lay a plastic warning tape 150mm above all underground pipes. This tape shall extend for the full length of pipes. Warning tapes shall be 200mm in width of approved durable plastic material with colour to comply with AS 1345 printed with the words "DANGER - BURIED PIPE BELOW" repeated continuously (see below for wording).

Warning tape shall contain a trace wire, which shall be secured and terminated above the surface of each end.

Text examples:

DANGER BURIED GAS MAIN BELOW

DANGER BURIED WATER MAIN BELOW
DANGER BURIED FIRE MAIN BELOW
DANGER BURIED NON POTABLE WATER MAIN BELOW
DANGER BURIED RECYCLED WATER MAIN BELOW
DANGER BURIED POTABLE WATER MAIN BELOW
DANGER BURIED SEWER MAIN BELOW
Recommend Seton – 1800 651 173 Chemical resistant, soil tolerant and rot resistant. Manufactured from Polyethylene for the band and homopolymer for the mesh. Up to two tracer wires are incorporated in the tape.

3.21 Electrical substation

No services shall be within the electrical substation and easement area.

Note: Fire Hydrants and Fire hydrant booster valves shall be 10 metres from the substation

This is a requirement of AS2419.1:2021

Ref: AS2419.1:2021 Clause 7.3.3

3.22 Protection of finished & polished surfaces

Throughout the project, all finished surfaces shall be adequately protected against damage. Provide suitable material for the protection of finished surfaces as required and as may be directed.

Remove protection from finished surfaces on completion of the project and leave surfaces in such a manner that final cleaning of dust only is required.

Scratched or damaged finished surfaces will not be accepted.

3.23 Protection of equipment, plant and pumps during construction

Throughout the project, all plants, equipment, pumps, etc., shall be adequately protected against damage from the environment. Provide suitable material for protection as required and as may be directed.

Remove protection on completion of the project and leave all items in working order.

Scratched or damaged finished surfaces will not be accepted.

Pumps and control panels shall not be exposed to the weather during construction.

3.24 Concrete

All concrete used in conjunction with this work shall have a minimum strength of 20 mPa at twenty-eight days when tested in accordance with AS 1012 Part 8 - Method for Making and Curing Concrete Compression, Indirect Tensile and Flexure Test Specimens in the Laboratory or in the Field.

3.25 Excavation in existing concrete slabs, pavements & roadways

If required by the Superintendent, trenches across existing roads shall be excavated and the pipeline constructed therein so that half the pavement & roadway are always maintained open to the public.

Slabs shall be saw cut and then the concrete removed with pneumatic tools.

Council road opening application and payment of all fees shall be made, and approval granted before any works being undertaken.

3.26 Core holes & sleeves

Refer also to the Core Hole Drawings clause.

Set out all core holes and sleeves in floors, walls, beams and columns in conjunction with the fixing of formwork and/or placing of concrete. To prevent the weakening of the building structure, all core hole locations shall be approved prior to placing concrete. Material for forming the core shall be of 0.7mm galvanised sheet metal or PVC-U and shall have a diameter of at least 75mm greater than the pipe it serves.

Determine the exact location of all penetrations through the structure tilt-up panels and provide the Builder with drawings indicating their locations. Supervise and approve the location of penetrations prior to the pouring of the panel.

Fill in all penetrations and core holes, including approved fire-stopping material.

3.27 Authorities' services search - dial before you dig

The Plumbing Contractor shall be allowed to undertake an Authorities services search and obtain all available information on Authorities services within the street and site area.

Dial Before You Dig PH: 1100

Web: www.byda.com.au

3.28 Excavation near authorities' services and easements

The Plumbing Contractor shall be allowed to hand dig when excavating the Authority's services and easements. The Authorities shall be given 48 hours' notice before any excavation takes place. If required or requested by the Authority, excavation shall take place only under their supervision.

3.29 Concealed piping

Unless otherwise approved or directed, pipes shall be concealed.

All chasing in walls for pipes and equipment shall be carried out with a mechanical saw. Chasing will not be allowed in concrete walls unless agreement in writing is obtained from the Superintendent.

Do not chase firewalls.

In general, chases in block work shall not exceed 1200 mm horizontally and not closer than 600 mm to supporting elements vertically. No face walls shall be chased. Where pipes cannot be chased, they shall be built in as the work proceeds. Discuss the location of services in masonry walls with the superintendent. Where pipes are concealed in walls, they shall be insulated. The insulation shall be placed on the pipes in such a manner that expansion and contraction will be unhindered, and the wall finish will not be cracked or otherwise damaged.

Where piping is to be concealed in timber or metal stud walls:

- Holes through studs, noggins, etc. shall be carefully drilled, with piping kept in straight lines.
- Copper hot, warm and cold water piping – 20mm and smaller shall be fixed with 'Plastic' sleeves/brackets. Each such sleeve/bracket to be screw fixed.
- Unless otherwise noted, each penetration through studs, noggins, etc. shall include a PVC sleeve with a wall thickness not less than 2mm.
- Ensure that all pipework is adequately fixed so that no water hammer occurs.
- Provide timber noggins, not less than 100mm x 50mm and adequately fixed to studs for fixing faucets/tapware, including ancillary pipe fittings.

3.30 Programming of work

Allow to comply with the program of the main contract.

3.31 Staging of work in accordance with builders' program

Work shall be carried out in accordance with the Builders Construction Program. Staging of the work shall be carried out so that the building can remain open for operation during construction. The plumbing contractor is to make themselves aware of the builders' proposed staging program and plan to keep existing services "alive" so that the building can function without major disruption. The change from the existing services to new services shall be accomplished 'out-of-operation hours' so as to minimise disruption. Notification of service changeovers shall be given 48 hours in advance to the Project Site Manager.

3.32 Out of hours' work

All work required to be completed after normal working hours shall be undertaken outside normal working hours, subject to the superintendent's and/or client's approval.

3.33 Protection of ground floor shops

Protect the ground floor shops and offices from debris during construction. Provide all drop sheets and plastic sheeting to allow for complete protection.

3.34 Existing authorities' connections to a site

All redundant existing authorities' connection points not being reused shall be located and disconnected at their point of connection to the authorities' service. Disconnections shall be to the approval of the relevant Authority.

Disconnection shall be in accordance with their respective guidelines and regulations.

Make all applications and pay all fees to the respective authorities before any disconnection taking place.

3.35 Existing disused services

All redundant existing services shall be investigated and disconnected at their connection point to the authorities' service. Disconnections shall be to the approval of the relevant Authority.

3.36 Disused sewerage service sealing

Allow to make an application and pay all fees to disconnect all disused private sewerage services and seal them at the point of connection to a Local authority sewer main.

This work must meet the Local authority's standards in the Plumbing Code of Australia and be done by a licensed drainer. The licensed drainer must arrange for an inspection of the work by a Local authority plumbing and draining inspector.

After the Local authority's inspector has looked at the work, the drainer can issue the Certificate of Compliance.

3.37 Existing services

The contract drawings indicate services within the site and have been prepared from available records and site investigations. Investigate the exact location, depth and size of services. Deviations of services other than that indicated are to be brought to the attention of the Superintendent prior to the commencement of any excavation and/or installation relating to such deviation.

3.38 Connection to existing services

It is the responsibility of the Plumbing Contractor to ensure that when connecting or diverting existing services, they have clearly located & ascertained that the services are, in fact, the services in question and have adequate allowance to take extra loads.

Dye tests shall be undertaken on all relevant, existing sewer and stormwater pipework connection/diversion points.

3.39 Site visit

Visit the site before submitting the final tender to assess "on-site" conditions. Failure to do so will forfeit any claim for not being aware of conditions affecting the tender.

3.40 Mechanical services co-ordination

Position of cold water, floor wastes and drainage points for mechanical services are indicative only. The plumbing Contractor shall coordinate with the Mechanical Services Subcontractor to ensure the correct position for the final choice of equipment and appliances.

3.41 Services co-ordination

The plumbing Contractor shall coordinate with the Services Subcontractor and the Builder to ensure all pipework and equipment coordination is undertaken.

Detailed coordination is the responsibility of the plumbing contractor; the Hydraulic drawings are only design coordinated.

3.42 Installation co-ordination

Various items of apparatus and equipment will be provided and fixed by others.

The Plumbing Contractor shall familiarise themselves with the requirements of the other subcontractor's works requiring coordination and shall examine the plans covering each of these services.

It shall be the responsibility of the Plumbing Contractor to schedule work so that it may be installed at the proper time and without delaying the completion of the entire project.

The Plumbing Contractor shall carefully check space requirements with other Subcontractors to ensure that the equipment, piping, etc., can be installed in the spaces allotted for same.

Where typical arrangements occur for multiple installations, such as core toilets on typical floors, the Plumbing Contractor shall complete one arrangement and obtain approval before duplicating the same.

When equipment and/or installations associated with the work of the contract are installed by other Subcontractors, the Plumbing Contractor shall attend and coordinate as necessary.

Associated fittings required to prevent clashes with other trades shall be deemed to have been included in the tender price. Check all dimensions on site before commencing work

The design drawings do not relieve the Plumbing Contractor of the co-ordination or buildability responsibility

All design changes occurring after the letting of the Plumbing Contract by other trades, including architectural, structural and mechanical services, shall be co-ordinated into the installation by the Plumbing Contractor as necessary

The Plumbing Contractor shall be responsible for the installation of all service elements associated with this subcontract, within the ceiling, duct and plant spaces provided in accordance and coordinated with all other services trade designs.

3.43 Acoustic pipe supports

Refer to the Fixture and Fittings section.

3.44 Acoustic lagging of pipe & fittings

Refer to the Fixture and Fittings section.

3.45 Inspection & flushing out of services – gravity systems

On completion of the installation of all services and before application for Certificate of Practical Completion, allow to lift all access chamber covers and major access gates and inspect gravity lines. Clear out any debris from lines and flush the entire system to ensure a

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clean bore. Replace access chamber covers and access gates after inspection, clearing and flushing. Provide the Superintendent with a minimum of 48 hours' notice prior to inspection and flushing procedures.

3.46 Inspection & flushing out of services - pressure systems

On completion of the installation of all pressure systems and before application for Certificate of Practical Completion, allow the flushing out of all piping installations to clean sand, copper filings, and foreign matter from the system.

Prior to flushing operation, disconnect all thermostatic mixing valves, tempering valves, filters, screens, ball float valve inlets, backflow preventers and other appliances that may be subjected to damage due to foreign materials. Ensure that the final installation is totally clear of any foreign matter and that all valves, filters, screens and outlets are not affected by such material. Replace any and all damaged items of equipment and make all necessary reconnections. Provide the Superintendent with a minimum of 48 hours notice prior to inspection and flushing procedures.

3.47 WELS scheme

All showers, taps, spouts, urinals, WCs etc. shall be rated under the Wels Scheme

WELS is the Federal Government's - Water Efficiency Labelling and Standards Scheme Proof of Wels authorisation shall be presented when requested by the superintendent.

3.48 AS5200 authorization

All plumbing and drainage products shall be approved in accordance with AS 5200 Technical Specification for Plumbing and Drainage products and shall be watermarked.

3.49 The plumbing code of Australia, AS/NZS 3500 and the BCA

All work shall be in accordance with the Plumbing Code of Australia, the relevant sections of AS/NZS 3500-Plumbing and Drainage and the Building Code of Australia (BCA).

3.50 Floor wastes – NCC requirement

Floor wastes shall be provided in accordance with the BCA

Volume 1 of the NCC - Clause F2D4 - "In a Class 2, 3 or Class 4 part of a building, the Floor of each bathroom and laundry located at any level above a sole-occupancy unit or public space must be graded to permit drainage to a floor waste".

3.51 Hot water – NCC Section J8D2 - Requirement

The Hot water for the Building shall be installed in accordance with the NCC - Section J8D2 Hot Water supply and Section 8 of AS/NZS 3500.4:2021

Extract from the NCC:

"A Hot water supply system for food preparation and sanitary purposes, other than a solar hot water supply system in climate zones 1, 2 and 3 must be designed and installed in accordance with Section 8 of AS/NZS 3500.4

3.52 Food premises

All work within the Kitchen / Food areas shall be in accordance with the relevant sections of AS4674 Design, construction and fit-out of food premises, the Building Code of Australia (BCA), the Local Council Guidelines/requirements and the New South Wales Health Department.

3.53 Development application

All work shall be in accordance with the relevant sections of the Council-approved development application (DA).

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3.54 Pipework Under Slab

Ref:AS3500.1:2021 Clause 5.5.4

Water services pipework located below ground under concrete floor slabs shall be installed in a conduit of compatible material so that the pipe can be removed and replaced.

Water service pipes located beneath concrete slabs on the ground shall be laid in a narrow trench on a bed of sand or fine-graded soil placed and compacted in a manner that will not damage the piping. A minimum distance of 75mm is required between the top of the pipe and the underside of the slab or slab-stiffening beam.

The ends shall be crimped or capped prior to pouring the concrete. The exposed pipework to be protected from damage. Any pipework that penetrates the slab shall be at right angles to the surface of the slab and lagged for the full depth of the slab penetration with –

An impermeable flexible material of not less than 6mm thickness; or

An impermeable plastic sleeve of conduit providing equivalent protection

3.55 Pipework head height – carpark

All pipework in the carpark shall be above 2.1m in height in accordance with the NCC.

Ref: NCC 2022 Volume 1: F5D2(3)

All pipework in the carpark above any vehicular path leading to or from a parking space for people with disabilities (wheelchair / Accessible parking bays) shall have a clearance of 2.2m.

Ref: AS 1428.1:1992 Clause 14.1

Ref: AS 2890.6: 2009 Clause 2.4

All pipework above a parking space for people with disabilities (wheelchair / Accessible parking bays) shall be above 2.5m in height.

Ref: AS 1428.2 Clause 14.2

Ref: AS 2890.6: 2009 Clause 2.4

3.56 CCTV inspection of pipework

To achieve practical completion, allow to undertake a closed-circuit television (CCTV) video inspection of all in-ground drainage lines.

The inspection shall be undertaken prior to practical completion to prove that all lines are clear of debris and any blockages.

The CCTV inspection is to be undertaken in the presence of the superintendent.

The inspection is to be carried out by an independent firm approved by the superintendent.

Supply a USB thumb drive to the superintendent to include the video and reports as part of the As-built documentation.

3.57 Future connection points

All connection points provided for future connections for water and gas shall be capped off with a threaded brass cap or plug. The connection shall be water and gas tight.

The gas connection shall be in accordance with AS/NZS 5601.1

3.58 Tender documents

The drawings, as scheduled, are issued as a guide only and shall be considered to be diagrammatic and approximate. The drawings and Specifications are intended to be mutually explanatory and complete, but all work called for by one, even if not by the other, shall be fully executed. Should the documents be in conflict, the Plumbing Contractor shall be deemed to have included the larger quantity and/or the more expensive components, as applicable.

Should the drawings and specifications indicate ambiguities the requirements of the Specification shall take priority. Unless the Plumbing Contractor advises regarding ambiguities or conflicts at the time of tender, the requirements of the Specification will be expected as fulfilment of the Plumbing Contract.

3.59 Documents before practical completion

Before practical completion, provide:

1. As-built drawings in PDF and AutoCAD / Revit Formats (matching Tender documentation format)
2. Operation and Maintenance Manuals
3. Updated sewerage services diagram for Local authority
4. Detailed certificates of compliance.
5. Test information and certificates for each
6. Thermostatic Mixing Valve
7. Backflow prevention device
8. Flow and Pressure Test including test information data
9. Fire Hydrant System
10. Fire Hose Reel System
11. Drenchers Wall Wetting Sprinklers
12. Fire seals, Fire Collars and sealing of all penetrations.
13. Access cover lifting keys
14. Hose tap operating keys

3.60 Protection of pipelines in frost affected areas

In regional climates where temperatures can reach sub-zero, design considerations must be given to services that will be affected if frozen and therefore inoperable or alternatively inaccessible due to extreme weather conditions.

External pipework shall be located below ground as far as is practicable. Where it is not possible to install pipework below ground, the provision shall be made to protect the pipework, if necessary, from freezing. Suitable insulation is specified at the rear of this document.

Wherever possible, pipes should be installed to avoid those areas of the building that are difficult to keep warm and where temperatures are likely to fall below freezing. These areas include:

- a. unheated roof spaces;
- b. unheated cellars;
- c. locations near windows, ventilators or external doors where cold drafts are likely to occur; and
- d. locations in contact with cold surfaces such as metal roofs, metal frameworks, or external metal cladding materials.

3.61 Protection of pipelines by encasement in polyethylene

Where installed underground, all new metallic pipelines, pipe fittings, valves and areas of existing branch pipelines of any material joining new pipelines for the distance specified shall be provided with a protective encasement of polyethylene tubing.

Such tubing shall be provided in addition to, but not as a substitute for, any corrosion protection that may be specified elsewhere.

3.61.1 Materials

Polyethylene - shall be manufactured of virgin polyethylene material conforming to the requirements of ASTM Standard Specification D-1248-74 Polyethylene Plastics moulding and Extrusion Materials.

Thickness - Polyethylene film shall have a minimum nominal thickness of 250 microns.

Tube Size or Sheet Width - Tube or sheet size for each pipe diameter shall be in accordance with manufacturer's recommendation for size of pipeline.

Adhesive Tape - Adhesive tape shall be plastic 50 mm wide adhesive tape not less quality than polyethylene film specified above and shall have the ability to bond to metal surfaces and also to polyethylene sleeving material.

Certification - Tenderers shall lodge certificates from approved N.A.T.A. Testing Laboratory or Laboratories that the materials proposed have been tested and conform to A.S.T.M. Standard Specification D1248-68 and the strengths and thickness requirements of this specification.

3.61.2 Installation

General - The polyethylene encasement shall prevent contact between the pipe and the surrounding backfill and bedding material but is not intended to be a completely air and water-tight enclosure. Overlaps shall be secured by the use of adhesive tape, which is capable of holding the polyethylene encasement in place until backfilling operations are completed. Polyethylene encasement shall be protected by precautions to prevent exposure to direct sunlight for any total period of more than 48 hours. Junctions - between wrapped and unwrapped pipe - Where polyethylene-wrapped pipe joins an adjacent pipe that is not wrapped, extend the polyethylene wrap to cover the adjacent pipe for a distance of at least 600mm.

Secure the end with three circumferential turns of tape.

Backfill - for polyethylene wrapped pipe - polyethylene sleeved pipe shall be backfilled for a height of 150mm over the pipe and fittings with sand as specified in clause "BACKFILLING". Every care shall be taken to prevent damage to the polyethylene wrapping when placing backfill.

Protection of Polyethylene encasement - Any damage to the polyethylene before or after backfilling shall be made good by the Plumbing Contractor to the satisfaction of the Superintendent.

3.62 Compaction schedule

When an area is to be filled or backfilled the soil is placed in layers called lifts. The ability of the first fill layers to be properly compacted will depend on the condition of the natural material being covered. If poor material is left in place and covered over, it may compress over a long period under the weight of the earth fill, causing settlement cracks in the fill or in any structure supported by the fill.

Consideration must be given to Site-specific conditions. The Contractor must consult the Super Intendant prior to the commencement of any groundworks to ascertain if there are any specific Geo-technical conditions or requirements.

Consolidation is a process by which soils decrease in volume. Consolidation is any process that involves a decrease in the water content of a saturated soil without the replacement of water by air. In general, it is the process in which reduction in volume takes place by expulsion of water under long term static loads. It occurs when stress is applied to a soil that causes the soil particles to pack together more tightly, therefore reducing its bulk volume. When this occurs in a soil that is saturated with water, water will be squeezed out of the soil. In the case

of excessive consolidation, the contractor will 'make good' the surrounding ground and ensure the correct compaction is achieved.

Service trenching backfill and compaction testing schedule

Location	Pipe bedding all services	Backfill to 300mm above crown of pipe or conduit	Backfill requirement	Test sample frequency	Level of compaction
Beneath or within 5 metres of a building	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Approved quarry rubble	Bedding: - 1 test per 25 metres or part thereof along the pipe route. Pipe surround material: - test per 50 metres or part thereof along the pipe route. Backfill material: - 1 test per 50 metres or part thereof for every 2nd layer along the pipe route.	Pipe bedding and surround: 90% Modified MOD to AS 1289.5.2.1 Backfill: 93% Modified MOD to AS 1289.5.2.1 for full depth from 300 mm above top of pipe to underside of the pavement.
Under vehicular pavement	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Approved quarry rubble	Bedding: -1 test per 25 metres or part thereof along the pipe route. Pipe surround material: - 1 test per 25 metres or part thereof along the pipe route. Backfill material:-1 test per 25 metres or part thereof for every 2nd layer along the pipe route.	Pipe bedding and surround: 90% Modified MOD to AS 1289.5.2.1 Backfill :95% Modified MOD to AS 1289.5.2.1 for full depth from 300 mm above top of pipe underside of the pavement.

Service trenching backfill and compaction testing schedule

Under non-vehicular pavement	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Approved quarry rubble	Bedding: - 1 test per 50 metres or part thereof along the pipe route. Pipe surround material: - 1 test per 50 metres or part thereof along the pipe route. Backfill material: - 1 test per 50 metres or part thereof for every 2" layer along the pipe route.	Pipe bedding and surround:- 90% Modified MDD to AS 1289.5.2.1 Backfill:- 93% Modified MDD to AS 1289.5.2.1 for full depth from 300 mm above top of pipe to underside of the pavement
Elsewhere	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Select fill to AS 1289 as applicable	Bedding: - 1 test per 50 metres or part thereof along the pipe route. Pipe surround material: - 1 test per 50 metres or part thereof along the pipe route. Backfill material:-1 test per 50 metres or part thereof for every 2nd layer along the pipe route.	Pipe bedding and surround:-90% Modified MDD to AS 1289.5.2.1 Backfill: - 95% Standard MDD to AS 1289.5.2.1 for full depth from 300mm above top of pipe to underside of the pavement.

Service trenching backfill and compaction testing schedule

Pipe Sizes exceeding 250mm or common services trenches	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm	Approved quarry rubble	Bedding: - 1 test per 50 metres or part thereof along the pipe route. Pipe surround material: - 1 test per 25 metres or part thereof along the pipe route. Back fill material: - 1 test per 50 metre or part thereof for every 2" layer along tile pipe route.	Pipe bedding and surround:- 90% Modified MDD to AS 1289.5.2.1 Backfill: - 93% Modified MDD to AS 1289.5.2.1 for full depth from 300mm above top of pipe to underside of the pavement.
Pits, pump stations etc.	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm or lean mix concrete	Packing sand – Washed naturally occurring sand with a maximum particle size of 3mm or lean mix concrete	Approved quarry rubble.	Bedding: - 1 test for the base of each pit. Backfill material:-2 tests per layer at alternating locations to ensure tests do not overlay the previous test location. Final: - 2 tests for the final layer alternating locations to ensure tests do not overlay the previous test locations.	Pip bedding and surround: - 90% Modified MDD to AS1289.5.2.1 Backfill: - 98% Modified MDD to AS 1289.5.2.1 for full depth from 300mm above top of pipe to underside of the pavement.
Over excavation of the base of the trench.	Concrete fill to within 75mm of the pipe invert or collar			As required by the above location references.	As required by the above location references.

Service trenching backfill and compaction testing schedule

All locations in unstable ground conditions	Seek advice from a professional qualified soils engineer and submit a written recommendation addressing all matters	As required by the above area references
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4. Shop drawings, as-built and manuals

4.1 Shop values

Drawings shall be submitted in PDF format and drawn to a minimum scale of 1:50 ratio for all wet areas and 1:100 for general areas. Quality shall be at least equal to the Tender/Construction drawings and show all co-ordination elements, such as structural elements and other services at, and adjacent to, the point of co-ordination, together with sufficient information to ensure that pipework will be installed within false ceiling and duct areas. The Plumbing Contractor shall have complete responsibility for the detailed drawings, coordination and final installation.

Drawings shall be submitted to the Building Contractor for checking prior to submission to the Superintendent for review and approval. As a guide in assessing the amount and extent of Workshop Drawings, the following requirements should be considered as minimum requirements to be detailed:

- a. All areas detailed in the contract documents
- b. All Plumbing Ducts
- c. All false ceiling areas where co-ordination with other services is required.
- d. Other areas deemed to require drawings for coordination and execution of the work.
- e. All access panel requirements, which are not already detailed or suitable on architectural drawings.
- f. Invert levels clearly indicated on plans
- g. Size and dimensioned set-out locations of all pipe penetrations.
- h. Fire Collar Matrix referencing each penetration through fire compartment/fire-rated wall.

Manufacture and/or installation, as applicable, shall not be commenced prior to the review of the drawings by the Superintendent.

Submit drawings to the Superintendent at least fourteen days prior to fabrication and/or installation of equipment and services.

Provide sufficient copies of the drawings to all other disciplines to ensure coordination with all other sections of the work.

In addition to those considered necessary for fabrication, attend all necessary meetings to coordinate services and prepare core hole and coordination drawings, based upon those meetings, for the whole of the Hydraulic Services work.

4.2 Core hole drawings

Provide core hole drawings of all slab, beam, wall, and tilt panel penetrations to the Superintendent and obtain approval from the structural engineer prior to the installation of any core holes, blackouts and pipes cast in beams.

4.3 Client training

At the completion of the project, the contractor is to conduct training at the agreed time at the system or equipment location. Use Items and Procedures listed in the final draft operation and maintenance manuals as the basis for instruction.

4.4 As built drawings & operation and maintenance manuals

A set of drawings shall be kept on-site and progressively marked up by the Plumbing Contractor as the work proceeds to record the locations, inverts and details of all installed services, equipment and valves.

Before applying for the Certificate of Practical Completion, a set of "As Built" drawings (comprising all floor plans, Diagrammatises and Details) prepared by the Plumbing Contractor

to a minimum scale greater than or equal to that of the contract drawings and relevant bound manuals indicating all items of service or equipment shall be submitted to the superintendent for review. The standard of drafting and minimum scales shall be equal to design documents, accurately plotted, showing dimensions of pipes, valves and services.

The hydraulic services drawings are to be provided in AutoCAD .dwg and/or Revit .rvt format (whichever is applicable to the project) and a separate PDF set.

During construction, when existing in-ground services are uncovered, their exact locations and ILs shall be measured and included in the As-Built documents.

4.4.1 Submission Requirements

Contract Documentation must include submission requirements as follows:

'As-Built' drawings must be progressively marked and completed and reviewed on a monthly basis during the construction stage. Progress prints must be submitted at 50% and 70% complete stages.

1. One final draft copy of As-Builts and Manuals must be submitted for final review by the Superintendent prior to Practical Completion.
2. Two copies of final documentation must be submitted for approval within 4 weeks of Practical Completion.
3. 10% of the Hydraulic Services Contract Value will not be paid until final approved 'As-Builts' & Operating & Maintenance Manuals documentation is received.

4.4.2 As-built Drawings – Content and Format

As-Built, Drawings shall show dimensions, types and locations of equipment, piping and ductwork in relation to permanent site features and other underground services. Show the 'as installed' locations of building elements, plant and equipment. Show off-the-grid dimensions where applicable. Include relationship to building structure and other services, and changes made during commissioning and the maintenance period. Include diagrammatic drawings of each system and all items of equipment.

Use the same border and title block as the contract drawings. Drawings to be submitted in AutoCAD .dwg and/or Revit .rvt format (whichever is applicable to the project), a USB Thumb Drive copy and 3 hard copies shall be supplied to the superintendent, with Architectural backgrounds updated to the latest architectural.

4.4.3 Operation & Maintenance Manuals

A PDF version provided on a USB Thumb drive and **three (3)** hard copies of the manuals are to be supplied to the superintendent upon completion of the works. Each manual is to be A4 size, loose-leaf, in commercial quality with hard covers, each indexed, divided and titled. The Operation and Maintenance manuals be compiled of the following segments.

Format	
▪	Cover and Spine: Identify each binder with typed or printed title "OPERATION AND MAINTENANCE MANUAL". Identify the title of the project, volume number, volume subject matter, and date of issue.
▪	Table of Contents for each Segment and sub-segment
▪	All Pages to be numbered consecutively
▪	Dividers: Durable divider for each separate element, with typed description of system and major equipment components. Clearly print short titles under laminated plastic tabs.

	▪ Drawings: fold drawings to A4 size and accommodate them in the binders so that they may be unfolded without being detached from the rings. Provide with reinforced punched binder tabs.
Segment 1 – Introduction And Contents	
a.	Directory: a. Plumbing Contractors information including Plumbers License number b. Hydraulic consultants' information c. Architects information d. Project managers information e. Any sub-contractors involved on the project Include: Name, address, telephone and facsimile numbers, email address and. Names of responsible parties.
b.	System description: Technical description of the system installed, written to ensure that the client staff fully understand the scope and facilities provided. Identify function, normal operating characteristics, and limiting conditions.
c.	Instructions for use of Lifting tools etc.
d.	Emergency procedures, including telephone numbers for service, and procedures for fault-finding.
Segment 2 – Fixtures And Fittings	
a.	Name, address, telephone & facsimile numbers, email address and Web Address of the manufacturer and supplier of items of equipment installed, together with catalogue list numbers.
b.	Installed fixtures and fittings list including warranties for each
c.	A copy of manufacturers technical literature, installation and commissioning instructions
d.	Detailed recommendations for preventative maintenance frequency and procedures which should be adopted by the principal to ensure the most efficient operation of the systems installed.
e.	Schedule of spares recommended to be held on-site, being those items subject to wear or deterioration and which may involve the principal in extended deliveries when replacements are required. Included complete manufacture and model numbers, and local sources of supply.
f.	Schedule of normal consumable items, local sources of supply, and expected replacement intervals.
Segment 3 – Plant And Equipment	
a.	Rheem
b.	Grundfos
c.	Name, address, telephone & facsimile numbers, email address and Web Address of the manufacturer and supplier of items of equipment installed, together with catalogue list numbers.
d.	Installed plant and equipment list including warranties for each
e.	Manufacturer's technical literature as appropriate. Register with the manufacturer as necessary. Retain copies delivered with equipment.

f.	A list of items that require annual service agreements with pump suppliers, filter suppliers, thermostatic mixing valves, backflow preventers, etc.
g.	Detailed recommendations for preventative maintenance frequency and procedures should be adopted by the principal to ensure the most efficient operation of the systems installed.
h.	Schedule of spares recommended to be held on-site, being those items subject to wear or deterioration and which may involve the principal in extended deliveries when replacements are required. Included complete manufacture and model numbers, and local sources of supply.
i.	Schedule of normal consumable items, local sources of supply, and expected replacement intervals.
Segment 4 - Valves And Meters Schedule	
a.	Prepare and show a comprehensive schedule of Meters- The schedule shall include A – Position B - Size C – Function D – Meter number
b.	Prepare and show a comprehensive schedule of control valves - Valve schedule shall include A – Position B - Size C – Function D - Valve number
Segment 5 – Approvals and Certificates	
a.	Water Authority Approval
b.	Sewer Authority approval
c.	Trade Waste Approval
d.	a combination of Plumbing Approval and an Infrastructure Charges Notice (or negotiated agreement) issued by the Rockhampton Regional Council.
e.	Fire Engineering Report (FER)
f.	Authority's approvals
g.	A copy of the completed commissioning and test report for all pipework and equipment.
h.	Fire Hydrant & Fire Hose Reels Flow Test results
i.	Certification of TMV's
j.	Certification of Backflow Devices
Segment 6 - Drawings	
a.	Updated Sewer service diagram
b.	Updated Hydrant Block Plan
c.	A paper set of the "As Built" drawings in A3 reduced size and in full scale size folded to fit into each Manual
d.	A USB Thumb Drive copy of the "As Built" drawings in PDF and AutoCAD (.dwg) or Revit (.rvt) formats in each manual

5. Works by other trades

5.1 Building contractor

The following works shall be undertaken by the building contractor:

5.1.1 Prior to works commencing on-site

1. Erosion & Sediment Control in accordance with Councils requirements and Department of Conservation and Land Management – Urban Erosion & Sediment Control Handbook. (Including Sediment Control Fences, wash bays, straw bales, etc.)
2. Dewatering of Site.
3. Stamping of plans – The Builder shall take to Local authority:
 - the original approved DA architectural drawings stamped by council for stamping
 - a copy of the structural drawings
 - a copy of the hydraulic drawings
 - Building over sewer (BOS) approval letter from Local authority
 - Peg out
4. The ground is water charged (refer to geotechnical report for further information). Any installation of sewer pipes in water charged or unstable ground requires a structural pipe support solution which is to be design and certified by a suitably qualified structural engineer.

5.1.2 General

5. **Supply and installation of Fire Extinguishers.**
6. Duct access panels and inspection panels (Plumbing Contractor to co-ordinate locations with builder).
7. Inspection panels for all plumbing traps in ceiling areas.
8. Inspection panels for all control valves in ceiling areas (size to be min 300mm x 300mm).
9. The construction of the plumbing ducts in accordance with Specification F 5.2 (Sound insulation for Building Elements) of the Building Code of Australia.
10. Supply and installation of the following Sanitary Fixtures:
 - a. Sink top and hole penetration
 - b. Basin top and hole penetration
 - c. Dishwasher machine
11. Ice machine
12. Coffee machine
13. Provide ventilation to:
 - i. Water Chiller Cupboard
 - ii. Boiling Water Unit Cupboard
14. Access for maintenance to boiling water unit, hot water heater and temperature and pressure relief valve on the hot water heater.

5.1.3 Acoustic

15. Acoustically line ceilings and ducts and provide acoustic insulation to the ceilings and ducts to achieve the appropriate rating in accordance with the BCA.

5.1.4 Waterproofing

16. Waterproofing of the shower area.

17. Waterproofing of all wet areas.
18. Waterproofing / membrane of all Planter Boxes.

5.1.5 Protection

19. Bollards to protect external hydrants, gas meters, water meters and hose reels.

5.1.6 Accessible Toilet

20. All accessible toilet & shower grab rails in accordance with AS 1428.
21. The construction and layout of the accessible toilet area in accordance with all parts of AS 1428 (Note: All dimensions in AS 1428 shall be strictly adhered to)

5.1.7 Overflows

22. Overflows from the Terrace / Balcony areas.
23. Overflows from the Roof through the parapet.
24. Sealing between all pipework that pass through the building façade / concrete tilt panel. Including box gutter overflows and downpipes

5.1.8 Plinths / Slabs

25. Plinths to pumps and equipment installations.
26. Concrete slab for hot water heater.
27. Plinths for Hydrant pumps, jacking pump and Cold water pressure set.

5.1.9 Hydrants / Sprinklers

28. Hydrant test sump in the Pump room
29. Hydrant Booster enclosure.
30. Provide strengthening to the fire hydrant and hose reel cupboard to support pipework dropping down the wall.
31. Hydrant Tank on level?? including water proofing.

5.1.10 Stormwater connection point inspection

32. In accordance with the DA and the Building Professionals Board (BPB) Act the stormwater connection point shall be inspected by the Principal Certifying Authority (PCA) prior to backfilling.

Allow to contact the PCA and arrange for the inspection.

5.1.11 Downpipes

33. All exposed downpipes are to be installed by the roofer. Refer to architects' specification and drawings.

5.1.12 Stormwater Detention System

34. Stormwater detention pit. The builder shall calculate the size of the detention tank prior to the construction of the tank. Any change in shape or depth shall be forwarded to the superintendent for checking and verification. It is critical that the depth from the overflow point to the centre line of the orifice is issued to the superintendent for approval before the orifice plate is made.
35. The surveying of the new installed detention system / tank and certification by the surveyor. The surveyor shall provide an accurate new drawing or mark up the design drawing indicating changes to the following in red:
 - All Levels
 - All measurements

- Tank or basin size (including water volume stored and taking into consideration the overflow level)
- Tank or Basin location
- Overflow Level
- Outlets
- Inlets
- Orifice Diameter size
- Orifice Centre line
- Contour levels
- Kerb Levels
- Pit Levels
- Pit Sizes
- Grate sizes
- Invert Levels
- Confirm presence of:
 - Step rungs
 - Debris Screen
 - Silt trap
 - High early discharge chamber (where required)
 - Flap Valve
 - Warning Signage for basin ponding (where required)
 - Tank access Signage (where required)

5.1.13 Signage

36. Allow to coordinate all signage with the plumbing contractor in accordance with the signage section of the hydraulic specification.

5.1.14 Authorities Sewer and Water Main Relocation

37. The relocation of the sewer and water mains in accordance with the approved documentation.
38. The coordination with the nominated sewer diversion contractor.

5.1.15 Authorities Sewer main encasement

39. Contact Rockhampton City Council to arrange for the pier inspections once the piers have been excavated.
40. Contact Rockhampton City Council to arrange for the sewer concrete encasement inspections.

5.2 Mechanical contractor

The following works by others shall be undertaken by the Mechanical contractor:

1. Drainage from condensate units to locations provided by the Plumbing Contractor.
2. Ventilation to the Hydrant pump room and hot water plant room.
3. Connection from the cold water control valve to the condensate unit in the computer room
4. Diesel – Provision of a connection in the carpark exhaust control panel to activate the carpark exhaust fan upon operation of the diesel pump. (The Plumbing Contractor will supply the connection from the FIP panel).

5. Provision of penetrations and sealing in air-conditioning ductwork for the hydrant diesel exhaust.

5.3 Building Management System (BMS)

For BMS connection requirements, refer to the electrical table below.

5.4 Electrical contractor

The following works by others shall be undertaken by the Electrical Contractor:

Item	Location	No off	Volts	Kw	BMS
Dishwasher			240V	GPO	No
Boiling Water Unit Chilled Water Unit Note: Section J6 of the BCA requires the power supply to be controlled by a timer.			240V	GPO	?
Automatic urinal controls in ceiling of each Male amenities			240V	GPO	No
Gas heaters ignition system (? off in? Plantroom)			240V	GPO	?
Gas Power Flue system (? off in? Plantroom). Connection from the power flue to the Heater.			240V	GPO	?
Each Basin – Sensor Taps			240V	GPO	No
Hot water heaters			240V or 415V	...k W	?
Water Filter for the Domestic Cold Water			240V	GPO	?
Cold water pressure set. Variable Speed Supply electrical supply to and connect to the respective control panels		Dual	240V or 415V	...k W	?
Rainwater re-use system. Supply electrical supply to and connect to the respective control panels		Dual	240V	...k W	?
Reverse Osmosis circulating pumps Supply electrical supply to and connect to the respective control panels		Dual or single	240V	GPO or ...k W	?
Hot water circulating pumps Supply electrical supply to and connect to the respective control panels		Dual or single	240V	GPO or ...k W	?

Item	Location	No off	Volts	Kw	BMS
Sewage pumps (Sewer Pump out pit) Supply electrical supply to and connect to the respective control panels		Dual	240V or 415V	...k W	?
Stormwater Submersible pumps (Stormwater pump out pit) Supply electrical supply to and connect to the respective control panels		Dual	240V or 415V	...k W	?
Fire Hose Reel Pump – Lockable isolating switch Supply electrical supply to and connect to the respective control panels Essential Service - when the following occurs: Extract from AS 3000 “where such hose reels form the sole means of fire protection, i.e. where fire hydrants and automatic fire-sprinkler systems are not installed.”			240V or 415V	...k W	?
Diesel Fire Hydrant Booster Pump – Power supply to be fire rated to its own circuit breaker. Essential Service Supply electrical supply to and connect to the respective control panels			240V	...k W	?
Fire Hydrant pumps and jacking pump. Supply electrical supply to and connect to the respective control panels			240V or 415V	...k W	?
Data Logger Panel (Local authority gas metering panel).			240V	GPO	?
Local authority modem access for Data panel.			240V	GPO	No
Electrical supply, 240V to gas meter housing for power and light. All electrical fittings to be flash proof pattern.			240V		No
Supply and fit light fitting to gas meter housing.			240V	-	No
Power supply to pressure switches fitted, (2) (by Local authority) to gas service in meter housing area			240V	-	No
Power supply to the Gas Shut off system in the Science Rooms / Laboratories???	Science Rooms under the teachers'		240V	60hz	?

Item	Location	No off	Volts	Kw	BMS
Hardwired connection	desk or laboratories				
Power supply to the Gas Solenoid shut off valve for the Hot Water Heaters. (sensors by Mechanical Contractor) Hardwired connection	Hot water Heater room.		240V	60hz	?
Installation of all wiring, conduits and associated equipment to relay fire hydrant booster pump operations from the Hydrant Pump control panel to the FIP (FDCIE) in the Fire Control Room and/or Fire Indicator Board. The following shall be indicated on the FIP (FDCIE) / mimic panel: Power supply to fire pump controller present Pump common Fault alarm Pump running			240v	-	?
Supply and fit light fitting to the Hydrant Pump Room.			240v	-	No
Power supply to the Oil Arrestor.			240v	...k W	No
Power supply to the Pot wash sink heater located??????????			240V or 415V	...k W	No
Water chiller			240v	GPO	No
Ice Machine			240v	GPO	No

5.5 FIP (FDCIE) connections

The following works by others shall be undertaken by the Electrical Contractor:

Item	Connection to FIP
Visual Alarm Device (VAD) (when required at booster AS2419.1:2021 Clause 7.3.2)	Required

5.6 Hydrant truck hard stand

The building contractor is to provide a hardstand for fire truck access. It shall be an identifiable and clearly marked trafficable all-weather pavement providing access and capable of supporting a fire brigade pumping appliance during firefighting operations.

NOTE: Examples of pavements are bituminous concrete, aggregate, concrete or similar surfaces.

The Hard Stand shall be a minimum of 14m x 4m and capable of supporting loads of 850kPa.

5.7 Lockable authorities' enclosures and cabinets

EM Utility Access Restricted Locking System can be keyed to allow access by all or any of the following utilities:

Authority	Padlock Type
Local authority	EM Assa Abloy
Local authority	EM Assa Abloy
QFD	003 Padlock

6. Materials

6.1 Supply of materials

Supply and fix all materials required to complete the work. All materials shall be first quality and the best of their respective kinds. Second-quality or inferior materials shall be rejected. All costs associated with the replacement of rejected materials shall be borne by the Plumbing Contractor. All materials shall conform to the latest Australian Standard Specification, Code or Interim Code. If no Australian Standard exists, they shall conform to the latest British Standard or the American Society for Testing and Materials in that order.

6.2 Samples

Supply samples of materials when requested. Samples shall be labelled showing the location of their installation. After approval, all subsequent work shall conform to the quality of the sample.

6.3 Measurement

Provide measuring or weighing equipment for materials when requested.

6.4 Rejection of unsatisfactory materials

Rejected materials shall be marked by the Plumbing Contractor and removed from the site.

6.5 Approval

All items in this section shall be approved by the local authority and watermarked.

6.6 Installations

All materials and fittings shall be installed in accordance with the manufactures' instruction and the relevant Australian Standards.

6.7 Cross-linked polyethylene

All Polyethylene pipe & fittings shall be equal to Rehau or where others are specified under Fixture and Fittings.

6.7.1 General

The system shall have a minimum Design Life of 50 years for hot water at 70°C & 1.0 MPa operating pressure for an applied safety factor of 1.5. Pipes shall achieve a pressure class of PN20 according to AS 2492 and shall be Cross-linked Polyethylene type a (PE-Xa) only, with a minimum degree of crosslinking of 70%, homogeneous across the wall thickness of the pipe to reduce the chance of stress concentration during bending and reducing the chance of pipe kinks. Pipe in contact with the water shall be uncoloured virgin material only.

6.7.2 System Components

Non-threaded fittings and sleeves up to size 40mm shall be lead-free polymer. 100% of polymer fittings shall be geometrically tested by the manufacturer against the tolerances.

Threaded fittings shall be DZR brass to AS 2345 with a maximum of 2.8% lead. 100% of all Brass fittings shall have a permitted dezincification depth of no greater than 100µm and, in addition, shall be marked with a visual indicator (yellow dot) to confirm a manufacturer pressure test was conducted on the individual fitting and that it passed.

Pipe and fittings shall be mechanically jointed using sliding compression sleeve technology only. No O-rings or detachable joints are permitted in concealed locations.

6.7.3 Manufacturer Quality Assurance Obligations

Manufacturer shall provide:

- A minimum 25 year' warranty against manufacturing faults, including 10 year' warranty that includes cover for consequential damages.
- Confirmation of undisrupted product traceability from the raw material suppliers to the finished product, for pipes and fittings.
- Proof of Annual Process Verification Tests for pipes and fittings, conducted by a NATA or ISO 17025 accredited 3rd party lab.

6.7.4 Installation Quality Measures

In order to reduce the likelihood of system components being installed incorrectly and to subsequently reduce the risk of a leak and/or safety incident:

- Contractors are to be manufacturer-trained and certified, with documented competency-based training, including provision of training records/certificates and refresher training courses at intervals of 2 years.
- All Pipe, fittings and installation tools shall be brand-matched, or otherwise manufacturer approved
- On request, the manufacturer shall conduct sample inspections of the installation quality and general compliance with the manufacturer's written installation and commissioning instructions. The Inspection shall be carried out by trained employees of the manufacturing company, with experience in the manufacturer's recommended installation and commissioning instructions and practices. A letter from the manufacturer shall be provided for every inspection conducted summarising conformance to the above requirements and outlining any recommendations Note: The Contractor remains responsible for compliance with the contract and with regulatory and standards requirements.

6.7.5 Limitations

Operating Temperatures:

90 °C – Maximum operation temperature

70 °C at 1,000 kPa – Continuous operation

Must not be used on Solar systems between the tanks and the Panels.

The maximum test pressure is 1000kPa.

6.7.6 Expansion

Allowance for expansion in pipework materials shall be in accordance with the installation catalogue, to be provided by the manufacturer, and any other written or verbal instructions provided.

6.7.7 Insulation

All PE pipework shall be provided with insulation as per AS3500.4.

6.7.8 Protective Tape

Installation of Rautitan systems within concrete/screed floors and walls are allowed, provided specific requirements from local installation standards are fulfilled and adequate protection from mechanical damage and chemical damage are provided. Fittings shall be wrapped with suitable tape to avoid direct contact with concrete. Nitto Denko 204E is approved for use with all Rautitan fittings.

PE-X pipework shall NOT be painted.

6.8 Cast iron pipes and fittings for sanitary plumbing

All cast iron pipe and fittings shall be:

- a. In conformity with AS 1631 – Cast grey and ductile Iron non-pressure pipes and fittings.

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- b. Spun material.
- c. Externally coated as required by AS 1631 – Cast grey and ductile Iron non-pressure pipes and fittings.
- d. Mechanically jointed.
- e. Heavy wall pipe in accordance with AS 1631 – Cast grey and ductile Iron non-pressure pipes and fittings.

Jointing

- f. Mechanical brass bolted joints with rubber rings.
- g. Epoxy resin joints only where specially approved.

6.9 Ductile iron pipes and fittings

All ductile iron pipe and fittings shall be in conformity with AS 2280 – Ductile iron pipes and fittings.

6.10 Copper tube & fittings

Copper tube shall:

- a. Conform to AS 1432 - Copper tubes for plumbing, gas fitting and drainage applications.
- b. Be installed in and commissioned in accordance with AS 4809
- c. Be type A where located under buildings
- d. Jointing - Be jointed with silver solder containing not less than 5% silver solder and 6 mm lapped joints.
- e. Jointing – This can also be achieved with the Viega Crimp fit ProPress system in accordance with manufacturers' requirements.
- f. Be Wrapped or Sleeved when installed inground.

Fittings for Copper Tube shall:

- a. Conform to Australian Standard AS 3688 - Water Supply - Copper and copper alloy compression and capillary fittings and threaded-end connectors.
- b. Be dezincified brass or copper suitable for jointing by silver solder.
- c. Minimum wall thickness shall be not less than the tube thickness it serves.
- d. Be first quality high-pressure fittings for rising mains.
- e. Note: Copper tube shall not be used on urinals, grease arrestor drainage or vents or any photographic equipment/machinery.

6.11 Copper press-fit fittings

All copper press fit pipes and fittings shall be equal to Viega or where others are specified under Fixture and Fittings.

6.11.1 Water

Connection with press-fit fitting made of copper or gunmetal. Inspection of un-pressed fittings utilises a positive leak path for the whole range 2.2kPa to 300kPa when a dry pressure test is performed and 100kPa to 650kPa with a wet pressure test. This warranted safety function is facilitated by a channel in the metallic body of the fitting.

After pressing, the connection is permanent and inseparable. The seal is made of EPDM, with performance characteristics allowing use in solar applications. The fitting body has a cylindrical pipe guide in front of the seal. Fittings carry Watermark Approval.

The manufacturer will conduct training in the use and installation of the copper press-fit pipe joining system (tools and fittings) and run an onsite project start-up session for the contractor prior to the commencement of the project. The training session will run approximately for an hour for users nominated by the contractor. At the end of the training session, accredited installer certificates will be issued for those who have completed the training session. A

training DVD will be provided for future reference. Continuous technical field support can be provided by the manufacturers' representatives.

6.12 Stainless steel press-fit fittings

All stainless steel press fit pipes and fittings shall be equal to Viega or where others are specified under Fixture and Fittings.

6.12.1 Water

Connection with press-fit fitting and tube made of Stainless Steel Grade 316. Stainless steel tube to conform to EN 10312 Series 2 sizing. Inspection of un-pressed fittings utilises a positive leak path for the whole range 2.2kPa to 300kPa when a dry pressure test is performed and 100kPa to 650kPa with a wet pressure test. This warranted safety function is facilitated by a channel in the metallic body of the fitting.

After pressing, the connection is permanent and inseparable. The seal is made of EPDM, with performance characteristics allowing use in solar applications. The fitting body has a cylindrical pipe guide in front of the seal. Fittings and tube carry Watermark Approval.

The manufacturer will conduct training in the use and installation of the stainless steel press-fit pipe joining system (tools and fittings) and run an onsite project start-up session for the contractor prior to the commencement of the project. The training session will run approximately for an hour for users nominated by the contractor. At the end of the training session, accredited installer certificates will be issued for those who have completed the training session. A training DVD will be provided for future reference. Continuous technical field support can be provided by the manufacturers' representatives.

6.13 HDPE

General - High-Density Polyethylene (HDPE) shall be approved to MP 52, Spec 005, covering pipes and fittings. Installation shall be carried out in accordance to AS/NZS 3500, AS 2033 and the manufacturer's recommendations.

Jointing - of Pipe and fittings shall be joined with electro fusion welded joints, ring seal sockets, expansion sockets, flanged joints and threaded joints to manufacturer's recommendations. Welded joints are not acceptable

Expansion and Contraction - shall be catered for as recommended by the manufacturer and relevant standards. Suspended installations shall have expansion sockets at a maximum of 6m fixing distance or deflection legs. In-ground installations can be installed rigid, without expansion sockets.

Fix points - shall be made at the location of every expansion socket or deflection leg, to direct movements into the desired direction. Fix points shall be designed to hold the pipe steady in position to ensure proper function of an installation. The strength of fix points shall be designed in accordance with the relevant standards and manufacturers guidelines.

Bracketing - distance shall be in accordance to AS/NZS 3500 and manufacturer's guidelines.

Storage - Pipes stored outside are to be protected against deformation by covering. Pipes may not be stacked to height, which could result in deformation.

6.14 Galvanised mild steel piping and fittings

Galvanised mild steel pipe and fittings:

- a. Pipes shall conform generally with AS1074 - Steel Tubes and Tubulars for Ordinary Service.
- b. Shall be Equal to Schedule 10.
- c. Shall be in accordance with AS 2419.1

- d. Shall be 2.11mm for sizes 15mm up to and including 20mm, 2.77mm for sizes 25mm up to and including 50mm, 3.05mm wall thickness for pipe diameters from 32mm up to and including 100mm and a minimum of 3.40mm for pipe diameters greater than 100mm and up to 150mm.
- e. Shall be hot dipped galvanized internally and externally and at their ends in accordance with AS/NZS 4792. – The Galvanized coating shall have a minimum zinc coated mass of 300g/m².
- f. Are to be certified or listed by and tested and certified by a recognised body as being fit for purpose in hydrant systems in accordance with AS 2419.1
- g. Shall not be On-site welded or modified.
- h. Any cutting that exposes the base metal or damages the galvanized protective coating shall be repaired in accordance with AS 2419.1
- i. Any disturbance of the protective coating by cutting, roll grooving or handling shall be repaired with a zinc-rich primer or equivalent in accordance with AS/NZS 4792
- j. Shall be jointed with approved galvanised roll grooved couplings
- k. Shall be protected with Petrolatum tape coating and over wrapped in accordance with AS2419 where located in ground. The pipe length shall not exceed 1.5m in total length in ground. (for risers only)
- l. Pipe marking shall be in accordance with AS 2419.1
- m. Pipework to be marked and identified in accordance with AS2419.1 and AS1345. Pipes shall have the manufacturers name, wall thickness and grade and galvanising specification ink mark in accordance with AS 2419.1

6.15 PVC-U pipes and fittings

All PVC-U pipes and fittings for sewer and stormwater shall be:

- a. Be sewer grade in conformity with AS 1260-PVC-U pipes and fittings for drain, waste and vent applications.
- b. Solvent weld jointed in accordance with the manufacturer's instructions.
- c. Incorporate fittings of similar manufacture to the pipe used.
- d. The Burning out of fittings for reuse is unacceptable.
- e. The Heating or annealing of PVC-U pipes and fittings is unacceptable.
- f. Jointing - Solvent weld cement (high bond solvent cement) as recommended by manufacturer and approved by authority.
- g. Jointing – use fully moulded fittings only. Glue on Boss or saddle type clamp or strapped on joints shall not be used.
- h. Be installed in accordance with:
 - AS/NZS 3500 Requirements
 - The Plumbing Code of Australia
 - AS 2032 Code of Practice for installation of PVC Pipe Systems

6.16 PVC pipes and fittings - pressure

All pressure PVC-U pipes and fittings for rising mains shall be:

- a. In conformity with AS/NZS 1477 - PVC pipes and fittings for pressure applications.
- b. Solvent weld jointed in accordance with the manufacturer's instructions.
- c. Incorporate fittings of similar manufacture to the pipe used.
- d. The Burning out of fittings for reuse is unacceptable.
- e. The Heating or annealing of PVC-U pipes and fittings is unacceptable.
- f. Jointing - Solvent weld cement (high bond solvent cement) as recommended by manufacturer and approved by authority.
- g. Be installed in accordance with:

- Manufacturers requirements
- AS/NZS 3500 Requirements
- The Plumbing Code of Australia
- AS 2032 Code of Practice for installation of PVC Pipe Systems.

6.17 PVC pressure pipe & fittings (rubber ring jointed)

All Vinyl Iron pipe and fittings used for potable water and pressure sewer shall be:

- a. PN16 for PVC-O & PN12 for PVC-M
- b. Rubber ring jointed
- c. Incorporate Ductile Iron fittings (epoxy coated)
- d. Installed in accordance with AS/NZS 3500.1

Note: PVC Pressure (rubber ring joint) shall not be installed under Buildings.

Thrust blocks shall be installed in accordance with manufacturer's recommendations.

6.18 Fibre reinforced cement pipe

Fibre reinforced cement pipe and fittings shall:

- a. Be manufactured and tested in accordance with AS 4139 Fibre Reinforced Concrete Pipes and fittings
- b. Incorporate fittings being fibre reinforced for non-pressure applications and cast iron for pressure applications.
- c. Be reinforced with Cellulose fibre.
- d. Be jointed in accordance with manufacturer's instructions – with rubber ring joints in accordance with AS 1646. Lubricant for the joints shall be approved by the manufacturer.
- e. Installation shall be in accordance with the manufacturers' recommendations.
- f. Be James Hardie FRC or approved equal

6.19 Valves

6.19.1 General

Unless otherwise indicated all valves shall be:

- a. In accordance with the respective Australian Standard as noted.
- b. Installed in an accessible position for means of operation and/or removal.
- c. Of bronze material for valves up to and including 80 mm diameter.
- d. Of screwed pattern with a union fitted to the outlet each side of valves up to and including 50mm diameter.
- e. Of non-rising spindle pattern for services other than fire services, with clockwise closing.
- f. Of rising spindle pattern for fire services, with clockwise closing.

6.19.2 Positioning of Valves - Finished Areas & Ducts

The position of Valves shall:

- a. Be at a height of approximately 400-mm above floor level adjusted to match the joints, where installed within a finished area.
- b. Be at a height of approximately 1500 mm above finished floor level where installed within a duct.
- c. Be centrally located within a duct where access is obtained from a small access panel. Valve position shall ensure ease of maintenance and co-ordination with other services.

6.19.3 Control Valves - (Cold Water Supply)

Unless otherwise indicated control valves shall be:

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- a. Loose jumper valve type fitted with "O" ring seals to the spindle.
- b. Manufactured and tested in conformity with AS 1718 - Water Supply - Copper Alloy Screw-Down Pattern Taps - Specified by Dimensions.
- c. Constructed of brass materials.
- d. Fitted with unions on each side of the valve for maintenance/removal. Alternatively, one union can be deleted provided an approved screwed joint is substituted.
- e. Recessed stop / control valves shall be chromium plated all brass construction, of pattern and finish equal to that specified under Taps and Valves.
- f. Valves on the existing meter set will need to be changed to lockable lever handle resilient seated ball valves (see AS2441, figure 6.1)

6.19.4 Non-Return Valves

Unless otherwise indicated non return valves shall be:

- a. Manufactured and tested in conformity with AS 1718 - Water Supply - Copper Alloy Screw-Down Pattern Taps - Specified by Dimensions.
- b. Horizontal pattern with a gunmetal swing check for services other than fire services fitted with limit stop to prevent sticking in the open position. [Have a non-water hammer characteristic incorporating spring loaded bronze valve and seat.]
- c. Body fitted with screw headed inspection cap.
- d. Constructed of brass materials.

6.19.5 Balancing Valves

Unless otherwise indicated balancing valves shall be:

- a. Of the globe type with facilities for measurement, regulation, isolation and venting,
- b. Constructed from a metal, arsenically inhibited brass alloy,
- c. Able to be pre-set and locked to the required flow rate,
- d. Able to be opened and closed without disrupting the pre-set calibration,
- e. Fitted with unions on each side of the valve for maintenance/removal.
- f. Incorporate setting indicators for reference purposes

6.20 Reconditioned valves

Reconditioned / second-hand valves of any type shall not be installed.

6.21 Fire hydrant landing valve

Landing Valves shall be:

- a. Have 2 valved outlets for external hydrants
- b. Have 1 valved outlet for internal hydrants
- c. Be in accordance with AS 2419.1 and AS 2419.2 - Fire Hydrant installations.
- d. Approved by the local water supply authority.
- e. Approved by the QFD
- f. Of loose jumper valve pattern.
- g. Of all bronze construction
- h. Fitted with brass/bronze cap and chain to the outlet thread.
- i. Of 65-mm diameter
- j. Installed at a height of 750-mm from the centre of valve to ground level, the valves shall be at an angle of 350 below the horizontal

6.22 Fire hose reels

Fire Hose Reels shall be:

- a. In accordance with AS 1221 Fire Hose Reels.
- b. Have 36m x 19-mm diameter elastomeric hose with 25-mm diameter nozzle.

- c. Have swinging hose guide complete with roller guides, galvanised nozzle steel mounting plates and 25-mm brass loose jumper valve and union.
- d. Be mounted at 1500-mm above floor level to centre of reel with 15-mm bolts.
- e. Be provided with additional, clear operating instructions inside cabinet or recess.

6.23 Clearouts

Where "CO" is indicated on the plan, provide and install clear-out inspection fittings.

Internal - riser from pipeline shall extend vertically to finished floor level and terminate with a stainless steel screw-out lid and frame.

External - Riser from pipeline shall extend vertically to ground level and terminate with a stainless steel screw-out lid and frame. A 100mm wide concrete surround should be installed around the clear-out.

6.24 Flexible connections

Flexible connections (braided stainless hoses) shall

- a. NOT be used in exposed locations
- b. Shall be minimum required length
- c. Shall be installed in accordance with manufacturer's instructions
- d. Be approved by Superintendent before installation

Shall be by one manufacturer throughout the project.

Should flexible connection be proposed – submit full details, including installation details – for approval.

6.25 Bushfire-prone areas

All above-ground exposed piping for the supply of water and gas shall be metal.

6.26 Fire stop collars & penetrations

Allow to supply and install approved fire stop collars to all pipe penetrations through slabs, walls and all fire-separated areas.

Fire Stop collars shall be provided on all PVC-U, HDPE and non-metal pipe penetrations through fire-rated reinforced concrete slabs.

All penetrations and core holes to be protected in accordance with NCC 2022 clause C4D15.

Fire Collars shall be:

Item	Brand
Pipework	Trafalgar Fire
Floor Wastes	Trafalgar Fire

Fire Collars to be indicated on Shop drawings and fire collar matrix to be provided.

All fire collars to have associated test reports provided.

Fire Collars indicated in the matrix below are to be confirmed/updated for the Shop Drawings by the plumbing contractor.

Refer to the hydraulic drawings cover sheet for service abbreviations.

FIRE COLLAR MATRIX – DRAINAGE SERVICES					
Service	Material	Penetration Size	Fire Collar (Slab 250>)	Fire Collar (Slab <250)	Fire Rating
FW	U-PVC	110	CIFWC-H100	CIFWC-100	-/240/240

FIRE COLLAR MATRIX – DRAINAGE SERVICES					
BTFW	U-PVC	110	CIFWC-H100	CIFWC-100	-/240/240
SHR	U-PVC	110	CIFWC-H100	CIFWC-100	-/240/240
ST 100	U-PVC	110	CHC-100	CLC-100	-/240/240
SD 100	U-PVC	110	CHC-100	CLC-100	-/240/240
DP 100	U-PVC	110	CHC-100	CLC-100	-/240/240
CVP 100	U-PVC	110	CHC-100	CLC-100	-/240/240
RV 65	U-PVC	69	CHC-65	CLC-65	-/240/240
WC	U-PVC	110	CHC-100	CLC-100	-/240/240
KS	U-PVC	56	CHC-50	CLC-50	-/240/240
SK	U-PVC	56	CHC-50	CLC-50	-/240/240
T/LT	U-PVC	56	CHC-50	CLC-50	-/240/240
TTD	U-PVC	56	CHC-50	CLC-50	-/240/240
TD	U-PVC	56	CHC-50	CLC-50	-/240/240
GV	U-PVC	43	CHC-40	CLC-40	-/240/240
BTH	U-PVC	43	CHC-40	CLC-40	-/240/240
B	U-PVC	43	CHC-40	CLC-40	-/240/240

FIRE COLLAR MATRIX – PRESSURE SERVICES				
Service	Material	Fire Rated Material	Fire Rated Wrap	Fire Rating
CW 50	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
CW 40	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
CW 32	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
CW 25	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
CW 20	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
HW 50	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
HW 40	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120

FIRE COLLAR MATRIX – PRESSURE SERVICES				
HW 32	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
HW 25	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
HW 20	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
GAS 50	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
GAS 40	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
GAS 32	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
GAS 25	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
GAS 20	COPPER	FyreFLEX Sealant	TWRAP 300MM	-/120/120
FH 100	STEEL	FyreFLEX Sealant	TWRAP 450MM	-/120/120
Multiple Services combined in single box	COPPER UP TO 50mm	Fire Box Maxi-350 350x125x250	TWRAP 450MM	-/120/120

7. Excavation and backfill

7.1 Trench excavation

Keep adequate dewatering equipment on the site at all times and keep the excavation free of water.

All excavation shall be in accordance with WorkCover QLD - Code of Practice for excavation.

All excavations shall be carried out in a safe, secure and prudent manner that will prevent any form of collapse, slip or failure of the sides of the excavation.

The ground shall be excavated as trenches to enable the various pipelines to be constructed at the locations shown on the drawings, with trench width increased where necessary to permit the construction of access chambers and pits. Trenches shall be excavated at uniform grades and in straight lines.

Upon commencement of excavations, a general examination of the trenches shall be made to determine the presence or otherwise of any material considered by the Plumbing Contractor to be detrimental to the life and future condition of the pipeline to be installed.

Wherever such material is located, the Plumbing Contractor shall advise the Superintendent immediately, together with a recommendation and the costs of remedial action to be taken. No work regarding this remedial precaution is to commence until written approval to proceed is obtained from the Superintendent.

Remove all excess excavation material from the site, and pay all cartage and disposal fees, wherever applicable, to provide a clean, neat site devoid of all debris created by this section of works.

7.2 Tree Protection and Trenching Requirements

Trenching within the vicinity of trees shall be undertaken using the minimum practicable trench width and depth to minimise damage to minor root systems. Works shall avoid major structural roots and remain outside the nominated Tree Protection Zone (TPZ) wherever possible.

All works within or adjacent to tree protection areas shall include, but not be limited to, the following requirements:

- Avoid unnecessary soil compaction within tree root zones.
- Do not place fill hard against tree trunks or root crowns.
- Utilise low-impact excavation methods (e.g. hand excavation, hydro-vacuum excavation) where practical.
- Provide arboricultural supervision for all works undertaken in close proximity to retained trees, particularly within or near the TPZ.

All works shall be carried out in accordance with the Arborist's Report and relevant standards, and any damage to trees or root systems shall be immediately reported to the Superintendent

7.3 Exceeding excavation

If the Plumbing Contractor has exceeded the section area of excavation, as shown, in consequence of any injudicious working, slips, falls, or for any cause other than by direction of the Superintendent then the Plumbing Contractor shall, at sole cost, remove such extra materials from the site, make good and fill in the extra excavation with concrete, sand or approved filling as may be directed.

No extra payment will be made for excavation in excess of that required by the drawings and specifications unless directed in writing by the Superintendent.

7.4 Timbering of excavation

Subject to any Act of Parliament, Ordinance or Regulation, the Plumbing Contractor shall satisfy himself as to the necessity of timbering any excavation and shall accept the sole responsibility as to its being required and to its use in the works.

Where necessary for safe and efficient completion of the work erect shoring and timbering of sufficient strength and quality to prevent earth and other material slipping or falling in or being shaken from the side of the excavation. As the work proceeds, all shoring and timbering shall be withdrawn except in the cases where the Superintendent has indicated in writing that such shoring and/or timbering shall be left in position. The supply, erection and withdrawal of timber work shall be considered to be included in the cost of excavation.

7.5 Precautions & safeguards

Carry out the work in a careful, secure, safe and tidy manner and take all precautions against damage whether arising from bad workmanship, breakage or machinery or plant, inefficient timbering, flooding or any other cause whatsoever. Provide, erect and maintain warning signs, temporary fences, barriers and warning lights adjacent to any works such as trenches and excavations or stacks of material which could be considered a danger to persons or traffic of any kind.

The Plumbing Contractor shall obey all directions given to him with regard to the provisions of lighting and barriers, but shall not be relieved of the responsibility to provide adequate lighting and barriers to prevent any accident or damage.

7.6 Use of explosives

The use of explosives will not be permitted.

7.7 Bedding material

Where trench base is of stable sand material pipe barrels shall be directly supported on the undisturbed natural ground. In all other cases, pipelines shall be supported on either a granular bedding or cement mortar bedding in accordance with AS/NZS 3500.2 clause 5.4 and the Plumbing Code of Australia.

Where compaction of backfill over the pipeline is to be made by the use of mechanical vibrator, a further sparging to a level equal to 1/3rd the pipe barrel shall be made using 6:1 cement mortar. The additional cement mortar shall be placed after the initial setting of the 4:1 sparging takes place.

Where a mechanical vibrator is employed, the lines shall be subjected to a second test upon completion of the compaction.

Where water-charged ground is encountered, as previously mentioned under TRENCH EXCAVATION, the plumbing contractor will allow for all pipe bedding requirements to satisfy the requirements of the authorities having jurisdiction over the works, including the local authority and compliance with AS/NZS 3500. These support methods will be deemed to be included in the subcontract tender.

7.8 Backfilling

7.8.1 General

Unless instructed otherwise where a length of trench has been inspected, the bedding material shall be placed and compacted. After a length of pipeline has been constructed, tested and inspected the trench where located under finished surfaces such as concrete, bitumen or paving shall be backfilled with granular material up to underside of the working surface. Granular material shall be in the form of sand, fine sized "run of river" gravel or similar, free of

clay and rock. Thoroughly compact the material around and above pipes by flooding, hand tamping or mechanically ramming to not less than 95% compaction as set out below.

In open spaces backfill with granular material 150 mm above pipe collars and then backfill with selected excavated material.

Selected excavated material shall be free from rocks or other hard particles, which would be retained on a sieve having 15 mm openings. It may contain compactable clay lumps up to 75 mm in size. The spoil from the excavation if it complies with the above, or can be sieved or broken up to comply, will provide a suitable material and shall be placed in successive layers and compacted to not less than 95% of the density obtainable from the Standard Compaction method in accordance with AS 1289 - Methods of Testing Soil for Engineering Purposes, shall be placed for the remainder of the trench depth up to 150 mm from the finished surface level.

Unless otherwise directed the remainder of the trench filling shall be carried out using the original top soil (where suitable) or approved sand - sandy / loam fill.

Backfilling shall be completed as soon as possible after laying to avoid damage or floating of pipes if trench is flooded by water.

7.9 Existing grass and garden bed resurfacing

The Grass areas that have been excavated shall be reinstated to match the existing levels. The grass shall match the existing grass on-site.

The Garden areas that have been excavated shall be reinstated to match the existing levels. The garden soil and mulch shall match the existing.

All damaged surfaces shall be repaired.

7.10 Trenches excavated in filled ground

Provide a schedule of rates in the tender for the items listed below, these items are a suggested means of support only. The local authority shall approve the type of support prior to installation. The plumbing contractor shall allow for a Structural Engineer to design and certify the support system:

Where Located Externally:

- a. Laid on concrete beams, with at least three steel rods of not less than 15 mm nominal size laid longitudinally within the beam or two layers of trench mesh (type F8TM) and be not less than 150mm thick with a minimum width of 450 mm for 100 mm and 150 mm pipework, 600 mm minimum width for 225 mm to 450 mm pipework and 750 mm minimum width for pipework over 450 mm. Beams shall be supported by concrete piers of not less than 230 mm x 230 mm founded to a depth of firm natural strata and placed at intervals on not more than 3000 mm with four steel rods of not less than 15 mm nominal size with a minimum of 50 mm cover and tied into the reinforcement of the beam for a distance of 300 mm.
- b. Where located internally: Laid on concrete beams as described in part (a) above and supported by tie rods of not less than 12 mm diameter reinforcing bar formed to fix the support beam steel to the concrete slab steel situated above the line. Tie rods shall be spaced at not more than 1500 mm and sleeved with 100 mm diameter PVC-U. Tube extending from within the support beam and into the concrete slab. The entire length of PVC-U tube shall be filled with concrete in such a manner that no section of the rod shall be exposed. Connect top of tie rod to slab steel with minimum of 300 mm length bend.
- c. Line the trench with a permeable fabric equal to "terra firma 700" then back fill to a depth of 450 mm with 10 mm blue metal gravel. The drainage line shall then be laid

on the bed of gravel and backfilled with 10-mm blue metal to a height of 150 mm above the collars of the pipework. The trench fabric shall then be turned over the gravel and the remainder of the trench backfilled as previously specified.

Generally, drainage lines are not permitted to be laid in filled ground without additional support. Consideration to waive the support may be given by the Superintendent and/or the governing authority where the fill is compacted to 98% dry density ratio (Australian Standard AS1289 - Methods of Testing Soil for Engineering Purposes E1.1).

Where additional treatment to achieve a compaction of 98% dry density is carried out using mechanical compacting equipment and the Plumbing Contractor proposes to install pipes and pits in the compacted fill without additional support, comprehensive compaction test results, location plans and "letters of adequacy" from the supervising structural engineer stating that the compacted fill is suitable for the support of pipes and pits is required to be submitted to the Superintendent and the governing authority. Approval in full shall be obtained from both parties before any work proceeds.

All trenches are to be inspected by the Superintendent and governing authority prior to laying of drainage lines; claims for redundant work as a result of failing to do so shall not be considered valid.

7.11 Trenches excavated in rock

It shall be deemed that excavation in rock has been allowed for in the tender where it should have been reasonably anticipated by the Plumbing Contractor, or where test bore results indicate the presence of rock and the test bore results are available for inspection by the Plumbing Contractor, or where inspection of the site and its surroundings would indicate the presence of rock.

Should the Plumbing Contractor encounter rock not previously anticipated during excavations, notify the Superintendent immediately so that the extent, quantity and type of rock excavation can be assessed by the Superintendent prior to commencement of rock excavation. Failure to observe this requirement shall forfeit the Plumbing Contractor's right to reimbursement of additional costs relating to the excavation of rock unless the extent, quantity and type of rock can be assessed later and there was not alternative to the excavated route.

Provide in the tender a schedule of rates, which will be the basis of assessing the cost of additional work associated with excavation in rock. The schedule shall be the Plumbing Contractor's price in cubic metre rates for additional cost (extra over) to excavate trenches in soft rock and in hard rock. The Plumbing Contractor shall also provide cubic metre rates for the additional cost to excavate pits in soft rock and in hard rock.

7.12 Sewer and stormwater drains in deep or wide trenches

Before laying sanitary drainage and stormwater drainage in a deep trench (i.e. depth greater than 3 metres) or a wide trench (i.e. width at top of pipe greater than outside diameter plus 600mm) the following is needed:

1. A certificate of structural adequacy for:
 - class of pipe and
 - bedding, haunch, pipe support and backfill issued by a practicing civil engineer
2. Permission of the relevant Authority.

Note: Pipes shall be installed in accordance with manufactures instructions / requirements.

8. Stormwater drainage

8.1 Generally

Supply and install all necessary pipes, junctions, bends, pits, inspection openings, excavation and sundry equipment to convey stormwater discharges from the **various downpipes**, to the point of discharge as shown on the drawings.

The location of pipes indicated on the drawings are diagrammatic. Pipeline positions shall be determined on site in conjunction with all other disciplines to ensure adequate co-ordination of all services and elements. Co-ordination shall be carried out prior to any setting out, excavation and pipe installation takes place.

The entire installation shall provide adequate drainage for the project and be to the approval of the Superintendent and Local Authority.

Pipe materials shall incorporate Best Environmental Practice (BEP) requirements and be as follows:

Location	Material
Rising Main (Pressure Service)	Cromford PE - Welded GRADE:PN16 With the words "Pressurised Stormwater" Printed continuously down the pipe
Stormwater Drainage – Below Ground	IPLEX Sewer Grade PVC-U DWV All joints shall be purpose-made PVC-U sewer grade - solvent-welded joints. Boss / saddle-type clamp / strapped-on joints shall not be used. Stormwater grade PVC-U pipe and fittings shall not be used.
Stormwater Drainage – Below Ground	IPLEX BLACK-MAX PE Sizes-225mm to 600mm
Stormwater Drainage – Below Ground	James Hardie Fibre Re-enforced Concrete Pipe (FRC) Sizes-225mm to 600mm The pipes are to be laid in accordance with AS 3725 Loads on Buried Concrete Pipes
Stormwater Drainage – Below Ground	Humes Concrete Pipe Sizes-225mm to 2100mm The pipes are to be laid in accordance with AS 3725 Loads on Buried Concrete Pipes
Stormwater Drainage – Above Ground	IPLEX Sewer Grade PVC-U DWV All joints shall be purpose made PVC-U sewer grade - solvent welded joints. Boss / saddle type clamp / strapped on joints shall not be used. Stormwater grade PVC-U pipe and fittings shall not be used.
Stormwater Drainage – Above Ground Fire Rated	IPLEX Crevit – Cast Iron Drainage System
Stormwater Drainage – Above Ground	ACO – Stainless Drainage System

8.2 Stormwater connection point

Ascertain the depth, position and suitability of the stormwater connection point prior to the commencement of any work (Pipe laying & excavation) and ensure that the stormwater can gravitate to the connection point. The Superintendent shall be advised immediately should any adjustment be required to execute the work.

No claims for redundant work will be considered due to failure to comply with this requirement.

No pipe laying or excavation shall be undertaken until this has been undertaken.

8.3 Downpipes

Item	Material	Supplied and installed by
Downpipes where not visible	PVC-U sewer grade material all as specified under "Materials".	Plumbing Contractor
Downpipes where visible	Aco Stainless Steel Pipe Grade 316 Available in Sizes 50/75/110/125/160/200	Building Contractor
Downpipes	Specified by Architect	By others
All internal downpipes and downpipes enclosed / cast in columns / bricked in	HDPE material all as specified under "Materials".	Plumbing Contractor

8.4 Downpipes cast in columns (HDPE)

Downpipes and Stormwater cast in slab or columns shall be HDPE material all as specified under "Materials".

Where downpipes are included within columns they shall be installed progressively with formwork.

All pipework shall be tested under maximum hydrostatic head conditions prior to the pouring of concrete and maintained totally full of water during pouring and curing of concrete. Inspection openings at the base of each downpipe shall be extended to the face of the column and provided with an access panel.

8.5 Downpipes connections to in ground drainage services

Where Downpipes and Stormwater join together the joints are to be fully sealed to withstand a minimum pressure of 50kpa. Stormwater grade PVC is not to be used.

Loose joints will not be permitted.

8.6 Charged downpipes and stormwater services

Downpipes and Stormwater where used in a charged situation must be installed to withstand the pressure they are subjected too. The table below nominates the maximum depth from the rainwater tank overflow to the lowest point of the inground drainage service.

Max Depth	Material
5m	Sewer Grade UPVC
100m	PN12 Pressure Grade UPVC

Where it is required to use Pressure pipe for a particular service the pipework is to be installed a minimum of one size larger than that nominated on the drawings.

8.7 Excavation and backfilling

Carry out all necessary excavation and backfilling as specified under "Excavation" and "Backfilling".

8.8 Gradients

Pipelines shall be laid true to line and bore from point to point.

Unless otherwise indicated on drawings pipelines shall be graded to a minimum of 1 in 100 slope.

8.9 Testing

In ground - Pipework of 225mm diameter and smaller shall be subjected to a hydrostatic test under maximum choke conditions for a minimum period of thirty minutes. Any leaks shall be repaired damaged pipe shall be removed.

Above Ground – Pipework shall be capped and subjected to a hydrostatic test.

Subsoil pipes in a drainage trench

Subsoil drainage pipes shall be provided adjacent to every inlet stormwater pipe at each pit and sump for a distance of 2 metres. The subsoil pipe shall be 100 mm diameter perforated corrugated Class 1000 or smooth wall PVC-U pipe conforming to AS 2439 - Perforated Plastics Drainage and Effluent Pipe and Fittings Part 1 and shall be fitted with a filter sock. The filter sock shall be appropriately tied at the upstream end of the subsoil pipe to preclude the entry of filter material. The subsoil pipeline shall be laid at the same grade as the stormwater pipeline.

8.10 Subsoil drainage

Subsoil drainage pipe shall be 100 mm diameter perforated smooth wall PVC-U pipe or PVC-U Class 1000 corrugated section, conforming to AS 2439 - Perforated Plastics Drainage and Effluent Pipe and Fittings Part 1.

The pipeline shall be surrounded by 150mm of blue metal (20mm). The blue metal shall be 150mm from the top bottom and either side of the pipe. The blue metal and pipework shall be wrapped in a geotech filter fabric. The fabric shall lap 100mm (minimum).

The entire subsoil trench shall then be backfilled to the bulk excavation level, or porous layer under the slab, with 20mm blue metal, to create a porous drainage layer directing water to the subsoil drain.

8.11 Stormwater connection point inspection

In accordance with the DA and the Building Professionals Board (BPB) Act the stormwater connection point shall be inspected by the Principal Certifying Authority (PCA) prior to backfilling.

Allow to contact the PCA and arrange for the inspection.

8.12 Stormwater certification

The Plumbing Contractor shall supply a certificate that the Box Gutters and overflows have been installed in accordance with the drawings and all overflows, sumps etc. are installed as per the drawings.

8.13 Detention tank

Allow to supply and install the following for the stormwater detention Tank:

- 2 x Grates.

- Step rungs
- Flap valve
- Orifice plate - 3mm thick stainless steel plate with machined orifice centred over outlet with 10mm Dyna bolts.
- Debris screen Maxi Mesh, with a handle to meet the council's requirements. To be installed at a 45° angle. Removable galvanised mild steel debris screen equal to Lysaght maximise rh3030 shaped around outlet & provided with lifting handle.

9. Sanitary plumbing and drainage

9.1 Generally

Supply and install all Sanitary Plumbing and Drainage from soil and waste fixtures to the Pipe materials shall incorporate Best Environmental Practice (BEP) requirements and be as follows:

Location		Material
Sanitary systems	Drainage/stack	IPLEX Sewer Grade PVC-U DWV Sizes – 40, 50, 65, 100, 150, 225mm
Sanitary systems	Drainage/stack	REHAU Raupiano Plus Sizes – 40, 50, 75, 110, 160mm
Sanitary Drainage	– Below Ground	IPLEX SewerMAX PE Sizes – 225, 300, 375, 450, 525, 600mm
Drainage – Above Ground		IPLEX Crevit – Cast Iron Drainage System
Drainage – Above Ground		ACO – Stainless Drainage System
Grey Water		IPLEX Sewer Grade PVC-U DWV
Black Water		IPLEX Sewer Grade PVC-U DWV
Trade Waste		Geberit HDPE Sizes – 50, 56, 63, 75, 110, 160mm
GAPO (Pressure Service) Terminate with an 80mm camlock fitting and cap at each end		Cromford – Black PE - Welded GRADE: PN16 With the words “Grease Arrestor Pump Out” Printed continuously down the pipe
Rising Main (Pressure Service)		Cromford – Black PE Welded GRADE: PN16 With the words “Pressurised Sewer” Printed continuously down the pipe

Provide all necessary materials, pipes, junctions, bends, offsets, branches, couplings, brackets, clear-outs, floor wastes, excavation, backfilling, testing and sundry equipment required for the installation as indicated on the drawings and to the regulations and requirements of the relevant authority and the satisfaction of the Superintendent

The location of pipelines indicated on the drawings is diagrammatic. Pipeline positions shall be determined on-site in conjunction with all other disciplines to ensure adequate co-ordination of all services and elements.

Co-ordination shall be carried out prior to any setting out, excavation and pipe installation taking place.

Execute the works, using only materials and structures as approved by the local Authority and to the satisfaction of the Superintendent.

9.2 Sewer connection

Make application to the Sewer Authority, ascertain the depth, position and suitability of the sewer drainage connection point prior to the commencement of any work (Pipe laying & excavation) and ensure that the new sewer drainage can gravitate to the proposed connection

point. The Superintendent shall be advised immediately should any adjustment be required to execute the work.

No claims for redundant work will be considered due to failure to comply with this requirement. No pipe laying or excavation shall be undertaken until this has been undertaken.

9.3 Sewer surcharge

Prior to the installation of any sewer drainage, allow to contact the local plumbing inspector and confirm that the Authority's sewer is not a surcharging sewer.

9.4 Traps

Traps shall be of 'P' type as per the following:

Material	Location
Copper	All exposed traps shall be chrome plated copper
PVC-U	All concealed traps (i.e. traps located in cupboards) shall be PVC-U
Copper, Brass or Stainless Steel	Traps subject to temperature discharges above 600c shall be chrome plated copper, brass or stainless steel
HDPE	All traps in the lab areas shall be HDPE

9.5 Sinks & basins

Plumbing Contractor will supply taps, plugs, and washers (brass chrome plated), as well as chrome plated traps and waste in kitchen/servery areas.

9.6 Excavation and backfilling

Carry out all necessary excavation and backfilling as specified under "Excavation" and "Backfilling".

9.7 Gradients

Pipelines shall be laid true to line and bore from point to point.

Unless otherwise indicated on drawings, pipelines shall be graded in accordance with the Authority's requirements.

9.8 Testing

Provide a hydrostatic test to choke level for a minimum period of 15 minutes and as required by the testing authority and the Superintendent.

9.9 Locating floor wastes, tundishes, drainage points

Coordinate the locations of floor wastes, tundishes and drainage points with the architect.

9.10 Floor wastes

Floor wastes with PVC-U pipe – an approved safe waste tray shall be used on all floor and shower wastes.

Use a 100 mm riser and a 50 mm or 65 mm outlet of self-cleansing pattern for sanitary plumbing and 100 outlets for sewer drainage.

Grate and surround to be chrome-plated brass.

Chrome-plated ABS grates or plastic grates are not acceptable

Ensure that all floor wastes are properly supported.

Ensure that all relevant floors are graded to floor wastes – coordinate with builder.

Ensure that all floor waste risers are watertight.

9.11 Expansion joints

Expansion joints should be installed in accordance with AS/NZS 3500 and for PVC-U in accordance with AS 2032 – Code of practice for installation of PVC-U pipe systems.

Supply and install approved expansion joints to all soil, waste, relief and main vent lines.

9.12 Inspection openings & gates

Install inspection openings in accessible locations so that each section of pipework can be cleaned. Inspection opening sizes shall be in accordance with the authorities' requirements.

Note: All inspection openings shall be brass.

9.13 Clear-out sewer access points

Install clear-out sewer access points in accessible locations so that each section of pipework can be cleaned. Clear-outs are to be raised and finished at surface level in an appropriate manor. Clear-out sewer access points sizes shall be equal to the size of pipework that it serves.

Note: All clear-out sewer access points shall be Stainless steel.

9.14 Overflow gully

Supply and install an overflow gully to provide a safe release from the connection point. Gully shall be constructed of 100mm diameter "P" trap and 100mm riser. Top of riser shall incorporate 200mm cast iron grate with concrete surround. Terminate gully top to comply with current Authorities regulations.

Provide inlet (fixture) or hose tap to charge gully

9.15 Swimming pool backwash

The pool backwash must connect to the sewer via a discharge pipe located over the sewer overflow gully. The connection shall be undertaken in accordance with the local authority requirements and AS 3500.2:2021 Clause 12.8.7

9.16 Vents

Vents – All vents penetrating the roof shall be painted to match the roof colour. Where the roof is a Colourbond colour or similar the paint batch number shall be matched and applied to the vents.

Provide approved cowl to each vent, terminate vents at minimum approved heights above roofs.

Flash each vent penetration through the roof with approved EDPM flashing – “Dektite” or approved equal. Install in accordance with manufacturer's instructions, ensure neat and watertight installation. Paint flashings to match roof colour.

9.17 Boiling water unit - drain from tundish

All pipework shall be copper type B as specified under Materials.

The connection between the copper trap and the tundish drainage line shall be a union connection to allow for ease of removal.

9.18 Pipework cast in slab and beams

All pipework cast in slabs shall be wrapped / protected in accordance with AS/NZS 3500, AS/NZS 2032 and Authority requirements.

All pipework shall be tested under maximum hydrostatic head conditions prior to the pouring of concrete and maintained totally full of water during the pouring and curing of concrete.

Pipework is to be tied into the structural steel and support to prevent any movement to structural engineer requirements.

9.19 Bath connections

All connections under baths shall be installed with an approved connection.

Jointing using sand & cement or silicon of pipework between the outlet of the bath and the drainage shall not be installed.

9.20 Accessible toilet & basin & shower

The accessible toilet & basin & shower shall be installed in accordance with all dimensions to AS1428 - Design Access and Mobility, Part 1 and Part 2

Refer to the sanitary fixtures and appliances schedule at the rear of the specification for the manufacturer and model of toilet suite, basin and tapware.

Note: All grab rails specified by architect and supplied by builder and fixed to wall in accordance with AS 1428

9.21 Trade waste application

It is the responsibility of the plumbing contractor to make the Trade waste application with Rockhampton City Council (or the Local Trade waste Authority) for the installation of the Grease arrestor

Coordinate the application with the client/building owner and the Trade Waste Inspector.

9.22 Joint sewer

The existing sewer and boundary trap on-site is part of a joint sewer, and it serves several properties upstream from the site. All the connected properties own this joint sewer.

The exact location of the joint sewer on-site is not known; the plumbing contractor shall be allowed to locate the pipe.

The joint sewer is to be re-laid and concrete encased (150mm all around) where it passes under the new building. The concrete encasing shall be supervised by the Superintendent.

The exact location and depth of the sewer shall be recorded on the as-built documents.

9.23 Local authority sewer concrete encasement

The encasement of the Local authority sewer must be:

- Carried out by an Accredited Constructor of minor Works (Sewer).
- Application to be raised with Local authority prior to commencement.
- Local authority shall inspect and approve the encasement.
- Allow to contact Local authority, make all applications.
- Provide certification of the encasement to the superintendent.
- Note: Local authority require that any pipes or collars that are cracked or damaged in any way are to be replaced prior to the concrete encasing

9.24 Co-ordination with bio-septic

Allow co-ordination with bio-septic for the excavation of the site and the installation of their equipment.

10. Cold water

10.1 Generally

Supply and install all cold water pipes from the **existing Authorities water main** to all fixtures, fittings and taps requiring cold water. Include for all pipework, bends, offsets, brackets, taps and sundry equipment required for the installation.

Pipe materials shall be as follows:

Location	Material
In-ground from the Authority water main to the Boundary Water Meter	Kembla - Copper Type B Note: When Installed in ground copper pipework to be sleeved. Refer to Specification Section: Cold & Hot Water Fixtures & Fittings
In-ground from Boundary Water Meter	Kembla - Copper Type B Note: When Installed in ground copper pipework to be sleeved. Refer to Specification Section: Cold & Hot Water Fixtures & Fittings
In-ground from Boundary Water Meter	Cromford – Blue Stripe PE GRADE:PN16
Cold Water Main Runs through building	Kembla - Copper Type B
Cold Water Main Runs through building	Viega – Stainless Steel - Sanpress Inox
Cold Water Rough-In (in wall)	Kembla Copper Type B
Cold Water Rough-In (in wall)	Rehau Rautitan Platinum

Note: All pipe sizing has been based on copper. If Rehau or Cromford is to be used, then a sizing adjustment must be made in accordance with the equivalent pipe size table below.

Copper	15	20	25	32
Rehau	20	25	32	40

Copper	32	40	50	65	80	100	125	150	200
Cromford PN16	40	50	63	90	110	125	160	200	250

Note: Check the water Main Maximum pressure – The Safe Working pressure of Type B Copper is:

50Ø - 1,810kPa, 65Ø - 1,440kPa, 80Ø - 1,620kPa,

100Ø - 1,210kPa, 150Ø - 1,000kPa

All pipes in masonry walls or within stud walls shall be prelagged copper tube.

Negotiate with Authority for the supply of all meters, take delivery from Authorities store and install on site as indicated on the drawings, complete with all valves and fittings required.

10.2 Water main connection

Make an application to the Water Authority, excavate it, and connect it to the Authority's 300mm diameter water main for the following services.

Service	Size
Cold Water	100mm
Fire Hydrant Service	100mm

and extend to the reticulation as indicated on the drawings.

The Water Authority should be contacted, and an application made to arrange for a quote for the connection to the water main. Allow to pay all fees for this work.

10.3 Valves - (in-line)

To pipelines supplying cold water to each unit supply and install a control valve.

To pipelines supplying cold water to each group of fixtures, supply and install valves for shutting down the system for isolation and maintenance purposes.

Locate valves behind access panels, within accessible ducts or as indicated on the drawings.

All valves shall be tagged, and the areas they serve must be clearly indicated. Valve location and function shall be referenced in the Operating Manuals.

Valves to be as specified in 'Materials'

10.4 Connection to fixtures

Provide unions at wall or floor surfaces and at fixtures and appliances to allow removal and replacement without the need to adjust connections.

The exposed basin and cistern connections are to be chrome-plated copper.

10.5 Flexible connections

Solid fixed pipework must be used in all circumstances where hot and cold pipework is not accessible for maintenance.

10.6 Testing

Provide a water pressure test of 1500kPa for a period of two hours.

Disconnect any equipment connected to the service not rated to the test pressure, before testing commences.

Note: Check the maximum test pressure of the pipe material prior to any pressure testing.

10.7 Taps

Unless indicated otherwise, taps shall be Australian-made and constructed of brass.

All taps, connections and cover plates shall be bright chromium-plated finish.

Taps shall be fitted with anti-splash nozzles, except for hose taps and/or where otherwise specified.

Hose Taps shall incorporate: loose Jumper Valve incorporating a neoprene washer

All other taps shall incorporate: Ceramic Disk Cartridges.

10.8 Mini stop valves

At each wall connection for both hot and cold water to all basin and sink mixers, supply and install a chrome-plated 15 mm right-angled, chrome-plated mini stop valves.

10.9 Schedule

Refer to the schedule at the rear of the specification.

10.10 Flushing & cleaning of water supply services

All as previously specified under "Flushing - Pressure Systems".

10.11 Sizing

The Plumbing Contractor shall size the internal cold and Hot water system in accordance with AS/NZS 3500.

10.12 External hose taps – vandal-proof

All external hose tap assemblies shall be vandal-proof with removable keys.

Before practical completion, provide 6 (six) operating keys to the Superintendent.

Provide an Isolation Valve prior to all Hose Taps.

10.13 Hose taps - external

All external hose taps within the garden and turfed areas shall be either screwed to the wall with three brass screws or supported on a hardwood stake. The hardwood stake is to be at least 50mm x 50mm in width and driven into the ground to a minimum depth of 300mm. The hose tap shall be secured to the stake with three brass screws. Unless otherwise stated in the drawings and details.

Provide an Isolation Valve prior to all Hose Taps.

10.14 Tap in toilets areas

Tap to be of brass construction and bright chrome plated. Allow to install a cold water hose tap in the toilet areas. Final location to be determined by the Architect.

10.15 Tap on verandas/balconies

Allow to install a Cold Water hose tap on all verandas/balconies.

Final location to be determined by the architect.

Note: Taps shall not be located within 1 metre of railings or edge.

10.16 Pipework identification

All non-potable pipework, fittings and hose taps shall be coloured copper tube. (Other than the rainwater re-use system)

11. Hot water

11.1 Generally

Supply and install the hot water service from the hot water heater to the fixtures and appliances. Include for all piping, fittings, supports, insulation, hot water heaters, valves and other sundry items of equipment required for the installation.

Pipe materials shall be as follows:

Location	Material
Hot Water Service	Kembla - Copper Type B
Hot Water Rough-In (in wall)	Kembla - Copper Type B
Hot Water Rough-In (in wall)	Rehau Rautitan Red

Note: All pipe sizing has been based on copper. If Rehau is to be used, then a sizing adjustment must be made in accordance with the equivalent pipe size table below.

Copper	15	20
Rehau	16	25

Where a hot water heater is installed the first 1m of hot water pipework from the heater must be installed in copper.

11.2 Testing

Provide a water pressure test of 1500kPa for a period of two hours.

Disconnect any equipment connected to the service not rated to the test pressure, before testing commences.

Note: Check the maximum test pressure of the pipe material prior to any pressure testing.

11.3 Connection to fixtures

Provide unions at wall or floor surface and at appliances to allow removal and replacement without the need to adjust connections.

Exposed basin connections to be chrome plated copper.

11.4 Flexible connections

Solid fixed pipework must be used in all circumstances where hot and cold pipework is not accessible for maintenance.

11.5 Control valves

Supply and install control valves to each group of fixtures. Locate behind access panels or within accessible ducts.

11.6 Expansion & contraction

Make adequate provision for expansion and contraction. Pipes located in walls and floors shall be provided with sufficient insulation so that expansion and contraction does not impose a strain on the pipework or finished surfaces.

Hot water supply pipes shall be installed with appropriate allowance for expansion and contraction as per AS/NZS 3500.4.

11.7 Cold water expansion valve

Supply and install on all inlets to the hot water heaters a cold water expansion valve, the valve shall be in accordance with the heater manufacturer's instructions. Extend 20mm copper drain to adjacent tundish.

11.8 Copper safe tray - hot water heater

Supply and install a copper safe tray 1.0mm thick for the hot water heater.

The safe tray shall be the same size as the heater cupboard minus 10mm all around for clearance. Grade tray to its outlet. Installation shall be in accordance with authorities' requirements. All exposed metal edges of safe tray to be folded minimum 10mm.

Tray shall not receive discharge from the TPR Valve in accordance with AS3500.4. The first 2m of drainage shall be in HDPE or Copper.

Provide approved PE water heater support between tray and water heater

11.9 Signage

In accordance with The Plumbing Code of Australia and AS/NZS 3500.4 Heated Water where hot water is being delivered to an existing outlet used for personal hygiene and the water exceeds 50°C a sign shall be installed.

11.10 Peats plugs

At the flow and return lines on the hot water system allow to supply peats plugs for temperature testing. The peats plugs should be installed at the outlet of the Heaters and the return line just before the hot water circulator.

11.11 Thermal insulation

Provide thermal insulation of all hot water pipework in accordance with the schedule below;

Material	Location	Thickness
Copper	Deadlegs	13mm closed cell
Copper	Circulated	25mm closed cell
PE – X	Deadlegs	13mm closed cell
PE – X	Circulated	25mm closed cell

11.12 Heat tracing cable

The following heater cables are approved for use on this project. The contractor shall be responsible for selecting the type of heating cable to be used for a given application and be equal to Thermon VSX-HT.

11.12.1 Low Temperature - Self-Regulating Heating Cable

- Heating cables shall be self-regulating, capable of maintaining process temperatures up to 66°C and a continuous exposure to pipeline temperature of 85°C while de-energized.
- Cable must be of parallel construction so that it can be cut to length without changing its power output per unit length.
- The heater cable assembly shall have a monolithic heating core construction consisting of two parallel 16 AWG nickel-plated copper bus conductors with a semiconductive PTC polymer extruded over and between these parallel conductors. A polyethylene dielectric insulating jacket is extruded over the heating element core.
- The semiconductive heating matrix and primary insulating jacket shall be cross-linked by irradiation.

- e. The basic cable will be covered by means of a metallic braid of tinned copper. The braid will provide a nominal coverage of seventy per cent (70%) and will exhibit a resistance not exceeding 0.0.0495 ohm/m.
- f. The cable shall be covered with a corrosion-resistant overjacket of thermoplastic elastomer (for possible exposure to aqueous solutions, mild acids or bases) or fluoropolymer (for possible exposure to organic chemicals or corrosives).
- g. For longer circuit lengths and higher heat loss requirements greater than 33 W/mt @ 10°C, the heating cable shall have 14 AWG nickel-plated copper bus conductors
- h. Long-term stability shall be established by the thermal performance benchmark test per IEEE 515 Std or IEC/IEEE 60079-30-1:2015 or CSA C22.2 No 130-16

12. Fire hydrant service

12.1 Generally

Supply and install a Fire Hydrant service to the locations shown on the drawings.

Include for all piping, fittings, valves, and hydrant valves to the locations shown on the drawings and other sundry items of equipment as required for the installation.

The Hydrant system shall be installed in accordance with AS 2419.1:2021, NCC 2022 E1D2, and the Building Code of Australia.

Pipe materials shall be as follows:

Location	Material
In-ground from the Authority Water Main to the Double Check Valve.	Kembla - Copper Type B Note: When Installed in ground copper pipework to be sleeved. Refer to Specification Section: Cold & Hot Water Fixtures & Fittings
In-ground from the Double Check Valve/Booster Assembly	Kembla - Copper Type B Note: When Installed in ground copper pipework to be sleeved. Refer to Specification Section: Cold & Hot Water Fixtures & Fittings
In-ground from the Double Check Valve/Booster Assembly	Cromford Red PE GRADE: PN20 With tracing tape over the top reading the words "Fire Hydrant" printed continuously down the pipe.
In-ground from the Double Check Valve/Booster Assembly	Blue Brute – GRADE:PN20 With tracing tape over the top reading the words "Fire Hydrant" running continuously in the trench. Note: uPVC shall not to be installed under Buildings
Above ground	Victaulic GMS with Victaulic fittings Note: In ground Risers to Hydrants - GMS shall be kept to an absolute minimum. The GMS shall be wrapped with Denso petroleum Tape

Note: All in-ground pipe sizing has been based on copper. If BlueBrute or Cromford is to be used, then a sizing adjustment must be made in accordance with the equivalent pipe size table below.

Copper	100	125	150	200
Cromford PN20	125	160	200	250

Copper	100	125	150	200
Blue Brute	125	125	150	200

Note: Check the water Main Maximum pressure – The Safe Working pressure of Type B Copper is: 80& - 1,620kPa, 100& - 1,210kPa, 150& - 1,000kPa

12.2 Hydrants

Supply and install Landing Valve hydrants in approved locations with the centre of the valve minimum of 750mm – a maximum of 1200mm above the floor.

All external Hydraulics shall be double-handed pillars and supported at their base with a concrete base of 450 x 450 x 450.

The valves shall be installed “at an angle of not more than 35° below the horizontal “as per AS 2419.1

Note: External hydrants shall be 10 metres from the building where the building is not sprinkler protected.

12.3 Landing valve covers

Supply and install covers for all landing valves.

12.4 Storz fittings

Supply and install Storz fittings and caps to all Landing Valve hydrants and the booster valve inlet & outlet.

The Storz fittings and caps shall be as per QFD requirements, which are noted below.

Cast Aluminium Storz connections are not permitted under AS 2419.2

12.5 Hose connections compatible with QFD

Hose connections on fire brigade booster inlets (AS 2419.3) are to be permanently fitted with a Storz 65mm permanent coupling complying with NEN 3374:1971 fitting H coupling having 2½ inch external gas thread i.e. British standard pipe (BSP) thread.

Hose connections on fire hydrant valves (AS 2419.2) shall be male Fire Brigade Thread semi-permanently fitted with a female Fire Brigade Thread Storz adapter. The Fire Brigade Thread Storz adapter shall have:

- a. A Storz 65mm connection conforming to NEN 3374:1971, Section 3, Dimensions and Coding, Parts, Part No. 5.
- b. An Fire Brigade Thread connection with female 64mm × 4.88mm pitch (21 1/2 x 51 1/5 TPI) to match the opposing thread on the fixed system.
- c. A rubber gasket (packing ring) between the male and female connection that conforms to NEN 3374:1971, Section 3, Dimensions and Coding, Parts, Part No. 49.

The Fire Brigade Thread Storz adapter should be manufactured as a single piece with sealing rings, using materials and processes that conform to NEN 3374:1971, Section 4, Material.

The Fire Brigade Thread Storz adapter must:

- a. Be forged aluminium alloy according to DIN 1725 Sheet 1 or later (AlMgSi with copper content not exceeding 0.05% by mass), unless being used in brackish environments
Note: Copper alloy or stainless steel should be used in brackish environments.
- b. Be fitted with a delivery sealing ring suitable for pressure operation made from nitrile rubber with a hardness of (75 ±5) Shore A according to NEN 5601:1977.
- c. Have a nominal waterway diameter of 58mm (refer to Figure 2).
- d. Have a lug spacing measured from lug to lug of 81mm (refer to Figure 2).
- e. Have a Storz connection/disconnection torque range of 3Nm to 8Nm.
- f. Not leak or permanently change shape when pressurised with water at 1600kPa.

The Fire Brigade Thread Storz adapter must be permanently marked in raised lettering with:

- The manufacturer's name, trademark or logo
- The manufacturer's forging mark.

The manufacturer should provide certification that the adapter has been type tested to a working pressure of 1600kPa from a body accredited by a member of the International Accreditation Forum, or test results to the same effect from a facility accredited by a member of the International Laboratory Accreditation Cooperation.

12.6 Testing of pipework

Test all pipework 1700kPa for a period of two hours in accordance with AS2419.1:2021

On completion, test the installation under full supply conditions all to the satisfaction of the Fire Brigade and the Superintendent. Record pressure and flow results and advise in writing to the superintendent.

Note: Check the maximum test pressure of the pipe material prior to any pressure testing.

12.7 Painting / identification

All Pipework shall be labelled in accordance with the Identification section of this specification.

All exposed pipework located within the pump room to be painted red in accordance with AS1345

12.8 Booster valve and double check valve with metered by-pass

Refer to the rear of the specification for these items.

12.9 Flow and pressure test

Flow and Pressure Test shall be undertaken in the presence of the superintendent.

Flow tests shall be with two most hydraulically disadvantaged hydrants operating.

Tests shall be in accordance with AS2419.1:2021 Appendix T

On completion of the flow test, supply the Superintendent with a Flow and Pressure Test results and a certificate with the following items:

Flow And Pressure Test For Hydrant Service At _____					
Flow rate	Pressure	Date	Time	Test Location and RL	Witness name & signature
Static	kPa				
5 l/s 1 hydrant flowing	kPa				
10 l/s 1 hydrant flowing	kPa				
20 l/s 2 hydrants flowing	kPa				
30 l/s 3 hydrants flowing etc	kPa				
Max flow	kPa				

Test undertaken by: of

Signed: Date:

12.10 Fire rating of pipework supports

External Pipework:

External pipework supported by the external wall or roof of building to be installed in accordance with AS2419.1:2021 Clause 8.4. The building shall be protected by an automatic sprinkler system in accordance with AS21181.1, AS2118.4, AS2118.6, FPAA101D or 25015 - Hydraulic Services Specification-REV_T1

FPAA101H or the walls, floors, beams, or roof of the building shall have an FRL of not less than 60/-/-

Internal Pipework:

Where internal above-ground pipework is installed, it shall be protected from damage by fire by one or more of the following

- a. An automatic fire sprinkler system that conforms to AS2118.1, AS2118.4, AS2118.6, FPAA101D or FPAA101H
- b. Support by walls, columns, floors, beams, or roofs having an FRL not less than 60/-/- and pipe supports and hangers having an FRL not less than 60/-/-
- c. Installation in a fire-isolated stair of fire-resisting shaft
- d. Protection by construction capable of resisting the effects of fire for not less than 60 min.

In a non-sprinkler protected building the pipe supports shall have a FRL not less than 60/-/-, while maintaining a pipe-support temperature of not less than 500°C when tested in accordance with AS1530.4, or other measures shall be taken to prevent its early collapse when exposed to fire. Reference AS2419.1:2021 Clause 10.4.1

12.11 Isolating valves

The isolating valves on the hydrant system shall be located in accordance with AS2419.1:2021 clause 8.7 AS2419.1:2021 clause 8.9

Note:

The Fire pump standard AS 2941 requires inlet valves to pumps to be OS&Y gate valves.

Butterfly valves etc. not allowed as turbulence could affect pump performance.

All other valves can be ball, butterfly or gate.

12.12 Block plan

A block plan shall be provided In accordance with AS 2419.1:2021 Clause 11.5 and in the following locations:

1. At the hydrant booster location
2. In the Hydrant pump room
3. Adjacent the FIP
4. As referenced in the FER

It shall be as per the following:

Minimum A2 in size

1. Engraved onto Aluminium
2. Block plan

12.13 Tagging

All Fire Hydrants shall be tagged in accordance with AS 1851.4.

The Plumbing Contractor shall maintain the Fire Hydrants for 12 months after practical completion and shall notify the superintendent that all Fire Hydrants have to be inspected and tagged in accordance with AS 1851.4

An installation log shall be kept on-site for the Fire Hydrants.

12.14 Temporary fire protection during construction

In accordance with the Building Code of Australia:

NCC 2022 Clause E1D15

After the building has reached an effective height of 12-metres

- i. the required fire Hydrants and fire hose reels must be operational in at least every storey that is covered by the roof or the floor structure above, except the 2 uppermost storeys: and
- ii. any required booster connections must be installed

Construct temporary fire protection during the construction process. As the building progresses, provide a 100 mm-diameter fire main, landing valves and hose reels up to within two levels of the highest poured concrete slab. Provide system fully charged with water during the construction period.

Supply and install all valves, pipework, hose reels, booster connections, and, where required, pumps to pressurise the system. Include one-way flow bypass around booster pumps. The highest landing valve in the temporary service is to be capable of maintaining a pressure of 250kPa during a 10-litres/second-flow rate through the valve.

The booster valve shall be located in a readily accessible location, identifiable from the street and shall have a list of emergency telephone numbers displayed adjacent to the valve. The list of telephone numbers is to include key site personnel and will be compiled in consultation with the Fire and Rescue QLD. The list is to be prepared on durable material, permanently fixed to the location.

12.15 Pipework identification – connection to a non-potable system

The hydrant pipework and fittings shall be coloured as the service is connected to a Non-Potable system.

Colour should be Lilac -P-23.

12.16 Hydrant truck hard stand

AS2419.1:2021 clause 3.3

The building contractor is to provide a hardstand for fire truck access. It shall be an identifiable and clearly marked trafficable all-weather pavement providing access and capable of supporting a fire brigade pumping appliance during firefighting operations.

NOTE: Examples of pavements are bituminous concrete, aggregate, concrete or similar surfaces.

The Hard Stand shall be a minimum of 14m x 4m and capable of supporting loads of 850kPa. The weight of a standard truck is approximately 15 tonnes and an aerial appliance is 28 tonnes.

Refer to Fire Brigade pumping appliance specifications and requirement

It shall be located:

1. Within 20m of a street fed fire hydrant booster
2. Within 4.5m of a tank fed fire hydrant booster
3. Within 50m of an attack fire hydrant
4. Within 20m of a feed fire hydrant
5. Within 20m of the access door to a hydrant Pump room
6. Within 20m of Hydrant Tanks

12.17 Fire brigade – fire and rescue QLD

Fire and Rescue QLD shall be notified at the completion of the installation to carry out a final inspection. Notification shall be provided 2 weeks before the project completion to allow for sufficient time for the Brigade to attend.

12.18 Fire control centre (for buildings over 50m effective height)

If the Building is over 50m effective height NCC 2022 clause S19C1 requires that the Fire Control Centre has an FIP, control switches and visual status indication for all required fire pumps.

Allow to supply control switches and visual status indication for all required fire pumps in the Fire Control Centre.

13. Hose reel service

13.1 Generally

The Hose Reel system shall be installed in accordance with NCC 2022 E1D2, AS 2441:2005 Amendment 1 and the Building Code of Australia.

Supply and install a Hose Reel service to the locations shown on the drawings.

Include for all piping, fittings, valves, hose reel, to the locations shown on the drawings and other sundry items of equipment as required for the installation.

Pipework shall be constructed of Copper Tube - Type B to AS 1432

Joints shall be silver brazed as per materials section.

Pipe materials shall be as follows:

Location	Material
In-ground	Kembla - Copper Type B Note: When Installed in ground copper pipework to be sleeved. Refer to Specification Section: Cold & Hot Water Fixtures & Fittings
Above-ground	Kembla - Copper Type B

Note: Fire hose reel service to be connected to a metered cold water service.

13.2 Hose reels

Supply and Install Hose reels of the type and pattern specified previously under "Materials". Locate centre of hub at height of 1500 mm above floor level and ensure that the installation upon completion permits the hose to run smoothly off the reel when pulled in any direction.

Separate operating instructions shall be permanently secured in a conspicuous upright location.

In accordance with AS 2441 where the hose guide of the arm cannot protrude beyond the recess an extended arm shall be installed.

Where a hose reel is installed on the potable water system a minimum of a single check valve (testable) must be installed as per AS3500.1:2021, Section 4, Cross connection and backflow prevention.

13.3 Testing of pipework

Test all pipework 1500kPa for a period of two hours. On completion, test the installation under full supply conditions all to the satisfaction of the Fire Brigade and the Superintendent. Record pressure and flow results and advise in writing to the superintendent.

Note: Check the maximum test pressure of the pipe material prior to any pressure testing.

13.4 Test point

Provide a 25mm tee and union at the most disadvantaged hose reel for testing. Cap off with brass plug.

13.5 Painting / identification

All Pipework shall be labelled in accordance with the Identification section of this specification.

13.6 Flow and pressure test for fire hose reels

A Flow and Pressure Test shall be undertaken in the presence of the superintendent.

Flow tests shall be with two most hydraulically disadvantaged hose reels operating.

On completion of the flow test, supply the Superintendent with Flow and Pressure Test results and a certificate with the following items:

Flow and pressure test for fire hose reel service at _____					
Flow rate	Pressure	Date	Time	Test Location and RL	Witness name & signature
Static	kPa				
.33 l/s 1 hose reel flowing	kPa				
.66 l/s 2 hose reels flowing	kPa				
.....l/s Max flow	kPa				

13.7 Tagging

All Hose Reels shall be tagged in accordance with AS1851.2

The Plumbing Contractor shall maintain the hose reels for 12 months after practical completion and shall notify the superintendent that all hose reels have to be inspected and tagged in accordance with AS1851.2

An installation log shall be kept on-site for the hose reels.

13.8 Valves

All system valves including the water meter, which can isolate flow in the hose reel water supply main, shall be secured in the open position by a padlocked metal strap and have attached an engraved non-ferrous metal tag with 8mm upper case wording:

FIRE SERVICE VALVE CLOSE ONLY TO SERVICE FIRE HOSE REELS.
--

The padlocks shall be a 003-type.

13.9 Temporary fire protection

Refer to Hydrant section of this specification and NCC 2022 Clause E1D16 when the new service is installed and commissioned.

13.10 Existing hose reels

Existing hose reels shall not be re-used.

13.11 Pipework identification – connection to a non-potable system

The hose reel pipework and fittings shall be coloured copper tube as the service is connected to a Non-Potable system.

Colour should be Lilac -P-23.

13.12 Signage other than doors

Where hose reels are not installed in a cupboard a sign shall be installed in accordance with AS 2441.

14. Extinguishers & fire blankets

14.1 General

Extinguishers and Fire Blankets shall be provided in accordance with AS 2444, NCC 2022 Clause E1D14

Supply and install all extinguishers including mounting brackets, location and identification signs in accordance with the requirements of:

Australian/New Zealand Standard 1841.2 - Portable fire extinguishers

Specific requirements for water type extinguishers, Australian/New Zealand Standard

1841.6 - Portable fire extinguishers - Specific requirements for carbon dioxide type extinguishers, Australian/New Zealand Standard

1841.5 - Portable fire extinguishers - Specific requirements for powder type extinguishers, Australian Standard 1850 - Portable Fire Extinguishers - Classification, rating and performance testing

Australian Standard 2444 - Portable Fire Extinguishers and Fire Blankets - Selection and location.

Final location of all portable extinguishers is to be determined by site inspection and approval from the QLD Fire Services prior to installation.

14.2 Signage

Extinguishers shall have their locations clearly indicated by placement of location signs, as shown in the diagram below.



Note: Also mentioned in works by others, confirm with the client who is providing the Extinguishers and Fire Blankets. Generally installed by the Builder or Sprinkler contractor. Due to the fact that they are the last item installed, when the plumber installs them, they are generally stolen by other trades etc.

15. Extinguishers & fire blankets

15.1 Generally

Supply and install all reverse osmosis water pipes in George Fischer Instaflex Pipework.

No metallic pipework or fittings are to come in contact with reverse osmosis water.

Pipe materials shall be as follows:

Location	Material
Reverse Osmosis Service	George Fischer PPH Piping System. Size to be 32mm for circulating loop and 15mm to taps.

George Fischer PPH piping systems manufactured using the beta grade raw material pipework. Where PPH is used, infrared welding is recommended.

15.2 Valves - (in-line)

To pipelines supplying water to each floor level provide a three-way valve on flow and return runs.

All valves shall be tagged and clearly indicate what areas they serve. Valve location and function shall be referenced in the Operating Manuals. 3-way valves with integrated stock position to prevent backflow into the membranes. Valves stem to have a break point and buttress threads to prevent loosening on vibration.

15.3 Connection to fixtures

Provide two three-way valves under each fixture to allow removal and replacement without the need to shut down the service.

15.4 Testing

The internal pressure test for water is performed at 1.3 times PN rating for the piping system for duration of 3 hours at 20°C.

15.5 Schedule

Refer to schedule at rear of specification.

15.6 Flushing & cleaning of water supply services

All as previously specified under "Flushing - Pressure Systems".

15.7 Sizing

The Reverse Osmosis pipework should be continuously installed in 32mm pipework.

15.8 Pipework identification

All pipework shall be labelled "Reverse Osmosis". All pipework shall be marked to identify the size, PN rating, material and other information to allow traceability to manufacture in accordance with the manufacturing standard specification.

15.9 System type

Fittings	All fittings shall be equal to George Fischer PPH Piping System or 316 Stainless Steel
Valves	3-Way Valves are to be used to prevent the build-up of bacteria. 3-way valves with integrated stock position to prevent backflow into the

	membranes. Valves stem to have a break point and buttress threads to prevent loosening on vibration.
All pipework Must conform with the following:	a) Be approved by Local Authority. b) Where PPH is used, infrared IR welding is recommended. c) Incorporate fittings of the same manufacture as the pipe used. d) Have Australian Standard's approval. e) Have Australian AS / NZS 4020 approval.
Installation	Systems of mixed manufacturing origin will NOT be accepted. All tradespersons assigned to the project are to be adequately trained by the manufacturer of the system proposed, prior to commencement of work on site. All pipework to be installed in accordance with the installation catalogue, to be provided by the manufacturer, and any other written or verbal instructions provided. All pipework to be tested in accordance with the installation catalogue, to be provided by the manufacturer, prior to concealment.
Bracketing	Only a clipping system approved by the manufacturer shall be used for the pipe system. Spacing to be in accordance with the installation catalogue, to be provided by the manufacturer, and any other written or verbal instructions provided. Bracketing – where metallic brackets are used with rubber support consideration needs to be given to the rubber formulation to ensure that it does not contain plasticers that will migrate into supporting pipework.
Expansion	Allowance for expansion in pipework materials shall be in accordance with the installation catalogue, to be provided by the manufacturer, and any other written or verbal instructions provided.

16. Tender break up

Plumbing Contractor:

Address:

.....
.....

This Tender is based on:

1. The Hydraulic Drawings H0..... to HIssue
2. The Hydraulic Specification Issue

ITEM	COST
Sewer Drainage	\$ -----
Grease Drainage in HDPE	\$ -----
Stormwater Drainage	\$ -----
Downpipes	\$ -----
On Site Detention Tank or Basin	\$ -----
Stormwater Detention Pit with items as specified	\$ -----
Council Stormwater Pit (????) and the Kerb inlet pit in street	\$ -----
Cold Water Booster Pump & Filtration Unit	\$ -----
Cold Water Service	\$ -----
Reverse Osmosis Service	\$ -----
Rainwater reuse Service	\$ -----
Hot Water Plant and Pumps	\$ -----
Hot Water - Flow and Return	\$ -----
Gas Service	\$ -----
Fire Hydrant Service & Pumpset	\$ -----
Hose Reel Service	\$ -----
Acoustic Lagging	\$ -----
Extinguishers	\$ -----
Shop Drawings	\$ -----
As Built Drawings, Operation and Maintenance Manuals	\$ -----
Cap off existing services to site	\$ -----
Sanitary Fixtures	\$ -----
Connection Charges from Authorities	\$ -----
Other items/Sundries/Contingency	\$ -----

Local authority Sewer Main - Concrete encasing (By an accredited minor works contractor) \$ -----

Tender price \$ -----

Including GST \$ -----

LABOUR RATES

PERSONNEL	COST (HRLY RATE)
Supervisor	\$
Foreman	\$
Tradesman	\$
Apprentice	\$
Labourer	\$

17. Sanitary fixtures & appliances.

17.1 Generally

Supply and install all sanitary fixtures and appliances connected to the hydraulic services unless specified as "connected only".

Where specified "supply and install" allow to take delivery, store as necessary and install. Provide all fixings, necessary cutting, and securing brackets to walls, levelling and connection to various services required for satisfactory operation.

Where specified "connect only" allow to connect to the various services fixtures and appliances installed by others.

All exposed brackets shall be white enamelled.

All exposed connections shall be chrome plated unless specified in white PVC-U.

Fixtures, appliances and brackets shall be first quality.

Allow for caulking joints around sanitary fixtures, pipes, brackets, bolts and nibs.

Obtain a written guarantee from the manufacturer for a period of two years. The guarantee shall cover all costs associated with replacement of a faulty article within the two-year period.

Note: All basin and sink traps are to be P Trapped

17.2 Tapware colour coding

All tap buttons or handles shall be colour coded as follows:

All tap outlet fixtures (including baths, showers and basins) shall be clearly identified for hot and cold water in accordance with the Plumbing Code of Australia (NCC Volume 3) and relevant Australian Standards.

Identification shall be by colour coding or markings, such as:

- Red or "H" for hot water; and
- Blue or "C" for cold water.

Tapware shall also comply with AS/NZS 3718 and AS/NZS 3662, including clear indication of direction of operation.

Tap	Water Temperature	Tap Indicator Lettering	Tap Indicator Colour	Tap Handle Colour
Hot	> 43.5°C	H	Red	Red
Warm	38.0°C to 43.5°C	W	Yellow	Yellow
Cold	----	C	Blue	Blue

17.3 Safety fixtures and fittings

- Safety Showers require a minimum of 75.7L(1.3L's)/min@210kPa
- Safety Eye/Face Washes require a minimum of 26L/min(0.5L's) @210kPa
- Safety Eye Washes require a minimum of 17L/min(0.3L's) @210kPa

17.4 Hose taps

17.4.1 Hose Taps – Amenities Bib Taps

Hose Taps – Amenities Bib Taps
Fixture

Hose Taps – Amenities Bib Taps	
Manufacturer	Enware
Contact Details	Ph: (02) 8536 4000
Model	Product Group: VP301 AntiVandal 15mm Hose Tap Chromed Brass with jumper valve and vacuum breaker. Sits a total of 100mm from face of wall.

17.4.2 Hose Taps – Garden And Carpark Areas

Hose Taps – Garden And Carpark Areas	
Fixture	
Manufacturer	Enware
Contact Details	Ph: (02) 8536 4000
Model	Product Group: VPK351 20mm Hose Tap & Removable Key Handle Brass with jumper valve and vacuum breaker. Sits a total of 80mm from face of wall.

17.4.3 Hose Taps – Rainwater Garden And Carpark Areas

Hose Taps – Rainwater Garden And Carpark Areas	
Fixture	
Manufacturer	Enware
Contact Details	Ph: (02) 8536 4000
Model	Product Group: VPK351 20mm Hose Tap & Removable Key Handle Brass with jumper valve and vacuum breaker. Sits a total of 80mm from face of wall.

17.4.4 Hose Taps – Stormwater/Recycled Water Re-Use Taps

Hose Taps – Stormwater/Recycled Water Re-Use Taps	
Fixture	
Manufacturer	L. W. Gemmell Pty Ltd
Contact Details	Ph: (02) 9647 2770
Model	20mm Hose Tap (LILAC) & Removable Key Handle. Brass with jumper valve.

17.4.5 Mini Taps – Wc's And Basins

Mini Taps – Wc's And Basins	
Manufacturer	Enware
Contact Details	Ph: (02) 8536 4000
Model	VP303, VP Cistern Tap R/A M&F 15mm
Location	Toilet cisterns and vanities

17.4.6 Cistern Cock – Recycled Water

Cistern Cock – Recycled Water	
Fixture	
Manufacturer	L. W. Gemmell Pty Ltd
Contact Details	Ph: (02) 9647 2770
Model	LWG-RWWC - Lilac colour for easy identification - Comes with C/P backing flange

17.4.7 Wash Down Hose

Wash Down Hose	
Manufacturer	Enware
Contact Details	Ph: (02) 8536 4000
Model	Enware retractable hose reel, 15m hose with dual action Gunna spray down gun. FS724
Location	Wash down

18. Acoustic

18.1 Acoustic pipe supports

Refer to bracketing section.

18.2 Acoustic lagging of pipe & fittings

Acoustic Lagging of pipes and Fittings	Location	All pipework in ducts, ceiling spaces and exposed areas (other than carpark). or All waste, stormwater & water pipework in ceiling spaces, and in ground floor shops and offices
	Manufacturer	Thermotec Australia Pty Ltd.
	Insulation	NuWrap 5
	Note	Refer to the Acoustic report for this project for all requirements In QLD the STC ratings are different.

Acoustic Insulation Requirements As Per The Building Code Of Australia (BCA)

BCA Requirement of $R_w + C_{tr}$ (airborne) of not less than:

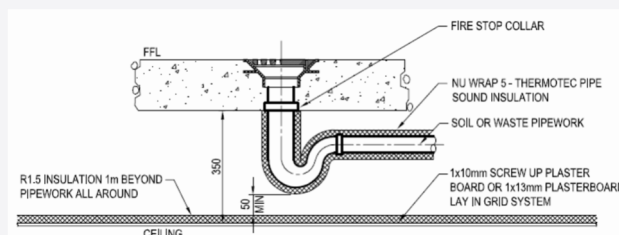
Achieved by:

“STC 40 if the adjacent room is a *habitable room* (other than a kitchen) “ NCC 2022 Clause F7D7 1(a)

Installing NuWrap 5 and when used in a system incorporating:

- 10mm plasterboard
- with ceiling insulation (R1.5 insulation blanket) 500mm either side of the pipe (installed by the builder in accordance with the manufactures specification)
- The gap between the Acoustic lagging and the ceiling insulation shall be 50mm.
- This will actually achieve a RW of 45 according to the Thermotec Brochure

Detail of STC 40



BCA Requirement of $R_w + C_{tr}$ (airborne) of not less than:

Achieved by:

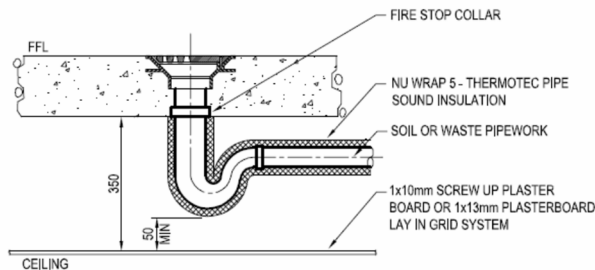
Acoustic Insulation Requirements As Per The Building Code Of Australia (BCA)

“STC 25 if the adjacent room is a kitchen or non-habitable room“

NCC 2022 Clause F7D7 1(b)

■ Installing NuWrap 5

Detail of STC 25



Installation

The material will be 5kg/m² loaded polymer barrier over 25mm nominal thickness convoluted polyurethane foam. The outer skin of the loaded polymer will be faced with a heavy duty reinforced aluminium foil, tested to achieve a fire performance of “four zeros” to AS1530.part 3.

When tested in a NATA (National Association of Testing Authorities) registered laboratory, the insertion loss of the material shall be as per the table above.

Material will be cut to size to neatly cover all pipe work. Straight lengths of pipe will incorporate a 50mm nom. overlap along the length. Fittings will be covered with NuWrap 5 carefully cut to ensure complete coverage. Use of factory precision cut NuWrap 5 (Flat Packs) for PVC-U fittings.

All insulation shall be installed around the pipework surface as tightly as possible without gaps, with the edges and ends tightly butted together.

All overlaps and joins will be sealed over using high tack reinforced aluminium foil self-adhesive tape. Tape will be pressurised correctly to ensure maximum adhesion at installation time. Tape will be tested to AS1530.3 to achieve acceptable fire performance to the Building Code of Australia.

Where necessary, the preformed sections shall be cut (using a sharp knife or scissors) and mitred, to ensure a tight fit around bends and at tees.

A professional lagger shall be employed to fit the insulation.

19. Maintenance access hatches and panels

19.1 Access Panels For Ducts

Access Panels For Ducts	
Manufacturer	Trafalgar
Contact Details	Ph: 1800 888 714
Model	AP-M-FE-300 – Feathered edge AP-M-WW-300 – Wet Wall 300mm x 300mm Opening/310mm x 310mm Cut Out
Note	Allow to Co-ordinate final location with builder on site <i>Trafalgar Rw34 Acoustic Access Panels have been designed for use where there is a noise transmission issue. The panels are constructed using a metal clad acoustic door and steel frame and feature a square key budget lock as standard. Trafalgar Rw34 panels have been independently tested and achieve a superior rating of 34 STC in suitable openings.</i>
Location	In all Hydraulic Ducts

19.2 Access Panels For Acoustic Ducts

Access Panels For Acoustic Ducts	
Manufacturer	Trafalgar
Contact Details	Ph: 1800 888 714
Model	Rw34 Acoustic Access Panels Rw34M-300-FE – Wet Wall Rw34M-300-WW – Feathered edge 300mm x 300mm Opening/310mm x 310mm Cut Out
Note	Allow to Co-ordinate final location with builder on site Trafalgar Rw34 Acoustic Access Panels have been designed for use where there is a noise transmission issue. The panels are constructed using a metal clad acoustic door and steel frame and feature a square key budget lock as standard. Trafalgar Rw34 panels have been independently tested and achieve a superior rating of 34 STC in suitable openings.
Location	In all Hydraulic Ducts

19.3 Access Panels For Fire Rated Ducts

Access Panels For Fire Rated Ducts	
Manufacturer	Trafalgar
Contact Details	Ph: 1800 888 714
Model	2 Hour Fire Rated Access Panels FR-W-CH-WW-300 – Wet Wall FR-W-CH-FE-300 – Feathered edge 300mm x 300mm Opening/310mm x 310mm Cut Out

Access Panels For Fire Rated Ducts

Note	Allow to Co-ordinate final location with builder on site Trafalgar Rw34 Acoustic Access Panels have been designed for use where there is a noise transmission issue. The panels are constructed using a metal clad acoustic door and steel frame and feature a square key budget lock as standard. Trafalgar Rw34 panels have been independently tested and achieve a superior rating of 34 STC in suitable openings.
Location	In all Hydraulic Ducts

19.4 Access Hatch For Ceilings

Access Hatch For Ceilings

Manufacturer	Trafalgar
Contact Details	Ph: 1800 888 714
Model	AP-M-WW-600 – Wet Wall AP-M-FE-600 – Feathered Edge 600mmx600mm Opening/610mmx610mm Cut-Out
Note	Allow to Co-ordinate final location with builder on site Trafalgar FR-C Fire Rated Ceiling Hatches have been designed for use where fire rated protection is requirement. The hatches feature concealed hinges and provide lockable access to services. Trafalgar FR-C hatches are a compact, lightweight and aesthetically pleasing means of providing access into fire rated ceilings. When installed they sit flush with the ceiling leaving a neat opening around the perimeter. The hatch panel is secured with a locking key.
Location	Where Hydraulic access hatches are required

19.5 Access Hatch For Acoustic Ceilings

Access Hatch For Acoustic Ceilings

Manufacturer	Trafalgar
Contact Details	Ph: 1800 888 714
Model	Rw34M-300-FE – Wet Wall Rw34M-300-WW – Feathered edge 600mmx600mm Opening/610mmx610mm Cut-out
Note	Allow to Co-ordinate final location with builder on site Trafalgar FR-C Fire Rated Ceiling Hatches have been designed for use where fire rated protection is requirement. The hatches feature concealed hinges and provide lockable access to services. Trafalgar FR-C hatches are a compact, lightweight and aesthetically pleasing means of providing access into fire rated ceilings. When installed they sit flush with the ceiling leaving a neat opening around the perimeter. The hatch panel is secured with a locking key.
Location	Where Hydraulic access hatches are required

19.6 Access Hatch For Fire Rated Ceilings

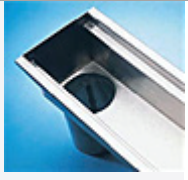
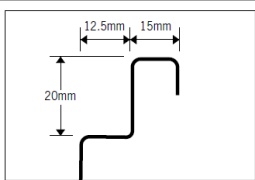
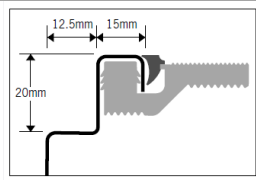
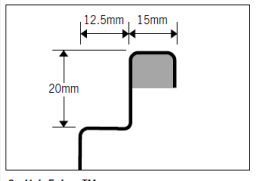
Access Hatch For Fire Rated Ceilings	
Manufacturer	Trafalgar
Contact Details	Ph: 1800 888 714
Model	2 Hour Fire Rated Access Hatch FR-C Fire Rated Ceiling Hatch 600mmx600mm Opening/610mmx610mm Cut-out
Note	Allow to Co-ordinate final location with builder on site Trafalgar FR-C Fire Rated Ceiling Hatches have been designed for use where fire rated protection is requirement. The hatches feature concealed hinges and provide lockable access to services. Trafalgar FR-C hatches are a compact, lightweight and aesthetically pleasing means of providing access into fire rated ceilings. When installed they sit flush with the ceiling leaving a neat opening around the perimeter. The hatch panel is secured with a locking key.
Location	Where Hydraulic access hatches are required

20. Stormwater drainage fixtures & fittings

Allow to supply and install the following items as indicated on the drawings.

20.1 Trench grates

Trench Grate - Balconies

Trench Grate - Balconies	
Manufacturer	ACO Polycrete Pty Ltd
Contact Details	Ph: 1300 765 226
Model	Modular 100
Grate	Class A – AntiSlip Stainless Steel Heal guard 
Channel	  Standard Edge An all purpose trough suitable for tiles, concrete and resin floors in pedestrian and light industrial applications  Vinyl Seal™ Provides superior seal between trough and vinyl sheet flooring applications. Also suitable for some resin floor applications  Solid Edge™ Provides strength in dense traffic areas, and when floor surround is poured, avoids having to fill beneath the edge.
Fall	Neutral Channel
Max Flow Rate	7.6L's

21. Signage

The following signage shall be provided (by the plumber) for the following:

21.1 Potable hot/cold water

Item	Signage Wording	Lettering/Location
All Water Meters	e.g. "WATER METER FOR SHOP 1" "COMMON AREA WATER METER" "WATER METER ROOM"	In white lettering not less than 50mm high, on a contrasting colour. Mounted adjacent to the meters.
All Pumps	e.g. "COLD WATER PUMP ROOM" "COLD WATER PUMP"	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the door to the room
Hot water	e.g. "DANGER: THIS TAP PRODUCES HOT WATER THAT MAY SCALD" <i>In accordance with The Plumbing Code of Australia and AS/NZS 3500.4 Heated Water where hot water is being delivered to an existing outlet used for personal hygiene and the water exceeds 50°C a sign shall be installed.</i>	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the wall behind the relevant fixture.
TMV'S	e.g. "This valve requires commissioning, maintenance and servicing as per AS4032.3 and the manufacturer's specifications every 12 months by a qualified person. Failure to do so may result in a reduction in valve performance that may result in the delivery of high temperature water"	Mounted on the cover of the TMV box or adjacent the valve as per the manufacturer's instructions
Valve tags	e.g. CW1 / HW1	38mm in diameter, 12.7mm high black filled numbers. Made from approx. 1mm brass with a 5mm hole and suitable attached to the Valve
Meter tags	e.g. CWM1 /HWM1	38mm in diameter, 12.7mm high black filled numbers.

Item	Signage Wording	Lettering/Location
		Made from approx. 1mm brass with a 5mm hole and suitable attached to the meter.

21.2 Rainwater re-use

Item	Signage Wording	Lettering/Location
All Pumps	e.g. "RAINWATER PUMP" "RAINWATER PUMP ROOM"	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the wall behind the pump
Rainwater Hose Taps	e.g. "RAINWATER"	Sign to be in accordance with AS1345, provided on an approved strong, durable, stain resistant material. Mounted adjacent to the rainwater hose taps.
Rainwater Pipework	All pipework from a rainwater tank shall be clearly marked at intervals not exceeding 500mm with the word "RAINWATER" in contrasting colour, in accordance with AS 1345.	
Valve tags	e.g. RW1	38mm in diameter, 12.7mm high black filled numbers. Made from approx. 1mm brass with a 5mm hole and suitable attached to the Valve.
Meter tags	e.g. RWM1	38mm in diameter, 12.7mm high black filled numbers. Made from approx. 1mm brass with a 5mm hole and suitable attached to the meter.

21.3 Recycled / stormwater / non drinking water

Item	Signage Wording	Lettering/Location
All Pumps	e.g. "STORMWATER RE-USE PUMP" "STORMWATER PUMP ROOM"	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the wall behind the pump
Recycled/Stormwater Hose Taps	e.g. "DO NOT DRINK – RECYCLED WATER"	Allow to supply and install signage to all outlets. Signs to be Purple / Lilac in accordance with clause 9.5.2 of

Item	Signage Wording	Lettering/Location
		AS/NZS3500.1 and AS1319 provided on an approved strong, durable, stain resistant material. Mounted adjacent to the recycled/stormwater hose taps.
Recycled/Stormwater Pipework	All pipework both above and below ground conveying recycled, reclaimed Non-Drinking water, shall be coloured Purple / Lilac in accordance with AS2700 and be clearly and permanently labelled 'RECYCLED / RECLAIMED – WATER – DO NOT DRINK' All buried pipes shall have an identification tape complying with AS2648.1 and marked with the following: (a) Contrasting purple lettering installed on top of the pipe, running longitudinally, and fastened to the pipe at not more than 3 m intervals. (b) The following statement in accordance with AS 1345: RECYCLED / RECLAIMED—WATER—DO NOT DRINK	

21.4 Non-drinking water – downstream of backflow device

Item	Signage Wording	Lettering/Location
Non-Drinking Water	e.g. “DO NOT DRINK – RECYCLED WATER”	Allow to supply and install signage to all outlets. Signs to be in accordance with clause 9 of AS/NZS3500.1 and AS1319 provided on an approved strong, durable, stain resistant material. Sign to be as follows: Mounted adjacent to non-drinking water outlets.
Non-Drinking Water Pipework	All pipework conveying water downstream of a backflow prevention device, installed for high or medium hazard protection, other than devices used for containment, shall be clearly and permanently labelled 'WARNING NOT FOR DRINKING' as per Clause 4.4.5 and clause 9 of AS/NZS3500.1	

21.5 Stormwater

Item	Signage Wording	Lettering/Location
All Pumps	e.g. “STORMWATER PUMPS”	In white lettering not less than 25mm high, on a contrasting colour.

Item	Signage Wording	Lettering/Location
		Mounted on the door to the control panel.
Stormwater Detention Tank	Shall be marked with the words “THIS IS AN ON SITE DETENTION TANK AND IS SUBJECT TO POSSIBLE FLOODING DURING HEAVY STORMS.”	In white lettering not less than 50mm high, on a contrasting colour.
Stormwater Detention Basin	Shall be marked with the words “THIS IS AN ON SITE DETENTION BASIN AND IS SUBJECT TO POSSIBLE FLOODING DURING HEAVY STORMS.”	In white lettering not less than 50mm high, on a contrasting colour.
Confined Spaces Entry	Shall be marked with the words “DANGER – CONFINED SPACE ENTER BY PERMIT ONLY”	As per relevant Australian standard

21.6 Sewer

Item	Signage Wording	Lettering/Location
All Pumps	e.g. “SEWER PUMPS”	In white lettering not less than 25mm high, on a contrasting colour. Mounted on the door to the control panel.
All Pumps	e.g. “SEWER PUMP ROOM” “SEWER PUMP”	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the door to the room.
Car wash bay	Shall be marked with the words “NO DEGREASING OR MECHANICAL REPAIR WORK IN THE CAR WASHBAY”	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the wall of the car wash.
Confined Spaces Entry	Shall be marked with the words “DANGER – CONFINED SPACE ENTER BY PERMIT ONLY”	As per relevant Australian standard

21.7 Fire hose reels

Item	Signage Wording	Lettering/Location
Hose Reel Cabinets	The door of the cabinet shall have the words “FIRE HOSE REEL”	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the door to the cabinet as per AS2441
Hose Reels in Factories	The wall behind the Hose Reel shall have the words “FIRE HOSE REEL”	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the wall behind the hose reel as per AS2441
All Pumps	e.g. “ FIRE HOSE REEL PUMP” “ FIRE HOSE REEL PUMP ROOM”	In white lettering not less than 50mm high, on a contrasting colour. Mounted on the door to the room
Isolation Valves	e.g. “VALVE nn FIRE SEVICE VALVE, CLOSE ONLY TO SERVICE FIRE HOSE REELS”	In lettering not less than 8 mm high, engraved on a non-ferrous metal tag. Mounted on the water meter and any isolation valves on the fire hose reel service
Valve straps	e.g. As Per Relevant Australian Standard	As Per Relevant Australian Standard Mounted on the water meter and any isolation valves on the fire hose reel service
Valve tags	e.g. FHR1	38mm in diameter, 12.7mm high black filled numbers. Made from approx. 1mm brass with a 5mm hole and suitable attached to the Valve. Mounted on each FHR
Testing tags	e.g. As Per Relevant Australian Standard	As Per Relevant Australian Standard Mounted on each FHR

21.8 Hydrants

Item	Signage Wording	Lettering/Location
Hydrant and Hose Reel Cabinets (Internal)	The door of the cabinet shall have the words "FIRE HYDRANT & HOSE REEL"	In white lettering; - not less than 50mm high, - on a contrasting colour. as per AS2441 & 2419
Hydrant and Hose Reel Cabinets (External)	The door of the cabinet shall have the words "FIRE HYDRANT & HOSE REEL"	In white lettering; - not less than 75mm high, - on a contrasting colour. as per AS2441 & 2419
Hydrant	The symbol 'FH'	Marked within a circular line of inside diameter 100mm, of the same thickness and colouring as the lettering Mounted on the wall behind. as per AS2419
Hydrant Isolation Valves	"VALVE ## FIRE MAIN VALVE – SECURE OPEN" Each Valve is to be individually numbered and listed on the block plan.	All Hydrant Valves shall be tagged. Tags shall be of Aluminium Engraved with Lettering 8mm High. Mounted on each hydrant Isolation Valve.
Hydrant Booster / Cabinets	Fire brigade booster connections shall be marked with the words "HYDRANT BOOSTER"	In letters; - not less than 50mm high, - in a colour contrasting with that of the background as per AS2419
Hydrant System Boost Pressure	'TEST PRESSURE 700kPa' OR 'TEST PRESSURE 250kPa' 'BOOST PRESSUREkPa'	In letters; - not less than 25mm high, - in a colour contrasting with that of the background as per AS2419 Displayed at the hydrant booster
Hydrant Block Plan	In accordance with AS2419.1 A block plan shall be provided in the following locations: 1. At the hydrant booster location 2. In the Hydrant pump room 3. Adjacent the FIP	Shall be; - engraved on aluminium, - minimum A2 in size.
Fire Control Room	The external face of the door to the fire control room must have a sign with the words – "FIRE CONTROL ROOM"	In letters of not less than 50mm high and of a colour which contrasts with that of the background.

Item	Signage Wording	Lettering/Location
		Mounted on the door to the room. As per The BCA specification E1.8 Fire Control Centres.
The Following Signage shall be Provided: At the Front of the Building and adjacent to the Fire indicator Panel.	Shall be marked with the words "FIRE HYDRANT BOOSTER LOCATED IN?????"	In letters not less than 50mm high, in a colour contrasting with that of the background. Adjacent to the Fire indicator Panel. Fire and Rescue VIC Requirement
Valve straps	e.g. As Per Relevant Australian Standard	As Per Relevant Australian Standard Mounted on the booster set and any isolation valves on the fire hydrant service.
Valve tags	e.g. FH1	38mm in diameter, 12.7mm high black filled numbers. Made from approx. 1mm brass with a 5mm hole and suitable attached to the Valve Mounted on each FH
Testing tags	e.g. As Per Relevant Australian Standard	As Per Relevant Australian Standard Mounted on each FH

21.9 Hydrants with tanks

Item	Signage Wording	Lettering/Location
Hydrant Tank/Sprinkler Tank	Fire brigade booster tanks shall be marked with the words "FIRE HYDRANT TANK"	In letters not less than 100mm high, in a colour contrasting with that of the background as per AS2419. Located external to the Tank
Hydrant Tank Water Level	Fire brigade booster tanks water level indicator valves shall be marked with the words "FIRE HYDRANT TANK WATER LEVEL INDICATOR"	In letters not less than 50mm high, in a colour contrasting with that of the background as per AS2419. Located external to the Tank adjacent the water level indicator

Item	Signage Wording	Lettering/Location
Hydrant Tank Quick Fill Valve	Fire brigade booster tank quick fill valves shall be marked with the words “FIRE HYDRANT TANK QUICK FILL VALVE”	In letters not less than 50mm high, in a colour contrasting with that of the background as per AS2419. Located external to the Tank adjacent the quick fill valve.

21.10 Fire extinguishers

Item	Signage Wording	Lettering/Location
Fire Extinguishers	Signage shall be Relevant to the type of extinguisher installed	Extinguishers shall have their locations clearly indicated by placement of location signs. Located on the wall behind each extinguisher.

22. General

22.1 Floor Penetrations

Floor Penetrations	
Manufacturer	The Wondercap Company
Contact Details	Ph: 1300 308 205
Model	Wondercap Kit (Base & Cap & Puddle Flange) WC100K – Toilets and Floor wastes WC50K - Future Basins and sinks etc.
Dimensions	100mm drain outlet WC100K 50mm drain outlet WC50K
Material	Body – UPVC
Location	To be used for all outlets
Note	Where fixtures are unlikely to be connected for a delayed period of time grate seals shall be installed inside each outlet to prevent sewer gases escaping. They can then be removed when fixture installation occurs.

22.2 Clear Outs – Internal

Clear Outs – Internal	
Manufacturer	SMC Stainless
Contact Details	Ph: (02) 4735 5666
Model	Code CO-4 100mm Push in Round Cleanout with rubber ring
Material	Stainless Steel

22.3 Clear Out – Garden Areas (100mm)

Clear Out – Garden Areas (100mm)	
Manufacturer	Plastec
Contact Details	Ph: (07) 5413 4444
Model	11676 – 100mm 15012 – 150mm
Description	100mm Bolted Trap Surround
Material	Plastic

22.4 Overflow Gully (100mm) – Standard

Overflow Gully (100mm) – Standard	
Manufacturer	Plastec
Contact Details	Ph: (07) 5413 4444
Model	11649
Description	100mm Overflow relief gully
Material	Stainless Surround with Plastic Grate

22.5 Overflow Gully (100mm) – Flood Prone Areas

Overflow Gully (100mm) – Flood Prone Areas	
Manufacturer	iPlex
Contact Details	Ph: 1300 047 539
Model	D128100DORC
Description	100mm Overflow relief cap.
Material	Plastic

22.6 Overflow Gully (100mm) – Flood Prone Areas

Swivel And Expansion Combination Joint	
Manufacturer	Plastec
Contact Details	Ph: (07) 5413 4444
Model	16365
Description	100mm Swivel and Expansion Combination Joint
Material	Plastic
Location	Connections to in ground tanks and Where sewer drainage runs through slabs/footings and/or beams.

22.7 Swivel And Expansion Combination Joint

Swivel And Expansion Combination Joint	
Manufacturer	Plastec
Contact Details	Ph: (07) 5413 4444
Model	16365
Description	100mm Swivel and Expansion Combination Joint
Material	HDPE
Location	Connections to in ground tanks and Where sewer drainage runs through slabs/footings and/or beams.

22.8 CLEAR OUTS – EXTERNAL (100mm)

CLEAR OUTS – EXTERNAL (100mm)	
Manufacturer	SMC Stainless
Contact Details	Ph: (02) 4735 5666
Model	Code CO-4 100mm Push In Round Cleanout with rubber ring
Material	Stainless Steel

22.9 CLEAR OUT – EXTERNAL (150mm)

CLEAR OUT – EXTERNAL (150mm)	
Manufacturer	SMC Stainless
Contact Details	Ph: (02) 4735 5666
Model	Code CO-6-D
Description	150mm Push In Round Cleanout with rubber ring
Material	Stainless Steel

22.10 Basket Trap Floor Waste – 150mm Round Vari-Level

Basket Trap Floor Waste – 150mm Round Vari-Level	
Manufacturer	SPS - Speciality Plumbing Supplies
Contact Details	Ph: (02) 9416 8031
Model	SPS 150mm Round Vari-Level Bucket Trap Includes Stainless Steel Sediment Basket and Secondary Strainer Disk. Height Adjustable Trap Assembly with Lift out Grate
Material	Stainless Steel Grate and Frame
Dimensions	150mm Round Grate 100mm Outlet – Suited to Copper, CI and HDPE
Note	Not Suitable for Vinyl Floor Finishes
Location	Commercial Kitchen???

22.11 Basket Trap Floor Waste – 150mm Square Vari-Level

Basket Trap Floor Waste – 150mm Square Vari-Level	
Manufacturer	SPS - Speciality Plumbing Supplies
Contact Details	Ph: (02) 9416 8031
Model	SPS 150mm Square Vari-Level Bucket Trap Includes Stainless Steel Sediment Basket and Secondary Strainer Disk.

Basket Trap Floor Waste – 150mm Square Vari-Level	
	Height Adjustable Trap Assembly with Lift out Grate
Material	Stainless Steel Grate and Frame
Dimensions	150mm x 150mm Square Grate 100mm Outlet – Suited to Copper, CI and HDPE
Note	Not Suitable for Vinyl Floor Finishes
Location	Commercial Kitchen???

22.12 Basket Trap Floor Waste – 150mm Square Vari-Level (Garbage/Bin Room)

Basket Trap Floor Waste – 150mm Square Vari-Level (Garbage/Bin Room)	
Manufacturer	SPS - Speciality Plumbing Supplies
Contact Details	Ph: (02) 9416 8031
Model	SPS 150mm Square Vari-Level Bucket Trap Includes Stainless Steel Sediment Basket and Secondary Strainer Disk. Height Adjustable Trap Assembly with Lift out Grate
Material	Stainless Steel Grate and Frame
Dimensions	150mm x 150mm Square Grate 100mm Outlet – Suited to Copper, CI and HDPE
Note	Not Suitable for Vinyl Floor Finishes
Location	Garbage Room???

22.13 Basket Trap Floor Waste – Vinyl Floor 100mm Round Push-In

Basket Trap Floor Waste – Vinyl Floor 100mm Round Push-In	
Manufacturer	SPS - Speciality Plumbing Supplies
Contact Details	Ph: (02) 9416 8031
Model	SPS 10mm Push-in Vinyl Floor Bucket Trap Includes Stainless Steel Sediment Basket and Secondary Strainer Disk. Height Adjustable Trap Assembly with Lift out Grate
Material	Stainless Steel Grate and Frame
Dimensions	100mm Round Grate 100mm Outlet – Suited to Copper, CI and HDPE
Note	Only be used in conjunction with sheet vinyl flooring.
Location	Commercial Kitchen???

22.14 Vinyl Floor Waste - Vinyl Floor 100mm Round Push-In – Retro Fit

Vinyl Floor Waste - Vinyl Floor 100mm Round Push-In – Retro Fit	
Manufacturer	SPS - Speciality Plumbing Supplies

Vinyl Floor Waste - Vinyl Floor 100mm Round Push-In – Retro Fit

Contact Details	Ph: (02) 9416 8031
Model	SPS 100mm Retro-Fit Vinyl Floor Bucket Trap Includes Stainless Steel Sediment Basket and Secondary Strainer Disk. Height Adjustable Trap Assembly with Lift out Grate
Material	Stainless Steel Grate and Frame
Dimensions	100mm Round Grate 100mm Outlet – Suited to Copper, CI and HDPE
Note	Only be used in conjunction with sheet vinyl flooring.
Location	Commercial Kitchen???

22.15 Sealed Floor Wastes - 100mm

Sealed Floor Wastes - 100mm

Manufacturer	SMC Stainless
Contact Details	Ph: (02) 4735 5666
Model	Code CO-4-W
Description	100mm Push In Round Cleanout Includes rubber ring with ability for future exchange with regular floor grate
Material	Stainless Steel
Location	All Sealed Floor wastes

22.16 Floor Wastes General

Floor Wastes General

Manufacturer	Ramtaps
Contact Details	Ph: (07) 3633 6000
Model	FWLCF100CP
Material	100mm round top leak control flange waste to suit 100mm PVC pipe
Note	Coated PVC-U grates and surrounds shall not be installed.
Location	All Areas

22.17 Floor Wastes Vinyl

Floor Wastes Vinyl

Manufacturer	Ramtaps
Contact Details	Ph: (07) 3633 6000
Model	VFWP100CP
Material	100mm vinyl floor waste set to suit 100mm PVC pipe
Note	Coated PVC-U grates and surrounds shall not be installed.
Location	All Areas

22.18 Anti-Sud Floor Waste

Anti-Sud Floor Waste	
Supplier	DrainFresh
Model	Drainfresh DFR100 – 100mm drop in grate (retrofit) DFRS100 – 100mm grate with round tiling surround DFSS100 – 100mm grate with square tiling surround
Material	Chrome plated brass
Location	Laundry Floor wastes

22.19 Floor Wastes – Anti Foaming Device

Floor Wastes – Anti Foaming Device	
Manufacturer	Grateseal
Contact Details	Ph: 1300 393 913
Material	Rubber
Location	Allow to install in all Laundry Floor Wastes

22.20 Reflux Valve - 100mm

Reflux Valve - 100mm	
Manufacturer	iPlex
Contact Details	Ph: 1300 047 539
Model	Iplex 100mm reflux valve with vertical extension riser to ground level to allow for removal
Material	PVC

22.21 Pipes Passing Through Footings And Beams

Pipes Passing Through Footings And Beams	
Manufacturer	Midlink
Material	25mm Styrolag foam lagging (Polystyrene)
AS/NZS 3500 requirement	Clause 3.8.2 (b) requires pipes passing through footings or walls, other than external walls below ground, it shall be installed with an annular space of not less than 25mm filled with a liner of Flexible material.
Location	Allow to install around all pipes that pass through footings and beams

22.22 Pipes Cast In Concrete

Pipes Cast In Concrete	
Manufacturer	Polilag
Model	100mm x 10 Meter Roll
Location	All PVC pipes cast in concrete

22.23 In-Sink Dry Basket Arrestor

In-Sink Dry Basket Arrestor	
Manufacturer	Vinidex
Contact Details	Ph: (02) 8278 0500
Model	IN-SINK DBATM
Material	Manufactured in 304 grade stainless steel
Location	
Note	This arrestor is a requirement of the Trade Waste Authority.

22.24 In-Sink Dry Basket Arrestor

In-Sink Dry Basket Arrestor	
Manufacturer	SMC Stainless
Contact Details	Ph: (02) 4735 5666
Model	WSCK-1-BV - IN-SINK Dry Basket Arrestor
Material	Manufactured in 304 grade stainless steel
Location	
Note	This arrestor is a requirement of the Trade Waste Authority.

22.25 Modular Grease Trap S Series - Mgts - Uv Stabilised, Polyethylene

Modular Grease Trap S Series - Mgts - Uv Stabilised, Polyethylene	
Manufacturer	Halgan
Contact Details	Ph: 1800 626 753
Model	MGTS 1000 - 1000 litre MGTS 1500 - 1500 litre MGTS 2000 - 2000 litre MGTS 3000 - 3000 litre MGTS4000 – 4000 litre MGTS5000 – 5000 Litre Refer installation Instructions 4.1 - for Moduling for 4000 and 5000 Prefabricated Inlet (HDPE) and Outlet Pipework–Code - HPWS
Lids	Above, Class B and D – Note Above Ground Requires Stand - - Concrete Infill Lids Available
Risers	Max 900mm Risers allowed
Above Ground	Above Ground require a Stand with an option of 300 mm high step can be installed near the 600 mm access lid to help with inspections inside the MGT
Backfill	Refer 5.2 – Installation Instructions 5.2.3.1 Excavation dimensions The excavated hole width shall be kept as narrow as practicable. The depth shall be not greater than 150 mm than the required depth. A 75 mm clearance is required at the sides of tank. 5.2.3.4 Backfill/Bedding material The backfill/bedding material shall be 1-part Portland cement to 4 parts clean sand. The backfill/bedding shall be thoroughly compacted by tamping at 300mm layers. The backfill/bedding material shall encase the whole tank.
Material	UV Stabilised, Polyethylene.
Pump out line	5.2.8 Vacuum Pumpout Line – (Installation Instructions) The vacuum pump out line is used by the cleaning contractor to pump out the MGT in restricted site applications. A 50 mm M.I. quick release coupling with dust caps is supplied with the unit. The vacuum line has to be extended to the external of the building for ease of access. The lines must be as straight and short as possible. Where bends are necessary, only long radius bends should be used. The size of the vacuum line pipe can be 50 mm for the first 6 metres and 80 mm thereafter. The piping and fittings material can either be class 12 pressure pipe or galvanised pipe (refer to material specification). A 600 mm long x 600 mm wide x 200 mm deep access area is required around the quick release coupling for ease of connection.
Location	

Modular Grease Trap S Series - MgtS - Uv Stabilised, Polyethylene

Note:

Installation to be in accordance with Manufacturers Guidelines.

Grease arrestor to be complete with all necessary inlets, outlets, lids, and other sundry equipment necessary for the installation as indicated on drawings.

22.26 Modular Grease Trap - Mgt - Uv Stabilised, Polyethylene

Modular Grease Trap - Mgt - Uv Stabilised, Polyethylene

Manufacturer	Halgan
Contact Details	Ph: 1800 626 753
Model	MGT550 – 550 litre MGT 1000 - 1000 litre MGT 1500 - 1500 litre MGT 2000 - 2000 litre MGT 3000 - 3000 litre MGT4000 – 4000 litre MGT5000 – 5000 litre Refer installation Instructions 3.2 - for Moduling Prefabricated Inlet (HDPE) and Outlet Pipework–Code - HPW
Lids	Above, Class B and D – Note Above Ground Requires Stand - - Concrete Infill Lids Available
Risers	Max 600mm Risers allowed.
Cleaning Platform	The MGT's are free standing and do not need a cleaning platform. A 300 mm high step can be installed near the 600 mm access lid to help with inspections inside the MGT.
Backfill	Refer 5.2 – Installation Instructions 5.2.3.1 Excavation dimensions The excavated hole width shall be kept as narrow as practicable. The depth shall be not greater than 150 mm than the required depth. A 75 mm clearance is required at the sides of tank. 5.2.3.4 Backfill/Bedding material. The backfill/bedding material shall be 1-part Portland cement to 4 parts clean sand. The backfill/bedding shall be thoroughly compacted by tamping at 300mm layers. The backfill/bedding material shall encase the whole tank.
Material	UV Stabilised, Polyethylene.
Pump out line	5.2.8 Vacuum Pumpout Line – (Installation Instructions) The vacuum pump out line is used by the cleaning contractor to pump out the MGT in restricted site applications. A 50 mm M.I. quick release coupling with dust caps is supplied with the unit. The vacuum line has to be extended to the external of the building for ease of access. The lines must be as straight and short as possible. Where bends are necessary, only long radius bends should be used. The size of the vacuum line pipe can be 50 mm for the first 6 metres and 80 mm thereafter. The piping and fittings material can either be class 12 pressure pipe or galvanised pipe (refer to material specification). A 600 mm long x 600 mm wide x 200 mm deep access area is required around the quick release coupling for ease of connection.
Location	

Note: Installation to be in accordance with Manufacturers Guidelines. Grease arrestor to be complete with all necessary inlets, outlets, lids, and other sundry equipment necessary for the installation as indicated on drawings.

22.27 Grease Arrestor - Halgan

Grease Arrestor - Halgan	
Manufacturer	Halgan
Contact Details	Ph: 1800 626 753
Model	Ecotec Grease Extractor S100
Material	UV stabilised, polyethylene.
Backfill	A 4 to 1 sand / cement mixture is required for backfill. The base shall be 150 mm thick concrete. The backfill around the side shall be 75 mm thick. Before backfilling the EGE must be filled with water
Location	
Note: Installation to be in accordance with Manufacturers Guidelines. Grease arrestor to be complete with all necessary inlets, outlets, lids, and other sundry equipment necessary for the installation as indicated on drawings. Lid shall be class – Above – Class B and D – Concrete Infill Lids Available Pump out line – Allow to install the pump out line from the grease arrestor to a location readily accessible for the vacuum truck. The line shall be pressure PVC-U Class 16 – 80mm. Provide KAMLOC couplings and Ball valve at each end in the horizontal position.	

22.28 Tundishes – Inwall

Tundishes – Inwall	
Manufacturer	SMC Stainless
Contact Details	Ph: (02) 4735 5666
Model	TU-RE-2
Material	Stainless Steel with clear Perspex face plate
Finish	Stainless Steel
Note	Air gap is at the font / bottom lip
Location	As indicated
Installation	Contractor is to install a 40mm DWV discharge pipe in wall and connect to the drainage above the Sink fixture trap. Pipes discharging over a tundish shall have an air gap of a size at least twice the internal diameter of the discharging pipe.

22.29 Tundishes For Boiling Water Unit

Tundishes For Boiling Water Unit	
Manufacturer	Zip
Contact Details	Ph: 1800 638 633
Type	Right angled Tundish
Installation	Contractor is to install a 20mm copper discharge pipe in wall and connect to the drainage above the Sink fixture trap. Pipes discharging over a tundish shall have an air gap of a size at least twice the internal diameter of the discharging pipe.

22.30 Tundish Under Bench

Tundish Under Bench	
Trap	
Manufacturer	Hepvo
Model	40mm waterless trap

Tundish Under Bench

Installation	The contractor is to co-ordinate the final location of the tundish to suit the requirements of the discharging fixture. Install the Hepvo trap complete with DWV PVC fittings to enable the entry point of the tundish to be finished with a 100mm tapered reducer. The waste pipe size, length and discharge point shall be in accordance with the table below.	
Note	Maximum permissible discharge temperature to the tundish is 65 degrees Celsius.	
Air Gap	Pipes discharging over a tundish shall have an air gap of a size at least twice the internal diameter of the discharging pipe.	
System Connected to	Minimum Discharge Pipe Size	Maximum Discharge Pipe length
Vented Drain	65mm	10 Metres
Fully Vented (Modified)	40mm	2.5 Metres
Single Stack	40mm	2.5 Metres
Single Stack (Modified)	40mm	2.5 Metres
Floor Waste Gully	40mm (No Trap Required)	10 Metres
Disconnecter Gully	40mm (No Trap Required)	6 Metres

22.31 Condensate Drainage Lagging - Thermotec

Condensate Drainage Lagging - Thermotec

Manufacturer	Thermotec
Contact Details	Ph: (02) 9771 6400
Model	Thermotec Quick Seal fire retarded closed cell polyethylene foam having a density of not less than 50kg/m3
Note: All condensate drainage shall be as per Sanitary Plumbing. All insulation shall be installed around the pipework surface as tightly as possible without gaps, with the edges and ends tightly butted together. All joints shall be taped using a 48mm adhesive backed PVC tape. Where necessary, the preformed sections shall be cut (using a sharp knife or scissors) and mitred, to ensure a tight fit around bends and at tees.	

22.32 Air Admittance Valve 6 FU's

Air Admittance Valve 6 FU's

Manufacturer	Studor Australia Pty Ltd
Contact Details	Ph: (02) 9698 7445
Model	Studor Premi-Aire 40 / Redi-Vent
Material	PVC-U
Air Flow	1.1 Litres per second
Fixture Units	6
Fixing	Screwed
Location	

22.33 Air Admittance Valve 60 FU's

Air Admittance Valve 60 FU's

Manufacturer	Studor Australia Pty Ltd
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Air Admittance Valve 60 FU's

Contact Details	Ph: (02) 9698 7445
Model	Mini-Vent
Material	PVC-U
Air Flow	7.5 Litres per second
Fixture Units	60
Fixing	Screwed
Location	

22.34 Air Admittance Valve For Lab Drainage 6 FU's

Air Admittance Valve For Lab Drainage 6 FU's

Manufacturer	Studor Australia Pty Ltd
Contact Details	Ph: (02) 9698 7445
Model	Studor Chem Vent
Material	PVC-U
Air Flow	1.1 Litres per second
Fixture Units	6
Fixing	Screwed
Location	It must be installed in a vertical position in an accessible location which permits free movement of air into the valve.

22.35 Air Admittance Valve 100 FU's

Air Admittance Valve 100 FU's

Manufacturer	Studor Australia Pty Ltd
Contact Details	Ph: (02) 9698 7445
Model	Maxi-Vent
Material	PVC-U
Air Flow	32 Litres per second
Fixture Units	1500 (branch drainage) 115 (stack vent)
Fixing	Push Fit rubber key sealed
Location	

22.36 Carbon Filter

Carbon Filter

Manufacturer	Studor Australia Pty Ltd
Contact Details	Ph: (02) 9698 7445
Model	Maxi-Filtra
Material	ABS
Air Flow	Pressure (Pa): 100 - Flowrate (l/s): 2 Pressure (Pa): 250 - Flowrate (l/s): 5 Pressure (Pa): 500 - Flowrate (l/s): 8
Temp Range	-20°C to +60°C
Fixing	Screwed
Location	

22.37 Boundary Trap - Lid

Boundary Trap - Lid

Manufacturer	SPS - Speciality Plumbing Supplies
Contact Details	Ph: (02) 9416 8031
Model	SPS 150mm Push In Round Cleanout with Concrete Surround.

Boundary Trap - Lid

	Code R200SCR/NS (Machines Stainless Steel, non-slip)
Material	Stainless Steel
Note	If not installed within a concrete slab area – a concrete surround shall be provided in accordance with as AS/NZS 3500
Location	As shown on the drawings

22.38 IPMF – Induct Pipe Mica Flap

IPMF – Induct Pipe Mica Flap

Manufacturer	ACE Gutters
Contact Details	Ph: (02) 9584 4777
Model	100mm Mica Flap
Location	As shown on the drawings

22.39 Combination Cooling Pit / Lint Trap – Single Inlet

Combination Cooling Pit / Lint Trap – Single Inlet

Manufacturer	SMC Stainless
Contact Details	Ph: (02) 4735 5666
Model	LT-MWM-AG-1
Material	Stainless Steel
Note	Above ground unit with a maximum 200 litre capacity LOCAL AUTHORITY APPROVED
Dimensions	600 x 800 x 700 deep
Location	

22.40 Combination Cooling Pit / Lint Trap – Dual Inlet

Combination Cooling Pit / Lint Trap – Dual Inlet

Manufacturer	SMC Stainless
Contact Details	Ph: (02) 4735 5666
Model	LT-MWM-AG-2
Material	Stainless Steel
Note	Above ground unit with a maximum 350 litre capacity LOCAL AUTHORITY APPROVED
Dimensions	1700 x 551 x 700 deep
Location	

22.41 Rubber Drainage Couplings – Sewer

Rubber Drainage Couplings – Sewer

Manufacturer	Fernco
Contact Details	Ph: (02) 9450 0766
Model	Stainless Steel Shear Rings – Available in sizes from 100mm to 300mm
Note	If required to be used, rubber drainage couplings must be the type nominated above

22.42 Tank Connection Rubber Seals

Tank Connection Rubber Seals

Manufacturer	Uni-Seal
Contact Details	Ph: (07) 3899 0611

Tank Connection Rubber Seals

Note	All tank connections are to be installed with the use of an approved Uni-Seal Device.
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22.43 Sewer Manhole

Sewer Manhole

Sewer manhole Precast type 600mm Diameter Access Chamber	
Base & Walls	Precast access chamber of reinforced concrete walls & base
Manufacturer	Humes
Type	Access Chamber
Material	Precast Concrete
Access Lid	Class B or Class D
Location	

22.44 Sewer Maintenance Shaft

Sewer Maintenance Shaft

600mm Diameter Access Shaft	
Base & Walls	Precast access chamber of reinforced concrete walls & base
Manufacturer	iPLEX
Type	EZIpit
Material	Plastic
Access Lid	Class B or Class D
Location	

22.45 Sealing Of Pipes Penetrating External Walls

Sealing Of Pipes Penetrating External Walls

Manufacturer	Projex / Link Seal
Contact Details	Ph: (02) 8336 1666
Note	Model Numbers may vary. Contact Projex for information.
Location	All pipe penetrations through external walls inground.

22.46 Stack Protector

Stack Protector

Manufacturer	Leda
Contact Details	Ph: 1300 780 450
Model	DG 41 F
Material	Pre galvanised mild steel
Note	Fasteners also supplied with Zinc or Galvanised coating for corrossions resistance Note: Care must be taken in the positioning of the Pipe Protectors so as not to interfere with the minimum building access requirements set out in the BCA.
Description	Can be finished in other colours Hot dip galvanised option Powdercoated 'Safety Yellow' standard

Stack Protector

Location	As required to protect services from damage by; ▪ Vehicular traffic ▪ Vehicle loading equipment ▪ Fork lifts ▪ Pallet trolleys ▪ Through traffic ▪ Doors and other structural elements
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23. Backflow protection

23.1 Backflow Cabinet

Backflow Cabinet	
Manufacturer	L. W. Gemmell Pty Ltd
Contact Details	Ph: (02) 9647 2770
Model	375 XLS complete (20mm – 50mm)
Specification	Maximum working water pressure 175 PSI Maximum working water temperature 180°F Hydrostatic test pressure 350 PSI End connections Threaded FNPT ANSI B1.20.1
Location	In the areas shown on the drawings

23.2 RPZD Backflow Preventer

RPZD Backflow Preventer	
Manufacturer	L. W. Gemmell Pty Ltd
Contact Details	Ph: (02) 9647 2770
Model	350 FSOSY complete (flanged) (65mm – 250mm) OR 350 XLS complete(screwed) (65mm – 250mm)
Specification	Maximum working water pressure 175 PSI Maximum working water temperature 140°F Hydrostatic test pressure 350 PSI End connections (Grooved for steel pipe) AWWA C606 (Flanged) ANSI B16.1 Class 12
Location	In the areas shown on the drawings

23.3 Backflow Cabinet

Backflow Cabinet	
Manufacturer	L. W. Gemmell Pty Ltd
Contact Details	Ph: (02) 9647 2770
Part Number	20-350/375CAB
	▪ Lockable Perspex or S/S front ▪ 40mm Drain ▪ Male connections ▪ Valves fully serviceable inside box ▪ ¼ turn ball valves and line strainer included ▪ Fits 2 x 20mm RPZD's ▪ Adjustable faceplate depth ▪ Adjustable brackets ▪ Vent slots to reduce condensation.
Material	Stainless Steel
Box	Width - 500mm Height- 630mm

Backflow Cabinet	
	Depth - 90mm
Face	Width – 555mm Height – 590mm

23.4 DCV Backflow Preventer

DCV Backflow Preventer	
Manufacturer	L. W. Gemmell Pty Ltd
Contact Details	Ph: (02) 9647 2770
Model	950XLT2 (20mm – 50mm)
Specification	Maximum working water pressure 175 PSI Maximum working water temperature 180°F Hydrostatic test pressure 350 PSI End connections Threaded ANSI B1.20.1
Location	In the areas shown on the drawings

23.5 RPZ Backflow Preventer

RPZ Backflow Preventer	
Manufacturer	All Valve Industries
Contact Details	Ph: (02) 8543 5111
Model	Apollo 4A-200 (15mm – 50mm)
Specification	Use Apollo 4A-200 series RPZ backflow prevention device with male BSP ends, rated to 82°C. Watermarked to AS2845.1. With self-contained check modules with individual accessible checks. 5-year warranty against manufacturers defects. Optional: with non-splash tundish attachment.
Location	In the areas shown on the drawings

23.6 RPZ Backflow Preventer

RPZ Backflow Preventer	
Manufacturer	All Valve Industries
Contact Details	Ph: (02) 8543 5111
Model	Apollo 4A-200 (65mm – 250mm)
Specification	Use Apollo 4A-200 series RPZ backflow prevention device flanged Table E, rated to 60°C. Watermarked to AS2845.1. With TriForce stem guided check valves with reversible silicone seats. 5 year warranty against manufacturers defects.
Location	In the areas shown on the drawings

23.7 RPZ Backflow Preventer

RPZ Backflow Preventer	
Manufacturer	All Valve Industries

RPZ Backflow Preventer	
Contact Details	Ph: (02) 8543 5111
Model	Apollo 4A-100 (65mm – 250mm)
Specification	Use Apollo 4A-100 series double check valve flanged Table E, rated to 60°C. Watermarked to AS2845.1. With TriForce stem guided check valves with reversible silicone seats. 5 year warranty against manufacturers defects.
Location	In the areas shown on the drawings

23.8 RPZ Backflow Preventer

RPZ Backflow Preventer	
Manufacturer	All Valve Industries
Contact Details	Ph: (02) 8543 5111
Model	Apollo 4A-600 (65mm – 250mm)
Specification	Use Apollo 4A-600 series double detector check valve flanged Table E, rated to 60°C. Watermarked to AS2845.1. With TriForce stem guided check valves with reversible silicone seats. 5 year warranty against manufacturers defects.
Location	In the areas shown on the drawings

23.9 DCV Backflow Preventer

DCV Backflow Preventer	
Manufacturer	All Valve Industries
Contact Details	Ph: (02) 8543 5111
Model	Apollo 4A-100 (20mm – 50mm)
Specification	Use Apollo 4A-100 series double check valve backflow prevention device with male BSP ends, rated to 82°C. Watermarked to AS2845.1. With self-contained check modules with individual accessible checks. 5 year warranty against manufacturers defects.
Location	In the areas shown on the drawings

23.10 DCV Backflow Preventer

DCV Backflow Preventer	
Manufacturer	Zurn
Contact Details	Ph: 1300 009 876
Model	LCB (65mm – 250mm)
Specification	Use LCB series double check valve backflow prevention device. Max working temp: 65C Max working Pressure: 1200kPa End Connections: BSPM
Location	In the areas shown on the drawings

23.11 SS Enclosures With Conbraco Apollo 4a RPZ

SS Enclosures With Conbraco Apollo 4a RPZ	
Manufacturer	All Valve Industries
Contact Details	Ph: (02) 8543 5111
Model	Apollo SSRE-P-4A (15mm - 40mm)
Specification	Stainless Steel Recessed Box w/Perspex Lid. Single or Double RPZ with lockable door. 40mm Plug and waste. Adjustable depth up to 45mm. Also available with Solid Lid.
Location	In the areas shown on the drawings

24. Pumps & control equipment

24.1 Installation of pumps

Pumps shall:

- a. be installed on a base to suit satisfactory operation of the pump;
- b. have vibration eliminators at the base of the pump, on the suction side and the delivery side of the pump, so as to minimize the transmission of noise into the building structure and along the piping system and to prevent undue stress being placed on the pump;
- c. have isolation valves on the delivery side and suction side of the pump;
- d. have a non-return valve on the delivery side of the pump before the isolation valve;
- e. have pressure gauges on the inlet and outlet of the pump; and
- f. have unions or flanges to enable the pump's removal.

24.1.1 Pump Controls

Pump control shall be provided to alternate between duty and standby pump operation over a 24-hour period.

24.1.2 Electrical Supply & Wiring

Electrical wiring between pumps and control panel shall be under the Hydraulic Services Sub-Contract. Power supply to the pump control panel shall be under the Electrical Services Sub-Contract.

Sewer Pumps And Station	
Duty	10 litres/second
Manufacturer	Sulzer
Model	Submersible Chopper Pump Type ABS XFP 100E
No off pumps	2
Quote	N/A
Conduits	Conduits between Control Panel and Pit shall be PVC-U electrical conduit to AS 3000. Conduit shall be 65mm diameter from pit to control panel with long radius bends.
Float Switches	Float Switches and wiring shall be suitable for corrosive marine environments.
Inverts	The invert levels of all connections shall be confirmed prior to ordering pump out pit.
Note	The vent from the pit shall be 100mm discharging to open air in accordance with AS 3500.1. Mica flaps shall not be installed. A pump data Plate must be installed within the control box door for inspection purposes. This is to be a minimum size of 75mm x 50mm square.
Location	As shown on site plan

The pumps shall be installed in the collection pit and shall be kept a minimum of 450 mm apart and clear of near walls. Each pump shall be complete with:

- Approved vibration eliminators between pump and rising main connection.
- Hose and electrical cables to be of sufficient length to allow removal of pump without entering pit.
- Approved brass unions at pump and rising main connection.

- Pumps shall be on guide rails with galvanised mild steel lifting chains. Lifting chains shall be located adjacent to access cover.

At completion of Building Contract, the Sub-Contractor shall remove pumps, clean, service and replace worn or damaged parts.

25. Plumbing contractor check list

The Following items are required to be supplied by the plumbing contractor to the Superintendent during the course of the project:

Project			
Project number			
Contractors Name			
Licences Name			
Start Date			
Finish Date			
No	Item	Submitted	Approved
	Copy of receipts for payment of authority application fees		
	Copy of all Authorities Applications for connection		
	Tender Break up / schedule of rates		
	Shop Drawings		
	Core Hole Drawings		
	As Built drawings		
	Operation and maintenance manuals		
	Survey of completed detention tank		
	Valve Schedule		
	Flow and Pressure Test For Fire Hydrant System		
	Flow and Pressure Test For Fire Hose Reel System		
	Preliminary Hydrant System Block plan for approval prior to manufacturing.		
	Contact Local Fire Brigade office. Inspected by: _____ from Fire and Rescue QLD		
	Trade Waste Inspection Inspectors Name: _____ Phone No: _____		
	Written guarantee for sanitary fixtures, fittings and all equipment from the manufacturer for a period of two years. The guarantee shall cover all costs associated with replacement of a faulty article within the two-year period.		
	Backflow registration with the water authority		
	Thermostatic mixing valve certification		

	Inspection and test plan signed off by the Superintendent and the Builder		
	Certification of the test pressures for Cold Water Fire Hose Reels Hot Water Fire Hydrant Gas		
	Certification of Fire Stop Collars in accordance with AS AS1530.4. and AS4072.1 in particular Clause 4.6.1, Clause 4.6.2, Clause 4.6.3, Clause 4.6.4. Manufacturer:..... Model:..... Fire Rating:		
	Certificate of currency for Public Indemnity and Public Liability insurance.		

26. Periodic maintenance procedures for hydraulic services

The following maintenance procedures and schedule identify the preventative and mandatory maintenance requirements. The maintenance procedures for each service or service component are identified with maintenance parameters and duration between services. The maintenance procedures also clarify if the works are preventative or mandatory. Preventative maintenance items ensure the correct operation of the hydraulic systems, mandatory maintenance satisfies Authorities statutory requirements for health and safety. The maintenance schedule is derived from the maintenance procedure list creating an annual program to monitor and maintain the hydraulic systems.

26.1 Stormwater

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Downpipes	Check drainage / Leaking joints	Monthly	No	Yes
Stormwater Pits / Grated Drains	Removal of any debris and silt / check grates are sound.	3 Months	No	Yes
RWO's	Remove any debris - check clear	Monthly	No	Yes
Stormwater Pump Stations	Routine service by Pump Suppliers staff (or authorised agent) 1. Check power supply is on and connections in control panel are secured. 2. Check isolating and non-return valves are in good condition and are not seized 3. Adjust pump coupling and mechanical seal spring tension 4. Check float switches for pump start, pump stop and high-level alarm 5. With pump running manually, check the following: ▪ Pump rotation ▪ Current draw ▪ Panel indicator lights ▪ Suction and discharge pressures ▪ Power supply voltages	6 Months	No	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	▪ Contactor overload settings (adjust to motor maximum current draw) ▪ High / Low level alarm 6. Switch pumps and duty selectors to automatic 7. Adjust pressure switch settings to desired pressures for duty and stand by pumps. Check pit condition and any presence of silt build up			

26.2 Sanitary plumbing & drainage

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Surcharge Gully	Check for water seal and gully clear.	6 Months	No	Yes
Pipework Generally	Check for leaks / damages or blockages.	12 Months	No	Yes
Basket Floor Waste	Remove any debris – check clear	Monthly	No	Yes
Sewer Pump Stations	Routine service by Pump Suppliers staff (or authorised agent) 1. Check power supply is on and connections in control panel are secured. 2. Check isolating and non-return valves are in good condition and are not seized 3. Adjust pump coupling and mechanical seal spring tension 4. Check float switches for pump start, pump stop and high-level alarm 5. With pump running manually, check the following: ▪ Pump rotation ▪ Current draw	6 Months	No	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	- Panel indicator lights - Suction and discharge pressures - Power supply voltages - Contactor overload settings (adjust to motor maximum current draw) - High / Low level alarm 6. Switch pumps and duty selectors to automatic 7. Adjust pressure switch settings to desired pressures for duty and stand by pumps. Check pit condition and any presence of silt build up			
Sewer Pump	Routine service by Pump Suppliers staff (or authorised agent) 8. Check isolating and non-return valves are in good condition and are not seized 9. Adjust pump coupling and mechanical seal spring tension 10. With pump running manually, check the following: - Current draw - Suction and discharge pressures - Power supply voltages	6 Months	No	Yes

26.3 Tradewaste

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Lint Arrestor	Basket arrestors should have the basket removed, scrapped and cleaned regularly to ensure proper continual operation of the unit.	Weekly	No	Yes
Grease Arrestor	1. Clean the grease arrestor regularly by using an EPA approved licensed liquid	Trade Waste Agreement	Yes	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	waste transporter. Regular cleaning will help to reduce odours and prolong the life of the arrestor. 2. Ensure that all drainage and trap vents are in good repair. 3. Check that lids, frames and concrete surrounds are in good order. 4. If required by the local authority, retain on site, in an accessible location, the tradewaste docket provided after every pump out. 5. Recycle or spent cooking oils. 6. Do not dispose of unwanted oil and grease down the drain. 7. The use of solvents, enzymes, mutant bacteria odour control agents or pesticides in grease interceptor traps is prohibited unless specifically approved by the Tradewaste Officer. 8. Do not add any cleaning agents or substances which may affect the contents of the arrestor causing undue corrosion. Do not attempt to clean the arrestor yourself.			

26.4 Dosed water

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Ozone Laundry System	Clean lint filter on micro ozone laundry support system	Weekly	No	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Ozone Laundry System	Routine Service be Ozone Laundry Support System (or authorised agent) 1. Monitoring of the wash load qualities each service interval: ozone titration for correct dosage of ozone, replacement of filters and non-return valves, replacement of ozone delivery hose as and when required. 2. Test all safety functions of the system 3. Clean inside of cabinets for lint and dust	12 Months	Yes	Yes

26.5 Cold water

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Exposed Pipework	Check for leaks and condition of pipework.	12 Months	No	Yes
Stop Valves	Shut down and open valve to ensure ease of operation.	12 Months	No	Yes
Fixtures / Taps Generally	Inspect for correct operation and soundness.	3 Months	No	Yes
Backflow Devices	As per 3500 and The Plumbing Code of Australia 1. Check pipework for leaks 2. Check and clean Y-type strainer as necessary 3. Check operation of valve 4. Carry- out statutory maintenance and re-testing, complete report and forward to Local authority Corporation	12 Months	Yes	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Domestic Pumpset	Routine service by Pump Suppliers (or authorised agent) undertaking system check of unit incorporating the following: 1. Check amperage and insulation resistance 2. Check control panel, panel lights, auto alternation, isolation valves, check valves, shaft seal, bearings, pump rotation, VFD + Transducer.	6 Months	No	Yes

26.6 Hot / warm water services

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Exposed Pipework	Check for all leaks and condition of exposed pipework in particular lagging	12 Months	No	Yes
Hot Water Heating Plant	1. Check temperature levels and settings. 2. Ensure area is clear of flammable material and other unwanted materials.	Weekly	No	Yes
Hot Water Heating Plant	Ease lever to check for operation	6 Months	No	Yes
Hot Water Heating Plant	Routine Service by Manufacturers staff (or authorized agent).	Quarterly	No	Yes
Boilerland Hot/Warm Water Heating Plant	1. Checks are per weekly Checklist. 2. Dose Expansion Tank with correct amount of Water Treatment. 3. Check operation of Expansion Tank Ball Valve. 4. Check temperature of warm water loop at a remote outlet and adjust set value PID control if necessary. 5. Check operation of all motorized valves and fail safe valves. 6. Move every valve from one extremity to the other to ensure free operation.	Monthly	No	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Boilerland Hot/Warm Water Heating Plant	1. Remove boiler Smoke Box Lid – Check for soot deposits. If necessary, use vacuum cleaner to remove soot from bottom of furnace, furnace walls and top of tube plate. Ensure that burner unit and other controls are covered during cleaning operation. 2. Clean out all dust and fluff from air passage ways of burner, fan blades and combustion head. 3. Clean the Flame Sensing Flame Rod or UV cell, paying particular attention to the glass window which in some cases takes the form of a narrow slot in the mounting tube. 4. Check the electrodes – high tension (HT) spark and Flame Rod if used, for cracks or other signs of deterioration. Make sure they are clean and tightly supported in the correct position. 5. Run Burner and carry out combustion checks as per “Commissioning” details. 6. Carry out normal Monthly Check.	Annually	No	Yes
UV Warm Water Disinfection	Isolate power and water supply to unit, remove tube and clean	6 months	No	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	with soft cloth and surgical alcohol			
UV Warm Water Disinfection	Replace UV tube	14 Months	Yes	Yes
Thermostatic Mixing valves	Temperature testing at fixture outlets to be recorded in log book	Monthly	No	Yes
Thermostatic Mixing valves	Test valve outlet temperature, correct fail safe operation, and ensure cartridge components are operable and replaced as required	12 Months	Yes	Yes
Thermostatic Mixing valves	Replace mixing valve cartridge	3 years	Yes	Yes
VIC Health Temperature Testing Requirements	Test water temperature at basin or shower outlet for minimum 20 seconds and record in log book, if temperature is below or above 40.5 – 43.5 degrees, isolate system and refer VIC Health Circular 2004/10 item 8.2	Weekly	Yes	Yes
VIC Health Water Quality Requirements	Microbial water sampling to confirm less than 10 Legionella colony forming units per millilitre of warm water	6 Months	Yes	Yes

26.7 Fire hydrant services

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Log book	All maintenance and testing to be recorded in Log Book	Monthly	Yes	Yes
Pump Service	Routine service by Pump Suppliers staff (or authorised agent) undertaking system check of unit incorporating the following: 1. Remove any corrosion from battery terminals 2. Clean, or replace as required, engine fuel sludge	6 Months	No	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	/ sediment trap and air filter elements 3. Remove and clean heat exchanger strainer(s)			
Pump Service	Yearly maintenance is the same as 6 months with the addition of the following: 1. Replace engine oil with new oil that meets the engine manufacturers specification 2. Replace engine oil filter and fuel filters 3. Check the condition of the fuel and replace if defective 4. Inspect Engine cooling hoses and replace where necessary 5. Flush engine cooling system and refill using manufacturers approved corrosion inhibitor 6. Pressure test engine cooling system 7. Inspect pump / driver coupling for wear and alignment 8. Grease pump bearings to the manufacturers' specifications 9. Ensure all non-return valves are operating freely and are seating correctly 10. Disassemble pump and repack pump glands	12 Months	Yes	Yes
Pump Service	Replace batteries on diesel pump	2 Years	Yes	Yes
Testing of Pumps	1. Check that the main isolating switch is in the in position and the green power supply lamps are illuminated and that no red warning lamps are on 2. Check that the fuel tank is full	Monthly	Yes	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	3. Check for any obvious signs of physical damage or deterioration 4. Test the float charge voltage of both the monitor and engine start batteries and record the result. 5. Start pumpset by reducing the applied water pressure to the starting device and run engine continuously for not less than 10min on the first automatic start and check that the driver achieves full speed within 15s of starting 6. Record the starting pressures, test run time and the hour meter reading at completion of the test 7. During and after the running period Check: ▪ Pump operates at correct discharge pressure. Record suction and discharge pressure ▪ Pump gland and drain operates efficiently ▪ Out of balance condition or abnormal noises are not evident ▪ Water, oil and fuel leaks are not evident and fittings on ancillary equipment are secure ▪ Battery charger or alternator / generator is operating correctly ▪ Battery charger power failure alarm operates correctly ▪ Cooling is discharging			
Testing of Pumps	Yearly maintenance is the same as 6 months with the addition of the following: 1. With the pump room door closed and the pump testing	12 Months	Yes	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	technician present, record results from the following tests: ▪ Run the pump set at shut off (zero flow) for 3 min allow all equipment to attain normal operating temperature ▪ Run the pump set at 130% of duty flow for 4 min and record the results. ▪ Reduce the flow to duty flow for sufficient time to record the water supply proving test results ▪ Further reduce the flow until shut off (zero flow) is achieved and continue to run the pumpset until total run time has reached 10min 2. Simulate an engine fail to start and ensure that engine start cycling requirements and alarm activations are satisfied			
Isolating Valves	1. Check all isolating valves including underground key-operated valves are in the correct operating position. 2. Check all above ground valves are secured	6 Months	Yes	Yes
Hydrant Landing Valves	Check all hydrant valves are: 1. Accessible. 2. Hand-wheels are securely fitted. 3. Blanking caps are in good condition.	6 Months	Yes	Yes
Booster Assembly	Check booster assembly: 1. Is accessible. 2. Hand-wheels are securely fitted. 3. Pressure gauges and blanking caps, where fitted, are in good condition	6 Months	Yes	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	4. For condition of washers on booster assembly connection inlets and replace if signs of deterioration. 5. For legible block plans incorporating working and test pressures. NOTE: The block plan should be appropriately located in the pump room and at the booster connections.			
Block Plan	Check block plan is up to date, legible and in an appropriate location.	6 Months	Yes	Yes
Test	Yearly Tests are the same as the 6 monthly with inclusion of the following: Undertake flow test on hydrant system ensuring minimum operating pressure and flow is achieved in accordance with AS2419 - 10litres/second @ 700kPa	12 Months	Yes	Yes
Test	5 Yearly Tests are the same as the 6 monthly with inclusion of the following: Pressure test on incoming water supply to ensure hydrant system design criteria is satisfied.	5 years	Yes	Yes

26.8 Fire hose reels

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Fire Hose Reels	1. Check that the operating instructions are legible. 2. Check for any damage or corrosion of components that could adversely affect the operation of the reel. 3. Check that all hose reel cabinets are accessible, clear of extraneous materials, clearly and correctly marked and in good repair.	6 Months	Yes	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
	4. With the nozzle closed and stop valve open, test that the hose reel can be unwound freely in its intended direction by unwinding at least 5m of hose. 5. With the nozzle closed, pressurize the reel by opening the stop valve. Determine if there are any leaks from the gland, nozzle, stop valve, hose or any fittings. 6. Test water to ensure it is able to flow through the reel by opening and closing the nozzle.			
Fire Hose Reels	Yearly maintenance is the same as 6 months with the addition of the following: 1. Check all hoses for kinking, excessive damage or wear, or collapse 2. With the stop valve and nozzle fully opened, measure the water flow rate from the most disadvantaged hose reel – 0.66L/s All maintenance and testing to be recorded in Log Book	12 Months	Yes	Yes
Combined Fire Hose Reels and Domestic Pump Set	Check all valves are secured in the open position, check and record operating pressure, check pump operation and ensure all indicator lights are operational and not indicating fault.	6 Months	Yes	Yes

26.9 Rainwater harvesting system services

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Tanks	Empty tanks of all water and debris, clean thoroughly with chlorine	3 Yearly	No	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Stormwater Re-use Pump	1. Check power supply is on and connections in control panel are secured. 2. Check pump set mountings are in good order and secure 3. Check water supply is on and bleed pump of any trapped air 4. Check isolating and non-return valves are in good condition and are not seized 5. Adjust pump coupling and mechanical seal spring tension 6. With pump running manually, check the following: ▪ Pump rotation ▪ Current draw ▪ Panel indicator lights ▪ Suction and discharge pressures ▪ Power supply voltages ▪ Contactor overload settings (adjust to motor maximum current draw) 7. Drain water from air vessels. Check and adjust air pressure in each vessel to be the same 8. Switch pumps and duty selectors to automatic 9. Adjust pressure switch settings to desired pressures for duty and stand by pumps.	6 Months	No	Yes
Filtration – Backwash	Clean plastic parts with a soft damp cloth only; do not use solvents, detergents, or acidic cleaning agents	6 Months	No	Yes
Filtration Bag	Clean plastic canister and replace filter cartridge	3 Months	No	Yes

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
UV Disinfection	Isolate power and water supply to unit, remove tube and clean with soft cloth and surgical alcohol	6 months	No	Yes
UV Disinfection	Replace UV tube	36 Months	Yes	Yes

26.10 Fire protection

Item	Maintenance	Period	Mandatory Maintenance	Preventative Maintenance
Log book	Weekly Recording in Log Book	Weekly	Yes	Yes
Sprinkler Pumps	These systems require a statutory inspection and testing procedure to be followed as per AS 1851.4	Monthly	Yes	Yes
EWIS	These systems require a statutory inspection and testing procedure to be followed as per AS 1851.4	Weekly	Yes	Yes
Sprinkler valves	These systems require a statutory inspection and testing procedure to be followed as per AS 1851.4	Monthly	Yes	Yes
Extinguishers	These systems require a statutory inspection and testing procedure to be followed as per AS 1851.4	6 Months	Yes	Yes

27. Periodic maintenance procedures for hydraulic services

HK Solutions Material Change Submission

No. _____

Project: _____

Job No: _____

Attention:

Date:

From:

Total Pages:

Subject:

1. Description of Material/Item to be changed:
2. What Material/Item has been specified by HK Solutions? **Attach Manufacturers Technical Data and specification*
3. What is the contractors preferred Material/Item? **Attach Manufacturers Technical Data and specification*
4. Reason the contractor wish's to change the specified Material/Item:
5. How does the Material/Item change benefit the client?
6. What is the cost benefit to the client of the Material/Item Change?
7. Does the product in question comply with all relevant codes and standards?

Based on the information supplied this submission is recommended/not recommended by HK Solutions

Superintendent:.....
(Signature) Date Name (please print)

MATERIAL CHANGE REQUEST

1. ACCEPTANCE OF MATERIAL CHANGE IS DEPENDANT ON CLIENT/SUPERINTENDANT APPROVAL.
2. THIS DOCUMENT HAS BEEN EXAMINED TO DETERMINE WHETHER OR NOT IT IS SUITABLE FOR USE FOR THE PURPOSES OF THE CONTRACT
3. ENDORSEMENT DOES NOT RELIEVE THE CONTRACTOR OF REPONSIBILITY FOR COMPLIANCE WITH THE TERM AND CONDITIONS OF THE CONTRACT AND CO-ORDINATION
4. ANY AMENDMENTS SHOWN DO NOT CONSTITUTE AN ORDER OR AUTHORITY FOR A PRICE VARIATION TO THE CONTRACT
5. THIS REVIEW EXTENDS TO GENERAL TECHNICAL DESIGN INTENT ONLY

If not approved, please indicate reason below:

****This application cannot be processed without the required attachments.***



Completed forms are to be emailed to Leo.campbell@HK_Solutions.com.au who will then forward to the client.

Appendix A – Hydraulic Drawing